

Nonlinear Singularities at Mixed Hyperelastic Corners: Operator Blow-up, Creasing, and Structural Bifurcation

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Abstract

This paper investigates the coupling between geometric stress concentrations and nonlinear structural instabilities in compressible hyperelasticity, focusing on a corner subject to mixed Dirichlet–Neumann boundary conditions. Existing literature treats fracture, cavitation, and flat-surface creasing in isolation, leaving the mixed-boundary corner unresolved. We establish a dichotomy in the asymptotic behaviour of the linearised tangent operator, dictated by the loading direction and the volumetric energy penalty. In tension, a mildly growing volumetric penalty yields a benign singularity where the tangent operator remains strongly elliptic and Fredholm: the standard Kondratiev edge exponents of linear elasticity govern the mesh-relevant intermediate scales, while at the vertex itself the extreme stretch screens the volumetric coupling and the increments are regular. Under compression, the local volume contraction drives an algebraic blow-up of the operator norm. We show that this breakdown is a structural bifurcation rather than a material loss of ellipticity. Using a tangent-cone blow-up reduction, we reduce the ordering between the localised surface creasing instability and pointwise material collapse to a single computable tip-disc criterion, under which creasing preempts collapse. An imperfect bifurcation analysis demonstrates that the quadratic energy coefficient vanishes by symmetry, whereas the subcritical cubic coefficient is tip-convergent yet explicitly dependent on the global stress-intensity factor, an outcome of the characteristic edge exponent satisfying $\alpha > 1/2$. The theoretical framework is supported by a scale-covariant Mellin–Chebyshev spectral formulation that resolves the singular exponents and buckling thresholds with exponential accuracy.

Keywords: Keywords go here.

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1 Introduction

C: The introduction body is still empty — the paper currently opens with §1.1 (literature) with no framing paragraph. Suggested content, one paragraph each: (1) the phenomenon (conditioning collapse at a compressed clamped–free corner, first seen numerically, claimed here to be a corner-localised creasing bifurcation); (2) the dichotomy result (tension benign / compression structural), with the two-scale tensile statement of Theorem 1; (3) the reduction chain (blow-up to the $(\nu, \hat{K})$ wedge, imperfect bifurcation, what is proved vs. hypothesised — the ledger of Table 1 is the honest map); (4) relation to Knowles–Sternberg, Ball, Biot/Hong, Kondratiev–Maz’ya *with actual citations* — §1.1–1.2 currently cite by name-and-decade in prose only, and `literature.bib` keys exist for several of these; (5) outline. Also: the running example is 2D plane strain, and several results (screening Lemma 4, existence Lemma 7) are genuinely two-dimensional (affine cofactor); the introduction should state up front where $d = 2$ is essential and where $d \geq 3$ survives via the edge reduction of Proposition 5(iii).

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1.1 Existing literature

The literature on singularities in finite and nonlinear hyperelasticity is organised around three major phenomena.

- **Crack tips in tension** (Knowles and Sternberg, 1970s). The classical work on nonlinear singularities treats a 360° corner (a crack) pulled in pure tension. In a neo-Hookean material the asymptotic displacement field departs markedly from linear fracture mechanics; the analysis is, however, confined to traction-free boundaries and pure tensile opening.
- **Cavitation** (Ball, 1982). Under severe hydrostatic tension the hyperelastic equations admit singular weak solutions in which a hole opens spontaneously in the interior of a solid domain — a point singularity with $\mathbf{F} \rightarrow \infty$.
- **Surface creasing** (Biot, 1963; Hong et al., 2009). A compressed flat half-space of neo-Hookean material becomes unstable at a critical stretch ($\lambda \approx 0.54$); this instability is now understood as a *crease*, a point singularity nucleating on an initially flat surface.

1.2 The gap

The existing nonlinear-singularity literature largely avoids the configuration of interest here: *a geometric corner (e.g. 90°) carrying mixed boundary conditions (Dirichlet meeting Neumann) under compression*. Each strand addresses only part of it:

- Linear theory (Kondratiev) handles the mixed-BC corner but ignores the large-deformation kinematics.
- Nonlinear theory (Knowles–Sternberg) handles large deformations but avoids the Dirichlet–Neumann condition and avoids compression.
- Instability theory handles compression but typically on flat domains or symmetric geometries, ignoring a pre-existing geometric corner singularity.

The open question this paper addresses is how an already-present stress concentration at a geometric Dirichlet–Neumann corner couples with a nonlinear structural instability (buckling/creasing), and how the resulting tangent operators can be bounded.

2 Problem formulation

2.1 Finite-strain elastostatics as energy minimisation

We consider a hyperelastic body occupying a reference d -dimensional domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, $\Gamma_D \cap \Gamma_N = \emptyset$. The deformation of the body is described by the displacement field $\mathbf{u} : \Omega \rightarrow \mathbb{R}^d$, the deformation map $\mathbf{y}(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$, and the associated deformation gradient

$$\mathbf{F} = \nabla \mathbf{y} = \mathbf{I} + \nabla \mathbf{u}, \quad J = \det \mathbf{F} > 0, \quad (1)$$

where ∇ denotes the gradient with respect to the reference configuration. The mechanical response is encoded in a strain energy density $\Psi(\mathbf{F})$, and the equilibrium configuration is the minimiser of the total potential energy

$$\mathcal{E}(\mathbf{u}) = \int_{\Omega} \Psi(\mathbf{F}) dV - \int_{\Omega} \mathbf{f}_0 \cdot \mathbf{u} dV - \int_{\Gamma_N} \mathbf{t}_0 \cdot \mathbf{u} dS, \quad (2)$$

over the space of admissible displacements $U = \{\mathbf{u} \in H^1(\Omega; \mathbb{R}^d) : \mathbf{u} = \mathbf{g} \text{ on } \Gamma_D\}$, where $\mathbf{f}_0$ is a body force, $\mathbf{t}_0$ a prescribed traction on the Neumann boundary, and $\mathbf{g}$ the prescribed boundary displacement. We seek $\mathbf{u} \in U$ such that $\mathcal{E}(\mathbf{u}) \leq \mathcal{E}(\mathbf{v})$ for all admissible $\mathbf{v}$.

2.2 Stationarity, stress, and tangent modulus

A minimiser satisfies the stationarity condition $D_{\delta \mathbf{u}} \mathcal{E}(\mathbf{u}) = 0$ for all admissible variations $\delta \mathbf{u}$ (vanishing on Γ_D), which is the weak form of equilibrium

$$\int_{\Omega} \mathbf{P}(\mathbf{F}) : \nabla \delta \mathbf{u} \, dV = \int_{\Omega} \mathbf{f}_0 \cdot \delta \mathbf{u} \, dV + \int_{\Gamma_N} \mathbf{t}_0 \cdot \delta \mathbf{u} \, dS, \quad (3)$$

where the first Piola–Kirchhoff stress is the first derivative of the energy with respect to the deformation gradient,

$$\mathbf{P}(\mathbf{F}) = \frac{\partial \Psi}{\partial \mathbf{F}}. \quad (4)$$

The natural boundary condition associated with Γ_N is $\mathbf{P}\mathbf{N} = \mathbf{t}_0$, with $\mathbf{N}$ the reference outward unit normal. Because (3) is nonlinear in $\mathbf{u}$, it is solved iteratively by Newton’s method. Linearising the residual about a current iterate $\bar{\mathbf{u}}$ in a direction $\Delta \mathbf{u}$ introduces the fourth-order tangent (elasticity) modulus $\mathbb{L}$, with components L_{iJkL} ,

$$\mathbb{L} = \frac{\partial \mathbf{P}}{\partial \mathbf{F}} = \frac{\partial^2 \Psi}{\partial \mathbf{F} \otimes \partial \mathbf{F}}, \quad L_{iJkL} = \frac{\partial^2 \Psi}{\partial F_{iJ} \partial F_{kL}}, \quad (5)$$

which possesses the major symmetry $L_{iJkL} = L_{kLiJ}$, so that each Newton step requires solving the linear problem

$$\int_{\Omega} \nabla \delta \mathbf{u} : \mathbb{L}(\bar{\mathbf{F}}) : \nabla \Delta \mathbf{u} \, dV = -R(\bar{\mathbf{u}}; \delta \mathbf{u}), \quad (6)$$

with $R(\bar{\mathbf{u}}; \delta \mathbf{u})$ the residual of (3) evaluated at $\bar{\mathbf{u}}$. The well-posedness and conditioning of (6)—and hence the convergence of the whole nonlinear solver—are governed by the coercivity and boundedness of the bilinear form induced by $\mathbb{L}(\bar{\mathbf{F}})$. The central object of this work is therefore the behaviour of $\mathbb{L}$ as a function of the local deformation state.

2.3 The neo-Hookean material

To fix ideas and to exhibit explicit constants we use, as a running example, the compressible neo-Hookean strain energy density adopted in the cut-finite-element study [?] that motivates this analysis,

$$\Psi(\mathbf{F}) = \frac{\mu_e}{2} (\text{tr}(\mathbf{F}^T \mathbf{F}) - d) + f(J), \quad (7)$$

where $\mu_e > 0$ is the shear modulus and $f(J)$ is a strictly convex volumetric penalty with $f(1) = 0$, $f'(1) = -\mu_e$, and $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$. Two standard choices are

$$f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J, \quad f_{\text{quad}}(J) = \frac{\kappa}{2} (J - 1)^2, \quad (8)$$

with λ_e the Lamé constant and κ a bulk modulus; (7) with $f_{\log}$ reproduces the model of [?]. The corresponding stress and tangent are

$$\mathbf{P}(\mathbf{F}) = \mu_e \mathbf{F} + f'(J) J \mathbf{F}^{-T}, \quad (9)$$

$$L_{iJkL} = \mu_e \delta_{ik} \delta_{JL} + (f''(J) J^2 + f'(J) J) F_{iJ}^{-T} F_{kL}^{-T} - f'(J) J F_{iL}^{-T} F_{kJ}^{-T}. \quad (10)$$

The first term $\mu_e \delta_{ik} \delta_{JL}$ is constant; the entire nonlinear part is controlled by J and the *inverse* deformation gradient $\mathbf{F}^{-1}$, yielding the structural bound

$$\|\mathbb{L}(\mathbf{F})\| \leq \mu_e + \Theta(J) \|\mathbf{F}^{-1}\|^2, \quad (11)$$

where $\|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1}$ is set by the smallest principal stretch and Θ is a scalar growth function fixed by f (computed explicitly in §3.4). Consequently $\mathbb{L}$ can become unbounded through volumetric collapse, $\lambda_{\min} \rightarrow 0$ (equivalently $\mathbf{F}^{-1} \rightarrow \infty$); whether a purely tensile blow-up of $\mathbf{F}$ (with $\mathbf{F}^{-1}$ bounded) leaves $\mathbb{L}$ finite depends on the growth of $\Theta(J)$, which turns out to discriminate sharply between the two penalties in (8). This dichotomy is the analytical backbone of the present paper.

2.4 The mixed-corner singularity

The configuration of interest is a right-angle junction where the two boundary-condition types meet: a Dirichlet face (Γ_D , prescribed displacement) abutting a traction-free Neumann face (Γ_N). In $\mathbb{R}^d$ this junction is a codimension-two *edge* Σ (a point for $d = 2$, a curve or surface for $d \geq 3$); its transverse geometry, in the two-plane normal to Σ , is the right-angle corner we analyse, and we use polar coordinates (r, θ) in that transverse plane. Such junctions arise generically wherever a clamped face meets a traction-free face — for instance along the base edges of a compressed body, the concrete setting taken up in Section 5. Classical Kondratiev/edge theory for the linearised problem predicts a transverse singularity

$$\mathbf{u}(r, \theta) \sim K r^\alpha \Phi(\theta), \quad \alpha \in (0, 1), \quad (12)$$

with r the distance to the edge Σ , K a generalised stress-intensity factor and α the (geometry- and Poisson-ratio-dependent) singularity exponent. The deformation gradient then scales as $\mathbf{F} = \mathbf{I} + \mathcal{O}(K r^{\alpha-1})$, so $\|\mathbf{F}\| \rightarrow \infty$ as $r \rightarrow 0$.

The question this raises—and which the remainder of the paper addresses—is how this kinematic singularity propagates into the nonlinear tangent operator (5). In view of (11), the answer hinges on which principal stretch diverges: a tensile corner keeps $\mathbf{F}^{-1}$ bounded and (for a suitable penalty) leaves $\mathbb{L}$ well behaved, whereas a compressive corner drives $\lambda_{\min} \rightarrow 0$ and blows up the *norm* of $\mathbb{L}$ (severe ill-conditioning). The tangent nonetheless remains strongly elliptic — the compressive instability is *structural* (a corner analogue of creasing), not a change of type of the PDE. The following sections analyse the two scenarios in turn (Section 3) and then show how a small imperfection regularises the compressive case through a structural bifurcation (Section 4).

3 Asymptotic behaviour of the tangent operator at a mixed corner

3.1 Geometry and linearised kinematics

Let the reference domain $\Omega \subset \mathbb{R}^d$ ($d \geq 2$) carry a right-angle Dirichlet–Neumann edge Σ of codimension two. Near a point of Σ split $\mathbb{R}^d = T\Sigma \oplus \Pi$ into the edge tangent space and the normal two-plane Π , and use polar coordinates (r, θ) in Π with $r \in [0, R]$, $\theta \in [0, \pi/2]$, where $r = \text{dist}(\cdot, \Sigma)$. Write B_ρ for the tubular neighbourhood $\{\mathbf{x} \in \Omega : \text{dist}(\mathbf{x}, \Sigma) < \rho\}$ of the edge. The leading singularity is governed by the transverse problem in Π , modulated smoothly along Σ ; for $d = 2$ the edge degenerates to a point and B_ρ to the corner disc.

- **Dirichlet boundary** (Γ_D): $\theta = 0$, deformation prescribed, $\mathbf{y}(r, 0) = \mathbf{g}(r)$.
- **Neumann boundary** (Γ_N): $\theta = \pi/2$, traction-free, $\mathbf{P}\mathbf{N} = \mathbf{0}$.

Classical Kondratiev theory for the *linearised* problem gives the leading-order singular displacement

$$\mathbf{u}(r, \theta) \sim K r^\alpha \Phi(\theta), \quad (13)$$

with K the generalised stress-intensity factor (set by the far field), Φ a smooth angular profile, and $\alpha \in (0, 1)$ the geometry- and Poisson-dependent exponent. Consequently

$$\mathbf{F} = \mathbf{I} + \mathcal{O}(K r^{\alpha-1}), \quad \|\mathbf{F}\| \rightarrow \infty \text{ as } r \rightarrow 0. \quad (14)$$

Whether this kinematic blow-up contaminates the tangent operator $\mathbb{L} = \partial^2 \Psi / \partial \mathbf{F} \partial \mathbf{F}$ depends not on $\|\mathbf{F}\|$ but on which principal stretches diverge, which is the content of the rest of this section.

3.2 Standing assumptions on the stored energy

The mechanism we describe is not specific to the neo-Hookean model; that model only serves to fix ideas and to exhibit explicit constants. We therefore phrase the analysis for a general frame-indifferent energy and isolate the structural properties that the argument actually uses. Throughout, $\|\cdot\|$ denotes the operator (spectral) norm on tensors, $0 < \lambda_{\min}(\mathbf{F}) \leq \dots \leq \lambda_{\max}(\mathbf{F})$ are the singular values of $\mathbf{F}$ (the d principal stretches), and we recall $\|\mathbf{F}^{-1}\| = \lambda_{\min}(\mathbf{F})^{-1}$.

Assumption 1 (Regularity and frame indifference). The stored energy $\Psi : \text{GL}^+(d) \rightarrow \mathbb{R}$ is objective, $\Psi \in C^2(\text{GL}^+(d))$, and $\Psi(\mathbf{F}) \rightarrow +\infty$ as $\det \mathbf{F} \rightarrow 0^+$.

Assumption 2 (Inverse-channel growth of the tangent). There exist a continuous, nondecreasing function $\Theta : (0, \infty) \rightarrow (0, \infty)$ and a constant $c_0 \geq 0$ such that the tangent modulus $\mathbb{L}(\mathbf{F}) = D^2 \Psi(\mathbf{F})$ satisfies, for all $\mathbf{F} \in \text{GL}^+(d)$,

$$\|\mathbb{L}(\mathbf{F})\| \leq c_0 + \Theta(\det \mathbf{F}) \|\mathbf{F}^{-1}\|^2. \quad (15)$$

Assumption 3 (Uniform strong ellipticity on bounded-distortion sets). For every $0 < c \leq 1$ and $M \geq 1$ there is $\gamma(c, M) > 0$ such that whenever $\lambda_{\min}(\mathbf{F}) \geq c$ and $\det \mathbf{F} \leq M$,

$$L_{iJkL}(\mathbf{F}) m_i N_J m_k N_L \geq \gamma(c, M) |\mathbf{m}|^2 |\mathbf{N}|^2 \quad \forall \mathbf{m}, \mathbf{N} \in \mathbb{R}^2. \quad (16)$$

The decisive feature of Assumption 2 is that the inverse-stretch factor $\|\mathbf{F}^{-1}\|^2 = \lambda_{\min}^{-2}$ is the *only* algebraic channel through which the Hessian can blow up; the dependence on $\|\mathbf{F}\|$ (equivalently on $\lambda_{\max}$) enters solely through the scalar $\Theta(\det \mathbf{F})$. Whether a tensile corner ($\lambda_{\max} \rightarrow \infty$, $\det \mathbf{F} \rightarrow \infty$, $\lambda_{\min}$ bounded below) is benign therefore reduces to the *growth rate* of Θ , which we quantify in §3.4. Materials with genuine tensile stiffening of the deviatoric part (Gent, Arruda–Boyce), whose tangent diverges through $\lambda_{\max}$ directly rather than through $\Theta(\det \mathbf{F})$, violate (15) and are excluded.

Lemma 1 (Assumptions for the neo-Hookean model). *Let Ψ be the neo-Hookean energy (7) with $\mu_e > 0$ and $f \in C^2(0, \infty)$. Then Assumption 2 holds with $c_0 = \mu_e$ and*

$$\Theta(J) = |f''(J)J^2 + f'(J)J| + |f'(J)J|, \quad (17)$$

and Assumption 3 holds on the convexity set $\{f''(J) \geq 0\}$ — in particular on all of compression $J \leq 1$ and moderate tension for $f_{\log}$, and globally for f_{quad} ($f'' = \kappa > 0$); it can fail under extreme dilation $J > J_$ for $f_{\log}$ (Remark 1). Assumption 1 additionally requires the barrier property $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$: this holds for $f_{\log}$ but fails for f_{quad} , since $f_{\text{quad}}(J) \rightarrow \frac{\kappa}{2} < \infty$ (see Remark 6 and Scenario B).*

Proof. Objectivity and C^2 regularity on $\text{GL}^+(d)$ are immediate; the barrier hypothesis $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$, when assumed, gives the blow-up required by Assumption 1. In (10) the first term is the scaled identity, of norm μ_e . Each of the remaining two terms is a rank-one, respectively transposition, contraction of $\mathbf{F}^{-T} \otimes \mathbf{F}^{-T}$; since $\|\mathbf{F}^{-T}\| = \|\mathbf{F}^{-1}\|$, each is bounded in operator norm by its scalar prefactor times $\|\mathbf{F}^{-1}\|^2$. The triangle inequality gives (15) with $c_0 = \mu_e$ and Θ as in (17). For Assumption 3, contracting (10) with $\mathbf{m} \otimes \mathbf{N}$ and writing $\xi := \mathbf{m} \cdot \mathbf{F}^{-T} \mathbf{N} = \mathbf{N} \cdot \mathbf{F}^{-1} \mathbf{m}$, the two volumetric terms combine $((f'' J^2 + f' J)\xi^2 - f' J \xi^2)$ to the single clean form

$$L_{iJkL} m_i N_J m_k N_L = \mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 + f''(J) J^2 \xi^2. \quad (18)$$

Wherever $f''(J) \geq 0$ the second term is nonnegative and (16) holds with $\gamma = \mu_e$, uniformly in $\mathbf{F}$ (no distortion bound needed); this covers f_{quad} ($f'' = \kappa > 0$) everywhere and f_{log} on $\{f''_{\text{log}} \geq 0\}$. $\square$ $\square$

Remark 1 (Strength and a tension caveat). Identity (18) is sharper than (16): the neo-Hookean tangent is strongly elliptic with constant *exactly* μ_e wherever $f''(J) \geq 0$, and the volumetric term only adds to it. The Hessian of the quadratic (Hookean) part is constant, which is why the *deviatoric* response never blows up; only the *volumetric* response, through Θ , can. One caveat for f_{log} : since $f''_{\text{log}}(J) = (\mu_e + 2\lambda_e - 2\lambda_e \log J)/J^2$, convexity is lost once $J > J_* := \exp(1 + \mu_e/(2\lambda_e))$, and then (18) can turn negative for ξ near its maximal value $|\mathbf{m}||\mathbf{N}|/\lambda_{\min}$. Hence Assumption 3 holds throughout *compression* ($J \leq 1$) and moderate tension, but f_{log} can lose strong ellipticity under *extreme dilation* ($J > J_*$)—a tension-side phenomenon distinct from the compressive analysis, and one more reason the bounded- J qualifier matters in §3.5. The globally convex f_{quad} is free of this.

3.3 The non-compressive hypothesis

The qualitative split between the two scenarios is captured by a single quantity: the smallest principal stretch near the corner. We call the corner singularity *non-compressive* when $\lambda_{\min}$ stays bounded away from zero, even though $\lambda_{\max}$ (hence $\|\mathbf{F}\|$ and $\det \mathbf{F}$) may diverge.

Assumption 4 (Non-compressive singularity, (NC)). There exist $\rho \in (0, R]$ and $c \in (0, 1]$ such that the equilibrium deformation satisfies

$$\lambda_{\min}(\mathbf{F}(\mathbf{x})) \geq c, \quad \text{equivalently} \quad \|\mathbf{F}^{-1}(\mathbf{x})\| \leq c^{-1}, \quad \text{for a.e. } \mathbf{x} \in B_\rho. \quad (19)$$

We stress what (19) does *not* say. It does not require incompressibility, and crucially it does *not* bound $\det \mathbf{F}$: in a tensile corner $\lambda_{\max} \sim r^{\alpha-1} \rightarrow \infty$, so $J = \det \mathbf{F} = \prod_i \lambda_i \sim r^{\alpha-1} \rightarrow \infty$ as well (one diverging stretch, the remaining $d-1$ bounded). The only quantity (NC) controls is $\|\mathbf{F}^{-1}\| \leq c^{-1}$. Physically, (NC) states that the material is pulled around the corner (tension/shear) rather than crushed into it. Because $\det \mathbf{F}$ is *not* bounded under (NC), the factor $\Theta(\det \mathbf{F})$ in (15) need not stay bounded; the next subsection makes its growth explicit, and this is precisely where the two volumetric penalties (8) part ways. Scenario B (§3.6) treats the complementary, compressive case in which (19) fails.

3.4 Explicit volumetric growth functions

The bound (15) exposes *two independent channels* through which the tangent can diverge, and they are kinematically decoupled:

$$\|\mathbb{L}(\mathbf{F})\| \leq \underbrace{\mu_e + \Theta(\det \mathbf{F})}_{\substack{\text{dilation channel} \\ J \rightarrow \infty}} \cdot \underbrace{\|\mathbf{F}^{-1}\|^2}_{\substack{\text{inverse channel} \\ \lambda_{\min} \rightarrow 0}}. \quad (20)$$

The *inverse channel* is the one usually blamed for trouble: it requires $\lambda_{\min} \rightarrow 0$, i.e. compression to zero volume, and is the subject of Scenario B. The *dilation channel*, $\Theta(\det \mathbf{F}) \rightarrow \infty$ as $J \rightarrow \infty$, is activated by *expansion* and is precisely what a tensile corner switches on, since $\lambda_{\max} \sim r^{\alpha-1} \rightarrow \infty$ forces $J \rightarrow \infty$ even with $\lambda_{\min}$ bounded below by (NC). Whether this matters is therefore not a property of the corner or of the sign of the loading, but of the *growth rate of Θ* —a property of the volumetric penalty f alone. We now evaluate it for the two penalties (8); the contrast shows that “tension is benign” is a model-dependent statement.

Logarithmic penalty. For $f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J$,

$$f'(J)J = -\mu_e + 2\lambda_e \log J, \quad f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J, \quad (21)$$

so that the combination collapses to a constant,

$$f''(J)J^2 + f'(J)J = 2\lambda_e, \quad (22)$$

and therefore

$$\Theta_{\log}(J) = 2\lambda_e + |2\lambda_e \log J - \mu_e| = \mathcal{O}(\log J) \quad \text{as } J \rightarrow \infty. \quad (23)$$

The volumetric stiffening grows only *logarithmically* in J .

Quadratic penalty. For $f_{\text{quad}}(J) = \frac{\kappa}{2}(J-1)^2$,

$$f'(J)J = \kappa J(J-1), \quad f''(J)J^2 = \kappa J^2, \quad f''(J)J^2 + f'(J)J = \kappa J(2J-1), \quad (24)$$

so

$$\Theta_{\text{quad}}(J) = \kappa J |2J-1| + \kappa J |J-1| = \mathcal{O}(\kappa J^2) \quad \text{as } J \rightarrow \infty. \quad (25)$$

The volumetric stiffening grows *quadratically* in J .

Consequence at the corner. We combine (15) with (NC) ($\|\mathbf{F}^{-1}\|^2 \leq c^{-2}$) and the tensile kinematics. Two kinematic regimes are possible at fixed angle: either exactly one stretch diverges, $\lambda_{\max} \sim |K| r^{\alpha-1}$ with the remaining stretches bounded, giving $J = \det \mathbf{F} \sim r^{\alpha-1}$; or two stretches diverge and the corresponding subdeterminant gives $J \sim r^{2(\alpha-1)}$. For the norm bounds of this subsection the distinction is immaterial — under Assumption 5 both give $\Theta(J) = \mathcal{O}((\log \frac{1}{r})^p)$ — and we quote the one-large-stretch scaling $J \sim r^{\alpha-1}$ as the representative case; the angular structure of the two regimes under the Williams ansatz, which turns out to *sharpen* the corner theory, is resolved in Lemma 4 below. The tangent norm near the corner then obeys

$$\|\mathbb{L}(\mathbf{x})\| \leq \mu_e + c^{-2} \Theta(\det \mathbf{F}(\mathbf{x})) \sim \begin{cases} \mu_e + \mathcal{O}(\log \frac{1}{r}), & f = f_{\log}, \\ \mathcal{O}(r^{-2(1-\alpha)}), & f = f_{\text{quad}}. \end{cases} \quad (26)$$

Thus the logarithmic penalty keeps the tangent bounded up to a logarithm, whereas the quadratic penalty produces a genuine *algebraic* blow-up of the tangent *even in pure tension*. This is the dilation channel of (20) acting alone, with the inverse channel inert ($\|\mathbf{F}^{-1}\|$ bounded); it is algebraically singular of a severity comparable to—though mechanistically distinct from—the compressive blow-up of Scenario B, which instead runs through $\|\mathbf{F}^{-1}\|^2$. The often-repeated statement that “tension is harmless for neo-Hookean” is therefore an artefact of the logarithmic volumetric model and is false for the quadratic one.

Physical reading: the large-dilation response. The dichotomy is, at bottom, a statement about how the bulk energy behaves under unbounded *expansion*, the loading a tensile corner imposes:

volumetric model	energy as $J \rightarrow \infty$	tangent $\Theta(J)$
$\lambda_e \log^2 J$	$\sim \log^2 J$ (soft)	$\mathcal{O}(\log J)$
tempered log (27)	$\sim \log J $ (softest)	$\mathcal{O}(1)$
$\frac{\kappa}{2}(J-1)^2$	$\sim J^2$ (stiff)	$\mathcal{O}(J^2)$

A penalty that is polynomial in J makes the material resist further dilation ever more stiffly, and the linearised modulus faithfully reports that stiffness as a blow-up; the logarithmic penalty is gentle about expansion and does not. Both penalties are designed to diverge correctly as $J \rightarrow 0$ (resisting collapse, the inverse channel); their large- J behaviour, which the dilation channel probes, was never physically constrained, and that latitude is exactly where they part. The exact cancellation (22) is the analytic signature of this softness. We make the benign case precise in Theorem 1 and flag the algebraic case in Remark 6.

3.5 Scenario A: the tensile (non-compressive) singularity

We work under the mild-growth condition isolated above.

Assumption 5 (Mild volumetric growth). The growth function satisfies $\Theta(J) = \mathcal{O}((\log J)^p)$ as $J \rightarrow \infty$ for some $p \geq 0$.

By (23), $f_{\log}$ satisfies Assumption 5 with $p = 1$; by (25), f_{quad} does not.

The benign corner case of Theorem 1 below will require Assumption 5 *together with* global convexity $f'' \geq 0$, and neither standard penalty satisfies both: f_{quad} fails mild growth, $f_{\log}$ fails convexity past J_* (Remark 1). Whether the combined class is nonempty — with both elastic moduli still free — deserves an explicit answer, and it has a clean one.

Remark 2 (The convex-mild class: forced sign, and an explicit two-parameter exemplar). Write $t = \log J$ and $h(t) := f'(J)J$, so that $f''(J)J^2 = h'(t) - h(t)$; convexity reads $h' \geq h$ and (17) becomes $\Theta = |h' - h| + |h|$. (i) *Forced sign.* If $f'' \geq 0$ and $\Theta(J) = \mathcal{O}((\log J)^p)$, then $f'(J) \leq 0$ for all J : were $f'(J_0) > 0$, convexity would give $f'(J) \geq f'(J_0)$ for $J \geq J_0$, hence $\Theta(J) \geq f'(J)J \geq f'(J_0)J$, algebraic growth. Every admissible material is therefore *never volumetrically tensile*: the bulk stress resists compression but never pushes back against dilation. This is exactly where $f_{\log}$ exits the class — its volumetric stress $h(t) = -\mu_e + 2\lambda_e t$ turns *positive* past $J_p = e^{\mu_e/(2\lambda_e)}$, and mild growth then forbids the convexity needed to sustain it. (ii) *The class is nonempty, with both moduli free.* Saturate the linear h of $f_{\log}$: for a temper scale $0 < T < \mu_e/(2\lambda_e)$ set $h_T(t) := -\mu_e + 2\lambda_e T \tanh(t/T)$, i.e.

$$f_T(J) = -\mu_e \log J + 2\lambda_e T^2 \log \cosh\left(\frac{\log J}{T}\right). \quad (27)$$

Then $f_T \in C^\infty(0, \infty)$, $f_T(1) = 0$, $f'_T(1) = -\mu_e$, $f''_T(1) = \mu_e + 2\lambda_e$ — the linearised moduli, hence the Poisson ratio, are *identical* to those of $f_{\log}$, and indeed $f_T = f_{\log} + \mathcal{O}(\log^4 J)$ near $J = 1$ — while globally

$$f''_T(J)J^2 = \mu_e - 2\lambda_e T \tanh\left(\frac{t}{T}\right) + 2\lambda_e \operatorname{sech}^2\left(\frac{t}{T}\right) \geq \mu_e - 2\lambda_e T > 0, \quad \Theta_T(J) \leq 3\mu_e + 2\lambda_e \quad (p = 0),$$

with the barrier $f_T(J) \sim (\mu_e + 2\lambda_e T)|\log J| \rightarrow \infty$ as $J \rightarrow 0^+$, and $h_T < 0$ everywhere (the forced sign realised). The family interpolates between the pure logarithmic barrier ($T \rightarrow 0$) and

$f_{\log}$ itself (the inadmissible limit $T \rightarrow \infty$); its energy grows like $|\log J|$ under dilation, *softer* than $f_{\log}$'s $\log^2 J$, which is the analytic mechanism by which it keeps both convexity and mild growth. (iii) *One-sided temper*. Since f_T and $f_{\log}$ agree to fourth order in $\log J$ at $J = 1$, one may glue $f_{\log}$ on $J \leq 1$ to f_T on $J \geq 1$ with C^3 regularity: the resulting penalty is *pointwise identical* to $f_{\log}$ throughout compression — every compressive statement in this paper is untouched — while the dilation tail is tempered, $f'' > 0$ holds globally, and the dilation core of Lemma 3 disappears ($r_* = 0$). The case split of Theorem 1 is thus a property of the untempered $\log^2$ tail of $f_{\log}$, in the large- J regime that (as §3.4 already observed) was never physically constrained, and it can be removed at no cost to the compressive analysis by this substitution.

Lemma 2 (Subordinate tangent). *Under Assumptions 1, 2, 5 and (NC), for every $\varepsilon > 0$ there is C_ε with*

$$\|\mathbb{L}(\mathbf{x})\| \leq C_\varepsilon r^{-\varepsilon} \quad \text{for a.e. } \mathbf{x} \in B_\rho. \quad (28)$$

In particular $\mathbb{L} \in L^q(B_\rho)$ for every $q < \infty$, and $\|\mathbb{L}\|$ is dominated by any negative power of r .

Proof. By (15) and (NC), $\|\mathbb{L}\| \leq \mu_e + c^{-2}\Theta(\det \mathbf{F})$. Assumption 5 and $\det \mathbf{F} = \mathcal{O}(r^{-2(1-\alpha)})$ (either kinematic regime) give $\Theta(\det \mathbf{F}) = \mathcal{O}((\log \frac{1}{r})^p)$, and $(\log \frac{1}{r})^p \leq C_\varepsilon r^{-\varepsilon}$ on $(0, \rho]$ for every $\varepsilon > 0$. $\square$ $\square$

Lemma 3 (Strong ellipticity outside the dilation core). *Let $f = f_{\log}$ and let $J_* = \exp(1 + \mu_e/(2\lambda_e))$ be the dilation threshold of Remark 1, beyond which $f''_{\log} < 0$. Under Assumptions 1, 3 and (NC), with the tensile kinematics $J \sim |K| r^{\alpha-1}$ of §3.4, define the dilation core radius*

$$r_* := (J_*/|K|)^{1/(\alpha-1)} = (|K|/J_*)^{1/(1-\alpha)}, \quad (29)$$

so that $J(\mathbf{x}) \leq J_$ for $r \geq r_*$. Then on the annular region $A_* := \{\mathbf{x} \in B_\rho : r_* < r < \rho\}$ the tangent satisfies the Legendre–Hadamard condition with the constant ellipticity modulus μ_e ,*

$$L_{iJkL}(\mathbf{x}) m_i N_J m_k N_L \geq \mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 \quad \text{for a.e. } \mathbf{x} \in A_*, \quad \forall \mathbf{m}, \mathbf{N} \in \mathbb{R}^2. \quad (30)$$

On the complementary core $\{r \leq r_\}$, where $J > J_*$, the constant-modulus bound (30) no longer holds, and strong ellipticity of $f_{\log}$ can fail (Remark 1): wherever the sharper condition $|f''(J)|J^2 \|\mathbf{F}^{-1}\|^2 > \mu_e$ is met, the directional modulus $\mu_e + f''(J)J^2 \xi^2 / (|\mathbf{m}|^2 |\mathbf{N}|^2)$ turns negative for ξ near its maximum. We do not claim the loss is pointwise throughout the core — only that the core is where it can occur, since $f''(J)J^2 < 0$ there is necessary for it.*

Proof. By the exact rank-one identity (18),

$$L_{iJkL} m_i N_J m_k N_L = \mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 + f''(J) J^2 \xi^2, \quad \xi := \mathbf{m} \cdot \mathbf{F}^{-T} \mathbf{N},$$

in which the volumetric prefactor is the single scalar $f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J$ (cf. (22)). This is nonnegative precisely when $\log J \leq 1 + \mu_e/(2\lambda_e)$, i.e. $J \leq J_*$. Since $J \sim |K| r^{\alpha-1}$ is decreasing in r (recall $\alpha < 1$), the threshold $J = J_*$ is crossed at the single radius (29), and $J \leq J_*$ throughout A_* . There the second term only adds to μ_e , giving (30). For $r < r_*$ we have $J > J_*$, hence $f''(J)J^2 < 0$; taking $\mathbf{m}, \mathbf{N}$ aligned so that ξ^2 attains its maximum $|\mathbf{m}|^2 |\mathbf{N}|^2 \|\mathbf{F}^{-1}\|^2$ (admissible since $\|\mathbf{F}^{-1}\| \leq c^{-1}$ is only an upper bound, with equality possible) drives the form below zero once $|f''(J)J^2| \|\mathbf{F}^{-1}\|^2 > \mu_e$. $\square$ $\square$

The picture is therefore sharper, and more honest, than “tension is harmless”: for the logarithmic penalty the tensile corner is strongly elliptic on the annulus A_* outside a *dilation core* $\{r \leq r_*\}$,

inside which $f_{\log}$ itself — not the corner — *may* cease to be strongly elliptic because the material is dilated past J_* (where the sharper condition $|f''(J)|J^2\|\mathbf{F}^{-1}\|^2 > \mu_e$ binds). The core radius r_* in (29) shrinks as the threshold $J_* = \exp(1 + \mu_e/(2\lambda_e))$ grows, i.e. as the penalty is made stiffer against extreme dilation (large λ_e/μ_e); a model with $f'' \geq 0$ globally would have $r_* = 0$ and no core. This is the tensile mirror of the model-dependence already seen for Θ in §3.4, and it confines the genuinely delicate behaviour to a controlled inner region.

The corner statements of Theorem 1 require one further input, and it is kinematic: the norm bound of Lemma 2 controls the *size* of the tangent but not its *limit* at the vertex, and it is the limit that fixes the corner pencil. We therefore impose the Williams ansatz in tensor form and record its consequence, which is stronger than one might expect: at the vertex the nonlinear tangent does not converge to the linearised elasticity tensor at all — it converges to the *scaled identity* $\mu_e \mathbb{I}$, because every nonlinear term of (10) carries two factors of $\mathbf{F}^{-T}$ and the extreme stretch drives $\mathbf{F}^{-1} \rightarrow 0$. The large deformation *screens* the volumetric coupling.

Assumption 6 (Williams-ansatz kinematics, (W)). On B_ρ the state satisfies $\bar{\mathbf{F}}(r, \theta) = \mathbf{I} + Kr^{\alpha-1}(\mathbf{G}(\theta) + o(1))$ uniformly in θ , with $\mathbf{G}(\theta) = \alpha \Phi(\theta) \otimes \hat{\mathbf{e}}_r + \Phi'(\theta) \otimes \hat{\mathbf{e}}_\theta$ the gradient profile of (13). The profile is *nondegenerate*: the angular determinant $\mu(\theta) := \det \mathbf{G}(\theta) = \alpha \Phi(\theta) \times \Phi'(\theta)$ is nonnegative on $[0, \pi/2]$ with finitely many zeros, each of finite order, and $\text{tr} \mathbf{G} \neq 0$ at each zero. (The Dirichlet condition $\Phi(0) = \mathbf{0}$ forces a zero of order two at the clamped face; nonnegativity of μ is the orientation consistency $J > 0$ of the ansatz itself.)

Lemma 4 (Vertex screening of the tangent). *Let Assumptions 1, 2, 5 and (W) hold, set $r_K := |K|^{-1/(1-\alpha)}$, and define the degenerate-ray layers*

$$\Lambda(r) := \left\{ \theta \in [0, \frac{\pi}{2}] : \mu(\theta) \leq 2 \|\mathbf{G}\|_\infty (r/r_K)^{1-\alpha} \right\}, \quad |\Lambda(r)| \leq C (r/r_K)^{(1-\alpha)/q_0}, \quad (31)$$

with q_0 the maximal order of the zeros of μ . Then, for r small:

- (i) Off the layers ($\theta \notin \Lambda(r)$) both transverse stretches diverge, $J = K^2 r^{2(\alpha-1)}(\mu(\theta) + o(1))$, the inverse gradient decays algebraically, $\|\bar{\mathbf{F}}^{-1}\| \leq C (r/r_K)^{1-\alpha}/\mu(\theta)$, and hence

$$\|\mathbb{L}(\bar{\mathbf{F}}) - \mu_e \mathbb{I}\| \leq \Theta(J) \|\bar{\mathbf{F}}^{-1}\|^2 \leq C (\log \frac{1}{r})^p (r/r_K)^{2(1-\alpha)} \mu(\theta)^{-2} \rightarrow 0. \quad (32)$$

- (ii) On the layers the one-large-stretch regime of §3.4 holds: $J \sim Kr^{\alpha-1} \text{tr} \mathbf{G}$, the smallest stretch is bounded below by a constant $c(\mathbf{G}) > 0$ — so (NC) holds there automatically — and $\|\mathbb{L}\| \leq \mu_e + C(\log \frac{1}{r})^p$ by Lemma 2.
- (iii) Consequently $\mathbb{L}(\bar{\mathbf{F}}(r, \cdot)) \rightarrow \mu_e \mathbb{I}$ in $L^q(0, \pi/2)$ for every $q < \infty$ at an algebraic rate, though not in L^∞ ; and off the layers the Legendre–Hadamard form $\mu_e |\mathbf{m}|^2 |\mathbf{N}|^2 + f''(J) J^2 \xi^2$ of (18) satisfies $|f''(J) J^2| \|\bar{\mathbf{F}}^{-1}\|^2 \rightarrow 0$, so strong ellipticity holds down to the vertex there even where $f'' < 0$.

Proof. In two dimensions the cofactor is affine, $\text{cof}(\mathbf{I} + \mathbf{M}) = \mathbf{I} + \text{cof} \mathbf{M}$, and $\det(\mathbf{I} + \mathbf{M}) = 1 + \text{tr} \mathbf{M} + \det \mathbf{M}$. With $\mathbf{M} = Kr^{\alpha-1}(\mathbf{G} + o(1))$ the determinant reads $J = 1 + Kr^{\alpha-1}(\text{tr} \mathbf{G} + o(1)) + K^2 r^{2(\alpha-1)}(\mu(\theta) + o(1))$. Off the layers the quadratic term dominates by the defining inequality of (31), giving the stated J and, from $\bar{\mathbf{F}}^{-1} = \text{cof} \bar{\mathbf{F}}^T / J$ with $\|\text{cof} \bar{\mathbf{F}}\| \leq 1 + |K| r^{\alpha-1} \|\mathbf{G}\|_\infty$, the decay of $\|\bar{\mathbf{F}}^{-1}\|$. The bound (32) then follows from (10): both nonlinear terms are quadratic contractions of $\mathbf{F}^{-T}$ with scalar prefactors dominated by $\Theta(J)$ (proof of Lemma 1), and $\Theta(J) = \mathcal{O}((\log \frac{1}{r})^p)$ by Assumption 5. On a layer, μ is small, the linear term $Kr^{\alpha-1} \text{tr} \mathbf{G}$ dominates J (using $\text{tr} \mathbf{G} \neq 0$ at the zeros of μ), $\mathbf{M}$ is a bounded perturbation of a large near-rank-one matrix, and $\lambda_{\min} =$

$J/\lambda_{\max} \rightarrow \text{tr } \mathbf{G}/\|\mathbf{G}\|$ -type constants: (NC) holds with a computable c . The measure bound in (31) is the finite-order condition on the zeros of μ . Part (iii) integrates (32) in θ against the layer bound, and the ellipticity statement is (18) with $|\xi| \leq |\mathbf{m}||\mathbf{N}|\|\bar{\mathbf{F}}^{-1}\|$. $\square$ $\square$

Lemma 4 replaces the naive expectation that the corner pencil of the nonlinear tangent should be that of linearised elasticity. It cannot be: for any $K \neq 0$ the state leaves every neighbourhood of $\mathbf{I}$ as $r \rightarrow 0$, so $\mathbb{L}(\bar{\mathbf{F}})$ never returns to $\mathbb{L}(\mathbf{I})$ at the vertex — smallness of $|K|$ only shrinks the crossover radius r_K below which the field is large. What the tangent *does* approach is $\mu_e \mathbb{I}$, the decoupled vector Laplacian, whose mixed Dirichlet–Neumann pencil on the quarter-plane has spectrum $s \in \{1, 3, 5, \dots\}$ (each Cartesian component separates, $\sin(s\theta)$ with $\cos(s\pi/2) = 0$): *no exponent in $(0, 1)$* . The corner operator is therefore a *two-scale* operator — linearised elasticity with the Williams exponents $\alpha(\nu)$ on the intermediate annulus $r_K \ll r \ll \rho$, the Laplacian pencil with Lipschitz-regular increments at the vertex itself — and since every inner exponent exceeds $1 > \alpha$, the r^α Williams field remains the dominant asymptotics visible from any fixed scale. This is the precise content we can prove of the slogan that the tensile corner *passes like linear elasticity*: it passes like linear elasticity on the mesh-relevant intermediate range, and *better* than linear elasticity at the vertex.

Remark 3 (Status of (W): verified above $\nu_W \approx 0.40$, fails below). Hypothesis (W) is a parameter-free, closed-form condition on the computed clamped-free profile, and checking it (§6.2) returns a sharp split of the physical range: $\mu \geq 0$ holds precisely for $\nu > \nu_W \approx 0.404$ (single order-two zero at the clamped face, $\text{tr } \mathbf{G} \neq 0$ there), while for $\nu < \nu_W$ the determinant is *negative* on a clamped-side sector $(0, \theta_z(\nu))$ that widens as ν decreases. Since the mode enters μ quadratically, the sign is intrinsic, and $J \sim K^2 r^{2(\alpha-1)} \mu$ at depth for either load sign: below ν_W the linear Williams field is orientation-reversing at depth on an $\mathcal{O}(1)$ sector, so the tensor ansatz cannot describe the state below the crossover scale there and the screened-vertex conclusion of Theorem 1(A) is established only on the complementary free-face sector. Nothing above the crossover scale is affected — the intermediate-annulus asymptotics, which is what the blow-up reduction and the numerics consume, holds for all ν — but a complete vertex theory for $\nu < \nu_W$ requires the self-consistent nonlinear inner state on the negative sector, the same open problem as the compressive inner region of Remark 4. The layer structure, by contrast, is confirmed across the whole range: $\text{tr } \mathbf{G} \neq 0$ at every zero of μ , so the one-large-stretch regime of §3.4 is exactly the layer regime, as Lemma 4 asserts.

For the weighted well-posedness we recall two standard inequalities in the Kondratiev space $V_\beta^{1,2}(B_\rho)$ —the closure of smooth functions vanishing on $\Gamma_D \cap \partial B_\rho$ under $\|\mathbf{v}\|_{V_\beta^{1,2}}^2 = \int_{B_\rho} (r^{2\beta} |\nabla \mathbf{v}|^2 + r^{2\beta-2} |\mathbf{v}|^2) dV$.

Lemma 5 (Weighted Hardy and Korn inequalities [?, ?]). *Let $\beta \in \mathbb{R}$ avoid the (discrete) set of exceptional exponents of the corner. There are constants C_H, C_K depending only on β and the opening angle such that, for all $\mathbf{v} \in V_\beta^{1,2}(B_\rho)$ vanishing on $\Gamma_D \cap \partial B_\rho$,*

$$\int_{B_\rho} r^{2\beta-2} |\mathbf{v}|^2 dV \leq C_H \int_{B_\rho} r^{2\beta} |\nabla \mathbf{v}|^2 dV, \quad (33)$$

$$\int_{B_\rho} r^{2\beta} |\nabla \mathbf{v}|^2 dV \leq C_K \int_{B_\rho} r^{2\beta} |\boldsymbol{\varepsilon}(\mathbf{v})|^2 dV, \quad \boldsymbol{\varepsilon}(\mathbf{v}) = \frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T). \quad (34)$$

Let $\beta_\star$ be the weight exponent for which the *linear* elasticity corner problem at this Dirichlet–Neumann junction is Fredholm (equivalently $\beta_\star$ avoids the singular exponents α). The existence of such a weight, and the resulting Fredholm property in the weighted space $V_{\beta_\star}^{1,2}$, is the standard

Kondratiev–Maz’ya–Nazarov–Plamenevsky theory for elliptic systems with conical/edge singularities [?, ?]: the operator is Fredholm precisely when β_* avoids the discrete spectrum of the corner operator pencil (the singular exponents), and the resolvent is then compact, so the spectrum is discrete and the critical mode ϕ of Assumption 8 is genuinely attained. We may write the Newton-step bilinear form as

$$a(\Delta \mathbf{u}, \delta \mathbf{u}) := \int_{B_\rho} \nabla \delta \mathbf{u} : \mathbb{L}(\bar{\mathbf{F}}) : \nabla \Delta \mathbf{u} \, dV. \quad (35)$$

Theorem 1 (Well-posedness of the linearised step, tensile corner). *Under Assumptions 1–5 and the non-compressive hypothesis (NC), the Newton-step form (35) is bounded and coercive modulo a compact perturbation: on any region Ω on which the tangent is strongly elliptic with the constant Legendre–Hadamard modulus μ_e (Lemma 3), there exist constants $\Lambda < \infty$, $\gamma_1 > 0$ and a compact form $k(\cdot, \cdot)$ such that, for all $\mathbf{u}, \mathbf{v} \in V_{\beta_*}^{1,2}(\Omega)$,*

$$|a(\mathbf{u}, \mathbf{v})| \leq \Lambda \|\mathbf{u}\|_{V_{\beta_*}^{1,2}} \|\mathbf{v}\|_{V_{\beta_*}^{1,2}}, \quad (36)$$

$$a(\mathbf{v}, \mathbf{v}) \geq \gamma_1 \|\mathbf{v}\|_{V_{\beta_*}^{1,2}}^2 - k(\mathbf{v}, \mathbf{v}). \quad (37)$$

The consequences split according to whether the vertex lies in the elliptic region.

(A) Globally convex-volumetric barrier ($f'' \geq 0$ together with the standing Assumptions 1–5; the class is nonempty with both moduli free — Remark 2 — the simplest exemplars being the pure logarithmic barrier $f(J) = -\mu_e \log J$ and the tempered penalties (27)). *Then the dilation core is empty, $r_* = 0$, ellipticity holds on the full corner neighbourhood $\Omega = B_\rho$ including the vertex, and (36)–(37) hold on $V_{\beta_*}^{1,2}(B_\rho)$. Hence the linearised corner operator is Fredholm of index zero in $V_{\beta_*}^{1,2}(B_\rho)$. Under the kinematic hypothesis (W) the vertex asymptotics are two-scale (Lemma 4): at the vertex itself the governing pencil is that of the screened limit $\mu_e \mathbb{I}$ — the mixed Dirichlet–Neumann vector-Laplacian pencil, with exponents $s \in \{1, 3, 5, \dots\}$, so the Newton increments are regular at the tip — while the Williams exponents $\alpha(\nu)$ of linearised elasticity govern the intermediate annulus $r_K \ll r \ll \rho$ and remain the dominant asymptotics visible from any fixed scale, every inner exponent exceeding $1 > \alpha$. This is the genuine corner result.*

(B) Logarithmic barrier $f_{\log}$ ($r_* > 0$). *The vertex sits inside the non-elliptic core $\{r \leq r_*\}$ (Lemma 3), so the estimates hold only on the vertex-free annulus $\Omega = A_* = \{r_* < r < \rho\}$. There the operator is a regular uniformly elliptic problem: (36)–(37) give Fredholmness of index zero in $V_{\beta_*}^{1,2}(A_*)$ and well-posedness on the annulus, but no corner-pencil asymptotics are asserted across the inner boundary $r = r_*$, which is a smooth interface carrying no vertex, not a conical point. The vertex behaviour of $f_{\log}$ is the open inner problem of Remark 4. In the mild-loading limit $r_* \downarrow 0$ (small $|K|$) the core shrinks below any fixed scale and the conclusion of (A) is recovered on every fixed sub-annulus down to mesh resolution.*

Proof. The bounds (36)–(37) are established on the elliptic region Ω ($= B_\rho$ in case (A), $= A_*$ in case (B)); the case split affects only whether the vertex is interior to Ω , hence whether the Kondratiev corner analysis applies.

Boundedness. By Cauchy–Schwarz and Lemma 2, on Ω the coefficient obeys the sharp bound $\|\mathbb{L}\| \leq \mu_e + c^{-2} \Theta(\log \frac{1}{r}) = \mu_e + \mathcal{O}(\log \frac{1}{r})$, a *logarithm*, not a power. Hence for every $\delta > 0$ there is C_δ with $\|\mathbb{L}\| \leq C_\delta r^{-\delta}$, and

$$|a(\mathbf{u}, \mathbf{v})| \leq \int_{\Omega} \|\mathbb{L}\| |\nabla \mathbf{u}| |\nabla \mathbf{v}| \, dV \leq C_\delta \left(\int_{\Omega} r^{2\beta_*} |\nabla \mathbf{u}|^2 \right)^{1/2} \left(\int_{\Omega} r^{-2\delta-2\beta_*} |\nabla \mathbf{v}|^2 \right)^{1/2}.$$

This exhibits a as bounded from $V_{\beta_\star}^{1,2} \times V_{-\beta_\star-\delta}^{1,2}$ (the dual pairing $V_{\beta_\star}^{1,2} \times V_{-\beta_\star}^{1,2}$ with a δ of weighted slack). The logarithm is subordinate to *every* algebraic weight, so the loss may be made smaller than the slack to the weight interval: let (β_-, β_+) be the maximal open interval of weights for which the *linear* corner pencil has no eigenvalue with $\Re$ -part in $[\beta_\star - \delta, \beta_\star]$ (such a δ -band exists because $\beta_\star$ avoids the discrete corner spectrum, hence sits at a positive distance δ_0 from it). Choose $\delta < \delta_0$, so $\beta_\star - \delta$ lies in the same admissible interval as $\beta_\star$. In case (B) the domain $\Omega = A_\star$ excludes the vertex, the two Kondratiev norms $V_{\beta_\star}^{1,2}$ and $V_{-\beta_\star-\delta}^{1,2}$ are then comparable (they differ only at the vertex, which $A_\star$ omits), and (36) follows directly with $\Lambda = \Lambda(\delta)$. In case (A) the vertex is interior to $\Omega = B_\rho$ and the two norms are *not* comparable there, so the same-space estimate is obtained by perturbation off a constant-coefficient symbol. The correct reference symbol at the vertex is *not* $\mathbb{L}_{\text{lin}} = \mathbb{L}(\mathbf{I})$: for any $K \neq 0$ the state leaves every neighbourhood of $\mathbf{I}$ as $r \rightarrow 0$, so $\mathbb{L} - \mathbb{L}_{\text{lin}}$ does not vanish at the vertex, and a non-vanishing (let alone logarithmically growing) principal-symbol perturbation shifts a corner pencil — the class $o(r^{-\delta})$ for every δ contains bounded perturbations and is *not* an admissible perturbation class for pencil stability. The correct reference symbol is the screened limit of Lemma 4: write $\mathbb{L} = \mu_e \mathbb{I} + \mathbb{R}$, so that under (W) the perturbation obeys $\|\mathbb{R}\| \leq \varepsilon(r) \rightarrow 0$ algebraically off the layers $\Lambda(r)$ of (31), and $\|\mathbb{R}\| = \mathcal{O}((\log \frac{1}{r})^p)$ on the layers themselves, whose angular measure vanishes algebraically. The off-layer part is a classical vanishing perturbation in the sense of Kondratiev–Maz’ya [?], under which the admissible weight interval of the reference (vector-Laplacian) pencil is unmoved and the *full* pair of estimates (36)–(37) transfers to $\mathbb{L}$ on $V_{\beta_\star}^{1,2}(B_\rho)$; the layer part is controlled in the dual pairing by the δ -slack exactly as above, and its absorption into the pencil statement is the perturbation step singled out at the end of the proof.

Gårding inequality. By Lemma 3 the symbol $\mathbb{L}(\mathbf{x})$ is strongly elliptic on Ω with the constant Legendre–Hadamard modulus μ_e . Freezing the (measurable, uniformly elliptic) coefficients on a dyadic partition $\{r \sim 2^{-j}\} \cap \Omega$ and applying the constant-coefficient Gårding inequality on each annulus—valid because the principal symbol is positive definite on the Legendre–Hadamard cone—yields, after summation against the weight $r^{2\beta_\star}$,

$$a(\mathbf{v}, \mathbf{v}) \geq \mu_e c_K^{-1} \int_{\Omega} r^{2\beta_\star} |\nabla \mathbf{v}|^2 dV - C \int_{\Omega} r^{2\beta_\star} |\mathbf{v}|^2 dV,$$

where the step from Legendre–Hadamard positivity to coercivity is the classical one for strongly elliptic systems: strong ellipticity with modulus μ_e (Lemma 3) gives the constant-coefficient Gårding bound for the symmetrised gradient $\varepsilon(\mathbf{v})$ on each frozen annulus, and the weighted Korn inequality (34) then converts control of $\varepsilon(\mathbf{v})$ into control of the full gradient $\nabla \mathbf{v}$. The remainder term has a strictly higher power of r than the Hardy weight $r^{2\beta_\star-2}$, hence by (33) it is relatively compact with respect to $\|\cdot\|_{V_{\beta_\star}^{1,2}}$; this is the compact form k , and we obtain (37) with $\gamma_1 = \mu_e c_K^{-1}$.

Fredholm property and exponents. Estimates (36)–(37) place the operator in the Kondratiev–Maz’ya framework [?]: it is Fredholm of index zero in $V_{\beta_\star}^{1,2}(\Omega)$ whenever $\beta_\star$ avoids the spectrum of the governing corner pencil. In case (A), where $\Omega = B_\rho$ contains the vertex, the governing pencil is fixed by the vertex limit of the coefficients, and by Lemma 4 that limit is $\mu_e \mathbb{I}$: the vertex pencil is the mixed Dirichlet–Neumann pencil of the decoupled vector Laplacian, with spectrum $\{1, 3, 5, \dots\}$ and hence no eigenvalue in the strip $0 < \Re \lambda < 1$; the energy weight is admissible with room to spare, and the elastic expansion (13) with the Williams exponents α is recovered as *intermediate* asymptotics on $r_K \ll r \ll \rho$, where $\bar{\mathbf{F}} \approx \mathbf{I}$ and $\mathbb{L} \approx \mathbb{L}_{\text{lin}}$. Two technical steps in this transfer are stated as such rather than hidden: the absorption of the degenerate-ray layers (a perturbation bounded, up to a logarithm, on angular sets of algebraically vanishing measure, to be handled by higher integrability of the gradient on dyadic annuli), and the transmission of the Dirichlet condition through the collapsing elastic boundary layer at the clamped face, inside

which the field stays moderate. Both are perturbation statements about a *vanishing* deviation from the screened limit — an entirely lesser order of difficulty than the pencil statement for the non-vanishing deviation from $\mathbb{L}_{\text{lin}}$ that a naive comparison with linear elasticity would require. In case (B), where $\Omega = A_*$ omits the vertex, the same estimates give Fredholmness and well-posedness on the annulus, but the inner boundary $r = r_*$ is a smooth interface, not a conical point, so the pencil eigenvalues describe no vertex asymptotics there; the corner expansion (13) is recovered only in the limit $r_* \downarrow 0$ on fixed sub-annuli, and the genuine vertex theory for $f_{\log}$ remains the open inner problem of Remark 4. $\square$ $\square$

Remark 4 (The dilation core). For $f_{\log}$ the conclusion of Theorem 1 is genuinely confined to A_* : the vertex itself sits inside the core $\{r \leq r_*\}$, where $J > J_*$ and $f_{\log}$ is not strongly elliptic. Four readings are consistent with the analysis. (i) *Mild loading*: for stress-intensity $|K|$ small the core radius $r_* = (|K|/J_*)^{1/(1-\alpha)}$ is so small that the corner is elliptic on all of B_ρ down to mesh scale, and the theorem governs the entire computationally resolved region. (ii) *Sharper model statement*: the tensile corner “passes like linear elasticity” on A_* , and the only obstruction is the material law’s own loss of ellipticity past J_* —a property of $f_{\log}$, not of the corner. (iii) *Open inner problem*: a complete vertex theory for $f_{\log}$ requires a separate analysis inside the non-elliptic core, which we leave open; a globally convex-volumetric barrier ($f'' \geq 0$, $r_* = 0$) removes the issue entirely and recovers the unrestricted statement. (iv) *Screened core geometry*: under the kinematic hypothesis (W) the core is not, in fact, a full disc. By Lemma 4(iii) the Legendre–Hadamard defect $f''(J)J^2\xi^2$ is damped by $\|\bar{\mathbf{F}}^{-1}\|^2 \rightarrow 0$ off the degenerate-ray layers, so strong ellipticity survives *down to the vertex* there even though $f'' < 0$; the genuinely delicate set shrinks to the layers $\Lambda(r)$ of (31) together with a bounded transition collar $r \sim r_K$ where the field is $\mathcal{O}(1)$ -nonlinear. One caution on the finite-extensibility reading sometimes offered here: a volumetric law that caps the dilation, $J < J_{\max}$, is *incompatible* with the assumed corner kinematics rather than a cure for it — under (NC) and $\lambda_{\max} \sim |K|r^{\alpha-1}$ one has $J \geq c\lambda_{\max} \rightarrow \infty$, so a capped material cannot carry the Williams blow-up at all and the state must reorganise, changing the ansatz rather than the tangent. The legitimate realisation of the finite-extensibility intuition is the one-sided tempered penalty of Remark 2(iii): a C^3 energy, identical to $f_{\log}$ throughout compression, whose softened dilation tail keeps $f'' > 0$ globally and so removes the core ($r_* = 0$) while leaving both moduli free. The dilation core is thus a feature of the specific large- J tail of $f_{\log}$, in a regime never physically constrained, not a physical catastrophe.

Remark 5 (Stress is singular, operator is not). The first Piola–Kirchhoff stress still blows up, $\mathbf{P} \sim \mu_e \mathbf{F} \rightarrow \infty$ as $r \rightarrow 0$, so the tensile corner is a genuine *stress* singularity. Theorem 1 asserts only that, on the annulus A_* , this singularity lives entirely in the *right-hand side* (the residual) of the Newton step, not in its operator: under a mild volumetric penalty the nonlinear problem inherits the regularity theory of the linear one, standard graded meshes restore optimal convergence, and no structural instability occurs. This is the precise sense in which a non-compressive corner “passes like linear elasticity” away from the dilation core.

Remark 6 (The quadratic penalty does *not* pass like linear elasticity). For f_{quad} , Assumption 5 fails: by (25) and (26) the tangent grows algebraically, $\|\mathbb{L}\| = \mathcal{O}(r^{-2(1-\alpha)})$, *even in pure tension*. The coefficients of (35) are then genuinely singular, of severity comparable to the compressive Scenario B, the boundedness step above fails for β_* , and the admissible weight interval—hence the effective corner exponents seen by the nonlinear operator—shifts relative to linear elasticity. A sharp analysis of this case (the correct weight, and whether coercivity survives) is governed by the volumetric blow-up rather than by the deviatoric singularity and is left open here.

We emphasise that this serves as a *cautionary comparator* rather than a model we advocate: as written, $f'_{\text{quad}}(1) = 0 \neq -\mu_e$, so $\frac{\kappa}{2}(J-1)^2$ does not combine with the deviatoric term $\frac{\mu_e}{2}(\text{tr}(\mathbf{F}^T \mathbf{F}) -$

d) to leave the reference configuration stress-free, and in practice it is used together with an isochoric (deviatoric/volumetric) split of $\mathbf{F}$. The point is that a standard-looking polynomial penalty already forfeits the benign tensile behaviour enjoyed by the logarithmic model (8) actually used in [?].

Remark 7 (Self-consistency of (NC)). Hypothesis (NC) is compatible with the singular kinematics (14): an angular profile Φ corresponding to opening/shear produces $\nabla \mathbf{u}$ whose dominant eigenvalue is positive (radial extension), so $\lambda_{\max} \rightarrow \infty$ while $\lambda_{\min} = \det \mathbf{F} / \lambda_{\max}$ need not approach zero. The angular structure is made quantitative by Lemma 4 and checked in §6.2: off the degenerate-ray layers both stretches diverge (so (NC) holds with room to spare), on the layers $\lambda_{\min}$ stays $\mathcal{O}(1)$, and the orientation clause of (W) holds for $\nu > \nu_W \approx 0.40$. The complementary profile, in which the dominant eigenvalue of $\nabla \mathbf{u}$ is negative, drives $\lambda_{\min} \rightarrow 0$ and violates (NC); that is exactly Scenario B below.

3.6 Scenario B: the compressive singularity (the pathological case)

Now assume the far-field loading pushes the material *into* the corner ($K < 0$, radial compression).

Kinematic consequence. The naive linear prediction gives $\nabla \mathbf{u}$ a negative dominant eigenvalue scaling as $-|K|r^{\alpha-1}$. Because $\mathbf{F} = \mathbf{I} + \nabla \mathbf{u}$, as $r \rightarrow 0$ (or as the pseudo-time load factor t increases) there is a critical threshold where the local contraction annihilates the material volume:

$$\det \mathbf{F} \rightarrow 0 \implies J \rightarrow 0, \quad \lambda_{\min} \rightarrow 0. \quad (38)$$

Operator blow-up (and its model dependence). Here it is the *inverse channel* of (20) that fires: $\|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1} \rightarrow \infty$. The dilation channel $\Theta(J)$ is now probed at $J \rightarrow 0^+$ rather than $J \rightarrow \infty$, and—exactly as in tension, but with the roles reversed—its behaviour discriminates between the two penalties. Writing $J = \prod_i \lambda_i \sim \lambda_{\min}$ near the corner (one collapsing stretch, the remaining $d-1$ bounded, cf. §3.4),

$$\Theta_{\log}(J) \sim 2\lambda_e |\log J| \rightarrow \infty, \quad \Theta_{\text{quad}}(J) \sim 2\kappa J \rightarrow 0 \quad (J \rightarrow 0^+), \quad (39)$$

so that the tangent norm $\|\mathbb{L}\| \leq \mu_e + \Theta(J)\|\mathbf{F}^{-1}\|^2$ behaves as

$$\|\mathbb{L}(\mathbf{x})\| \sim \begin{cases} \mathcal{O}(\lambda_{\min}^{-2} \log \frac{1}{\lambda_{\min}}), & f = f_{\log}, \\ \mathcal{O}(\lambda_{\min}^{-1}), & f = f_{\text{quad}}. \end{cases} \quad (40)$$

Thus the logarithmic penalty—benign under tension—produces here the *stronger* blow-up, $\lambda_{\min}^{-2}$ *amplified* by a logarithm, whereas the quadratic penalty gives the milder rate $\lambda_{\min}^{-1}$. This reversal is not a redemption of the quadratic model: $f_{\text{quad}}(J) \rightarrow \frac{\kappa}{2} < \infty$ as $J \rightarrow 0^+$, so it provides *no energy barrier against volume collapse* and violates Assumption 1; the milder tangent growth merely reflects this missing barrier (the material is free to interpenetrate at finite energy), which is a worse pathology, not a better one. The logarithmic penalty’s stronger blow-up is, conversely, the analytic price of a genuine barrier $f_{\log}(J) \rightarrow +\infty$ —and it is precisely that barrier which forces, and gives physical meaning to, the bifurcation rescue of Section 4.

Ellipticity is *not* lost — the instability is structural. It is tempting to attribute the breakdown to a loss of strong ellipticity (an indefinite acoustic tensor). For this neo-Hookean model that is *false*, and it is worth stating the fact precisely, as it shapes the rest of the analysis.

C: Same kinematic caveat as (W) on the tensile side: at depth the Williams field has $J \sim K^2 r^{2(\alpha-1)} \mu(\theta)$ 0 for $\nu > \nu_W$ regardless of load sign, so the compressive collapse $J \rightarrow 0$ lives in the transition annulus $r \sim r_K$, not at the vertex — consistent with the tip-disc picture of Prop. 8, but worth one sentence here.

Lemma 6 (Compressive strong ellipticity of the neo-Hookean tangent). *For the energy (7), the rank-one form of the tangent is exactly (18), $L_{iJkL}m_iN_Jm_kN_L = \mu_e|\mathbf{m}|^2|\mathbf{N}|^2 + f''(J)J^2(\mathbf{m} \cdot \mathbf{F}^{-T}\mathbf{N})^2$. Wherever $f''(J) \geq 0$ — in particular throughout compression, $J \leq 1$, for both penalties (8) — the tangent is strongly elliptic with constant $\geq \mu_e > 0$, uniformly in $\mathbf{F}$, and the acoustic tensor $Q_{ik}(\mathbf{N}) = L_{iJkL}N_JN_L$ is positive definite and never degenerates under compression.*

Proof. The identity is (18) (Lemma 1). As $J \rightarrow 0^+$, $f''(J)J^2 \rightarrow +\infty$ for $f_{\log}$ (cf. (65)) and $f''(J)J^2 = \kappa J^2 \geq 0$ for f_{quad} ; in both cases the second term only *adds* to the constant μ_e . $\square$ $\square$

Thus the compressive corner is *not* a material instability: the volumetric stiffening that blows up $\|\mathbb{L}\|$ is exactly the term $f''(J)J^2$ in (18), and it is positive — it strengthens ellipticity while destroying conditioning. What fails is the *boundedness* of the operator, not its positivity.

Conclusion for Scenario B. The assumption that the deformation smoothly follows the fundamental path into the corner therefore produces, not a loss of ellipticity, but two distinct effects: (i) a blow-up of the tangent *norm*, $\|\mathbb{L}\| \rightarrow \infty$ (eq. (40)), which renders the Newton step catastrophically ill-conditioned even though it remains elliptic; and (ii) for a barrier penalty $f_{\log}$, an unphysical infinite accumulation of energy ($f(J) \rightarrow \infty$ as $J \rightarrow 0$), signalling that the smooth fundamental path is not the physical solution. The genuine mechanism that intervenes is a *structural* bifurcation of the compressed state — the corner analogue of *creasing*, an instability of a material that remains everywhere strongly elliptic (Lemma 6), driven by the geometry and boundary conditions rather than by a change of type of the PDE. To capture it we introduce in Section 4 an imperfect perturbation that regularises the bifurcation and bounds the singularity; we work from here on with the barrier model $f_{\log}$, for which collapse is genuinely resisted.

4 Imperfect bifurcation analysis and asymptotic bounds

The previous section showed that the perfect compressive path drives $\lambda_{\min} \rightarrow 0$ and blows up the tangent norm while the material remains strongly elliptic (Lemma 6), so the breakdown is a structural instability rather than a loss of type. The premise of this section is that this scenario is an idealisation: a real system always carries small disturbances—geometric imperfections, load eccentricities, material inhomogeneity—and these select a nearby *regular* solution branch on which the collapse never quite happens. We model the disturbance by a single scalar amplitude ϵ , carry out an imperfect-bifurcation analysis, and quantify how the tangent blows up as $\epsilon \rightarrow 0$. The punchline is that the blow-up is confined to a vanishingly small neighbourhood of the corner and proceeds at a controlled algebraic rate, so that for every $\epsilon > 0$ the problem is regular.

4.1 The regularised problem and the fundamental path

We localise to the corner neighbourhood B_ρ of §3, regard the loading as transmitted through data on the outer arc $\partial B_\rho \setminus (\Gamma_D \cup \Gamma_N)$, and introduce a pseudo-time load parameter $t \in [0, 1]$ that monotonically scales the compressive part of that data. Following the answer to the question posed in Scenario B, we add a small, fixed volumetric body force $\epsilon \mathbf{p}(\mathbf{x})$ with $\|\mathbf{p}\|_{L^2(B_\rho)} = 1$ and $\epsilon \ll 1$ as the imperfection (we write the field as $\mathbf{p}$ to avoid collision with the penalty $f(J)$). The regularised equilibrium is $G(\mathbf{u}, t, \epsilon) = 0$,

$$\langle G(\mathbf{u}, t, \epsilon), \mathbf{v} \rangle = \int_{B_\rho} \mathbf{P}(\mathbf{F}) : \nabla \mathbf{v} \, dV - \epsilon \int_{B_\rho} \mathbf{p} \cdot \mathbf{v} \, dV - \ell_t(\mathbf{v}) = 0 \quad \forall \mathbf{v} \in V, \quad (41)$$

with $\mathbf{P} = \partial\Psi/\partial\mathbf{F}$, V the admissible test space, and ℓ_t the (linear) load functional at pseudo-time t . Let $\mathbf{u}_0(t)$ denote the *fundamental path*, i.e. the branch of (41) with $\epsilon = 0$ that continues the low-load compressive solution — constructed in Lemma 7 and regulated by Assumption 7 below. We use the barrier penalty $f_{\log}$ throughout, so the collapse $J \rightarrow 0$ is genuinely resisted (Assumption 1); along $\mathbf{u}_0(t)$ the corner stretch $\lambda_c(t) := \lambda_{\min}(\mathbf{F}_0(t))$ decreases as $t \uparrow 1$ but, on the bifurcating branch we construct, never reaches zero.

The fundamental path deserves an existence statement rather than a definition by fiat. Existence is variational and, in two dimensions, genuinely provable — for a *globally convex* volumetric penalty, which is where the tempered class of Remark 2 pays a second dividend; what resists proof is the *regularity* of the path, and we collect exactly the regularity that the analysis below consumes into one standing assumption, stated here, before its first use, rather than discovered midway.

Lemma 7 (Existence and energy bound for the fundamental state). *Let $d = 2$, let the loading be displacement-controlled (ℓ_t realised as boundary data $t\mathbf{g}$ on the Dirichlet part, $\epsilon = 0$), and let the volumetric penalty f be convex on $(0, \infty)$ with the barrier $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$ (the class of Remark 2; in particular the one-sided tempered penalty, which coincides with $f_{\log}$ throughout compression). Extend $f \equiv +\infty$ on $(-\infty, 0]$. Then for every t the energy $\mathcal{E}_t(\mathbf{u}) = \int_{B_\rho} \Psi(\mathbf{I} + \nabla\mathbf{u}) dV$ attains a global minimiser $\mathbf{u}_0(t)$ over the admissible class $\{\mathbf{u} \in H^1 : \mathbf{u} = t\mathbf{g} \text{ on } \Gamma_D \cup \text{outer arc}\}$, and every minimiser satisfies $J_0 > 0$ a.e. The minimum value $E_{\min}(t)$ is continuous in t and obeys $E_{\min}(t) \leq Ct^2$ for any smooth extension competitor that stays orientation-preserving on $[0, 1]$.*

Proof. Coercivity: $\Psi \geq \frac{\mu_e}{2}|\mathbf{F}|^2 + f(J)$ with f convex, bounded below by an affine function of J and $|J| \leq \frac{1}{2}|\mathbf{F}|^2$, so minimising sequences are bounded in H^1 ; pass to a weak limit. In $d = 2$ the cofactor is *affine* in $\mathbf{F}$, whence for $\mathbf{u}_k \rightharpoonup \mathbf{u}$ in H^1 the identity $\int J_k \varphi = -\frac{1}{2} \int \mathbf{u}_k \cdot (\text{cof } \nabla \mathbf{u}_k) \nabla \varphi + \dots$ pairs a strongly L^2 -convergent factor (Rellich) with a weakly convergent one: $\det \nabla \mathbf{u}_k \rightarrow \det \nabla \mathbf{u}$ in $\mathcal{D}'$, and since the J_k are bounded in L^1 with a distributional limit that is itself the L^1 function J , no singular part can form. Lower semicontinuity: the quadratic term is convex; for the penalty, $f = \sup_\ell \ell$ over its affine minorants, $\int \ell(J_k) \rightarrow \int \ell(J)$ for each ℓ , and monotone approximation gives $\liminf \int f(J_k) \geq \int f(J)$ (Goffman–Serrin; the barrier side is Fatou). Finite energy forces $J_0 > 0$ a.e. The bound $E_{\min}(t) \leq \mathcal{E}_t(t\mathbf{G}) \leq Ct^2$ is Taylor expansion of Ψ at $\mathbf{I}$ along the smooth competitor; continuity of $E_{\min}$ follows by comparing minimisers at nearby loads after correcting the boundary data with the same extension and using the lower semicontinuity just proved. $\square \quad \square$

Three caveats calibrate the lemma. For $f_{\log}$ itself the direct method is obstructed by the same large- J nonconvexity that creates the dilation core of Lemma 3 — one more reason to temper the tail (Remark 2); minimisers need not be unique, so a *selection* $t \mapsto \mathbf{u}_0(t)$ continuous in t is not automatic; and minimisers of polyconvex energies carry essentially no regularity theory, so nothing in the lemma bounds $\lambda_{\min}(\mathbf{F}_0)$ below or gives the state a corner expansion. Those are precisely the unprovable inputs, and we now name them. Define the pointwise-collapse load $t_{\text{coll}} := \inf\{t : \text{ess inf}_{B_\rho} \lambda_{\min}(\mathbf{F}_0(t)) = 0\}$.

Assumption 7 (Regular fundamental path, (FP)). There is a selection $t \mapsto \mathbf{u}_0(t)$ of solutions of (41) with $\epsilon = 0$, continuous into H^1 with $\mathbf{u}_0(0) = \mathbf{0}$ (for a convex penalty, a selection of the minimisers of Lemma 7), such that:

- (i) *Pre-collapse non-degeneracy.* $t_{\text{coll}} > 0$, and for every $\bar{t} < t_{\text{coll}}$ there is $c(\bar{t}) > 0$ with $\lambda_{\min}(\mathbf{F}_0(t)) \geq c(\bar{t})$ a.e. on B_ρ for all $t \leq \bar{t}$; consequently $t \mapsto \mathbb{L}(\mathbf{F}_0(t))$ is bounded and continuous into L^∞ on $[0, \bar{t}]$.

- (ii) *Corner expansion.* For $t < t_{\text{coll}}$ the state admits the expansion $\mathbf{u}_0 = K(t) r^\alpha \Phi(\theta) + \mathbf{u}_0^\sharp$ with the pointwise gradient envelope $|\nabla \mathbf{u}_0^\sharp| \leq C_\sharp(t)(r/\rho)^{\alpha'-1}$, $\alpha' > \alpha$, and $C_\sharp$, K bounded on compact subsets of $[0, t_{\text{coll}})$.
- (iii) *Monotone load-to-stretch map.* Condition (M): at each material point the relevant stretches $t \mapsto \lambda_\parallel$, $\lambda_{\min}$ are nonincreasing along the path.

Remark 8 (What (FP) costs, and how it removes a circularity). Part (i) is the compressive analogue of (NC) and cannot be discharged by the tensile screening of Lemma 4: under compression the inverse channel $J \rightarrow 0$ has no analogous cancellation, so the coefficient field of the state genuinely may degenerate, and (i) is an assumption about where — namely, not before t_{coll} , which (i) merely asserts is positive. Part (ii) is likewise a hypothesis, not a citation: the linear Kondratiev theory applies to the *nonlinear* state only through a frozen-coefficient reading whose vertex behaviour is exactly the open inner problem of Remark 4. The gain from stating (FP) here is logical hygiene downstream: previously, continuity of the second variation for $t < t_c$ silently consumed a $\lambda_{\min}$ floor that only the later buckling-precedes-collapse condition (BC) supplies — a circle. Under (FP) the circle opens into a line: Lemma 8 needs (i) only on $[0, t_*]$, where t_* is the load at which a destabilising field first exists; if $t_* < t_{\text{coll}}$ — the single quantitative input, a tip floor at the *Biot* load, not at collapse — then the first crossing satisfies $t_c \leq t_* < t_{\text{coll}}$, and the ordering (BC) is a *conclusion*. The residual open content is thus one lower bound on t_{coll} , quantified in the concrete setting by the tip-disc estimate there, and nothing else.

Remark 9 (The instability is structural, not material). By Lemma 6 the tangent stays strongly elliptic throughout compression, so the bifurcation here is a *structural* one (a buckling of the boundary-value problem), not a material change of type. We work with the global form on the bounded domain B_ρ . The operator $\mathcal{L}(t) = \partial_{\mathbf{u}} G(\mathbf{u}_0(t), t, 0)$ has discrete spectrum and attains $\mu_1(t)$ as a genuine eigenvalue *provided* the fundamental stretch stays bounded away from collapse up to the crossing, $\inf_{B_\rho} \lambda_{\min}(\mathbf{F}_0(t)) \geq c > 0$ for $t \leq t_c$: then $\mathcal{L}(t)$ has *bounded* strongly-elliptic coefficients on the bounded domain, hence (Gårding, Korn and the Rellich embedding) compact resolvent — we make this precise as Lemma 18 in the concrete setting. The reservation is essential: what makes the problem delicate is not a loss of ellipticity but the *norm* blow-up $\|\mathbb{L}\| \rightarrow \infty$ (eq. (40)), and it is exactly this blow-up, through the scale-invariant (creasing) corner, that would push the bottom of the spectrum into the *essential* spectrum and destroy attainment. Crucially, that blow-up occurs only on the unphysical continuation past t_c , where $\lambda_{\min} \rightarrow 0$; at the crossing itself the spectrum is discrete *so long as buckling precedes collapse*. The coercivity constant thus stays positive while the boundedness constant diverges, and the crossing eigenvalue $\mu_1(t)$ and the conditioning are controlled by different mechanisms. Assumption 8 records the spectral facts (attainment and simplicity) we use; §5.10 shows that, for the concrete model, both follow once buckling is known to precede collapse.

4.2 What is provable, and what must be assumed

It is worth separating the genuinely provable facts about the critical point from the irreducible hypotheses. Let $Q_t(\mathbf{v}) := \int_{B_\rho} \nabla \mathbf{v} : \mathbb{L}(\mathbf{F}_0(t)) : \nabla \mathbf{v} dV$ be the second-variation form on the fundamental path and $\mu_1(t) := \inf_{\|\mathbf{v}\|_V=1} Q_t(\mathbf{v})$.

The critical load exists (conditionally provable). A clarification is essential here. By Lemma 6 the neo-Hookean material remains strongly elliptic throughout compression, so $\mu_1(t)$ *cannot* be driven negative by a rank-one (material) instability: any crossing must be a *structural*

loss of positivity of the global form Q_t , i.e. a buckling of the boundary-value problem in which the destabilising test field is not rank-one. This is the creasing mechanism.

Lemma 8 (Existence of the critical load, and the ordering it inherits). *Assume (i) coercivity at zero load, $\mu_1(0) > 0$; (ii) a structural instability of the compressed state: there exist $t_* \in (0, 1)$ and $\mathbf{v}_* \in V$ with $Q_{t_*}(\mathbf{v}_*) < 0$; and (iii) the destabilisation precedes collapse, $t_* < t_{\text{coll}}$, with (FP)(i) in force. Then μ_1 is finite and continuous on $[0, t_*]$, there is a first $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$ and $\mu_1(t) > 0$ for $t < t_c$, and the buckling load automatically satisfies the ordering $t_c \leq t_* < t_{\text{coll}}$.*

Proof. At $t = 0$, $\mathbf{F}_0 = \mathbf{I}$ and $\mathbb{L} = \mu_e \mathbb{I} + (\text{volumetric at } J=1)$ is positive definite, so with Korn's inequality $\mu_1(0) > 0$. Hypothesis (ii) gives $\mu_1(t_*) \leq Q_{t_*}(\mathbf{v}_*) / \|\mathbf{v}_*\|_V^2 < 0$. By (iii) and (FP)(i) the coefficients $\mathbb{L}(\mathbf{F}_0(t))$ are bounded in L^∞ uniformly on $[0, t_*]$ and continuous in t — this is where the finiteness of Q_t comes from, and it is not free: without the floor $\lambda_{\min} \geq c(t_*)$ the form need not even be bounded — so μ_1 is continuous, and the intermediate value theorem gives the first zero $t_c \leq t_* < t_{\text{coll}}$. $\square$ $\square$

Hypothesis (i) is provable for neo-Hookean (positive-definiteness at $\mathbf{F} = \mathbf{I}$ plus Korn). Hypothesis (ii) — the existence of a structurally destabilising field — is the genuine content and is *not* elementary: it is the corner counterpart of the Biot/creasing instability [?, ?] and we do not establish it here.

The critical mode (assumed).

Assumption 8 (Critical mode). At t_c the infimum $\mu_1(t_c) = 0$ is attained at a mode $\phi \in V$, $\|\phi\|_V = 1$, spanning a one-dimensional, isolated kernel of $\mathcal{L}_c$ (the rest of the spectrum bounded away from 0), with ϕ supported in the edge neighbourhood B_ρ .

This bundles two spectral facts: *attainment* (the minimising sequence does not escape into the corner as essential spectrum) and *simplicity*. Simplicity is generic — in the precise sense of holding for almost every point of the wedge parameter family $(\nu, \hat{K})$, the only sense in which *generic* carries content for a fixed body — and, for the model bodies, can be argued from the lateral *face-preserving* reflection R of §5.6 (not the corner bisector $\theta \mapsto \frac{\pi}{2} - \theta$, which swaps Γ_D and Γ_N and is no symmetry of the mixed problem); attainment is the borderline-elliptic difficulty flagged in Remark 9. Neither is irreducible: in the concrete setting attainment is *proved* (Lemma 18) and simplicity is reduced to genericity *within* the symmetry sector carrying the critical mode (§5.6), the only residue being the cleaner geometric condition that buckling precede pointwise collapse.

The crossing is transversal (provable under monotone loading).

Lemma 9 (Transversality). *Suppose the loading is monotonically destabilising at t_c ,*

$$\langle \phi, \dot{\mathbb{L}}(t_c) \phi \rangle := \int_{B_\rho} \nabla \phi : \partial_t \mathbb{L}(\mathbf{F}_0(t_c)) : \nabla \phi \, dV < 0. \quad (\text{M})$$

Then, under Assumption 8, μ_1 is differentiable at t_c with $a := -\mu_1'(t_c) = -\langle \phi, \dot{\mathbb{L}}(t_c) \phi \rangle > 0$.

Proof. For a simple eigenvalue, first-order perturbation theory gives $\mu_1'(t_c) = \langle \phi, \dot{\mathbb{L}}(t_c) \phi \rangle$; hypothesis (M) fixes the sign. $\square$ $\square$

Condition (M) is the precise meaning of “more compression $\Rightarrow$ less stiffness in the buckling direction” and holds for any monotone compressive load path; it is an input on the *loading*, not on the solution.

The imperfection is active (provable, generic).

Lemma 10 (Generic activation). *For every $\mathbf{p}$ outside the closed codimension-one subspace $\phi^\perp$, the projection $\eta := \int_{B_\rho} \mathbf{p} \cdot \phi \, dV \neq 0$. Hence $\eta \neq 0$ for generic imperfection.*

Remark 10 (Universality of the imperfection). The volumetric body force $\epsilon \mathbf{p}$ is chosen only for concreteness; the analysis depends on the imperfection *solely* through the scalar $\eta = \langle \mathbf{p}, \phi \rangle$, so any perturbation with a nonzero ϕ -projection produces the identical reduced equation (44). Indeed, write a generic imperfect equilibrium as $G(\mathbf{u}, t, 0) = \epsilon \mathbf{h}$ with $\mathbf{h} \in V'$ the (linearised) imperfection functional. A boundary-traction defect on Γ_N contributes $\mathbf{h}(\mathbf{v}) = \int_{\Gamma_N} \mathbf{t}_1 \cdot \mathbf{v} \, dS$; a geometric imperfection — rounding the corner with radius $\sim \epsilon$, or a small shape variation $\mathbf{x} \mapsto \mathbf{x} + \epsilon \boldsymbol{\theta}(\mathbf{x})$ — enters, after pulling back to the reference domain, as a fixed load functional $\mathbf{h}(\mathbf{v}) = \int_{B_\rho} \mathbb{S}[\boldsymbol{\theta}] : \nabla \mathbf{v} \, dV$ linear in the shape field $\boldsymbol{\theta}$. In every case the kernel projection $\langle G, \phi \rangle = 0$ yields the same normal form with η replaced by $\langle \mathbf{h}, \phi \rangle$, nonzero for $\mathbf{h}$ outside the closed codimension-one subspace $\phi^\perp$ (Lemma 10). The Koiter $\frac{2}{3}$ law (52) and all subsequent bounds are therefore universal across generic imperfections, not an artefact of the body-force model.

The nonlinearity is explicit; only the sign of its orthogonal-mode feedback is open.

For the neo-Hookean model the reduced coefficients are not opaque. By Lemma 16 the deviatoric energy is quadratic and drops out, so every coefficient of order ≥ 3 is purely volumetric; moreover the quadratic coefficient $c_2 := \frac{1}{2} \langle \phi, \partial_{\mathbf{u}}^2 G[\phi, \phi] \rangle$ vanishes *identically* for a mode antisymmetric under the lateral, face-preserving reflection R of §5.6 — not the corner bisector $\theta \mapsto \frac{\pi}{2} - \theta$, which swaps Γ_D and Γ_N and is no symmetry at all. What these structural facts do *not* settle is the *sign* of the cubic and quintic coefficients; indeed, by Lemma 16(ii), the explicit single-mode (frozen) parts point *against* subcriticality, $b_{\text{dir}} > 0$ and $d_{\text{dir}} < 0$ in compression. The subcritical pattern therefore hinges on the geometry-dependent orthogonal-mode feedback overturning these signs; we record it as a residual hypothesis to be verified numerically (Remark 11) rather than asserted from the creasing analogy [?, ?].

Assumption 9 (Subcritical sign pattern (residual, geometry-dependent)). With $c_2 = 0$ by Lemma 16(i), the full reduced cubic and quintic coefficients

$$b := \frac{1}{6} \langle \phi, \partial_{\mathbf{u}}^3 G[\phi, \phi, \phi] \rangle, \quad d := \frac{1}{120} \langle \phi, \partial_{\mathbf{u}}^5 G[\phi, \dots, \phi] \rangle, \quad (42)$$

each the frozen part of Lemma 16 corrected by the orthogonal-mode feedback, satisfy $b < 0 < d$ (subcritical with quintic saturation). This is a condition on the corner geometry, not on the material model.

Remark 11 (Status of the subcritical signs: geometry-decided, checked numerically). Assumption 9 is the one genuinely residual hypothesis of the construction, and Lemma 16 localises exactly where its content lies. The deviatoric response and the quadratic coefficient are settled ($c_2 = 0$, no feedback correction); the cubic and quintic decompose as $b = b_{\text{dir}} + b_\psi$ and $d = d_{\text{dir}} + d_\psi$, with the single-mode parts explicit and, in compression, of definite but *adverse* sign ($b_{\text{dir}} = 2 \int_{B_\rho} f''(J_0) d_\phi^2 \, dV > 0$, $d_{\text{dir}} = \int_{B_\rho} f'''(J_0) d_\phi^3 \, dV < 0$). The corrections b_ψ, d_ψ are the Lyapunov–Schmidt feedback through the orthogonal complement $V^\perp$: quadratic forms in the second-order field $\boldsymbol{\psi}$ that depend on the opening angle, the Dirichlet–Neumann arrangement and the mode ϕ — on the geometry, not on f . Subcriticality thus requires b_ψ to overturn the explicit positive b_{dir} (and likewise for d). The reason this does not reduce to a single universal corner number is a homogeneity count (Proposition 6, §5.7): the clamped–free edge exponent satisfies $\alpha > \frac{1}{2}$, so the cubic integral is convergent at the

tip but *not* dominated by it, leaving an $\mathcal{O}(1)$ dependence on the normalised stress-intensity $\hat{K}$; the sign of b is therefore a function $b(\nu, \hat{K})$ on the model wedge rather than an inheritance from flat creasing. The *direction* of the cubic feedback can in fact be signed analytically: $b_\psi \leq 0$ always, by the variational structure of the first crossing (Lemma 11 below), so the genuinely open content is the magnitude comparison $|b_\psi| > b_{\text{dir}}$; the quintic feedback d_ψ we do not sign. The magnitude is evaluated numerically from the eigenmode and reduced operator of the two-dimensional model wedge of Proposition 5 as a function of $(\nu, \hat{K})$, which is the proper arbiter of a geometry-dependent sign. Were the verdict $b > 0$ instead, the bifurcation would be supercritical and *more* benign — the physical branch smooth through t_c with no snap at onset — and the regularisation conclusions of §5.10 and Proposition 4 would hold a fortiori; the complementary transcritical case $c_2 \neq 0$ is covered in Remark 13.

4.3 Lyapunov–Schmidt reduction

Proposition 1 (Normal form). *Under Assumption 8, Lemma 9 and Assumption 9, there is a neighbourhood of $(\mathbf{u}_0(t_c), t_c, 0)$ in which the solutions of (41) are exactly*

$$\mathbf{u}(t, \epsilon) = \mathbf{u}_0(t) + w\phi + \psi(w, t, \epsilon), \quad \psi \in V^\perp := \{\phi\}^\perp, \quad (43)$$

where ψ is smooth with $\psi = \mathcal{O}(w^2 + \epsilon)$, and the amplitude w solves the scalar imperfect bifurcation equation, retained to quintic order because the cubic is destabilising ($b < 0$),

$$a(t_c - t)w + bw^3 + dw^5 = \epsilon\eta + \mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t|w^2 + w^6), \quad (44)$$

with $a > 0$ (Lemma 9), $b < 0 < d$ (Assumption 9) and η as in Lemma 10.

Proof. Let P be the V -orthogonal projection onto $\text{span}\{\phi\}$ and $Q = I - P$. Write $\mathbf{u} = \mathbf{u}_0(t) + w\phi + \psi$ with $\psi \in V^\perp$. The range equation $QG = 0$ has invertible linearisation $Q\mathcal{L}_cQ$ on $V^\perp$ by Assumption 8, so the implicit function theorem yields a unique smooth $\psi = \psi(w, t, \epsilon)$ with $\psi(0, t_c, 0) = 0$ and $\psi = \mathcal{O}(w^2 + \epsilon + |t_c - t|)$. Substituting into the kernel equation $\langle G, \phi \rangle = 0$ and Taylor-expanding G about $(\mathbf{u}_0(t_c), t_c, 0)$ gives (44): the linear term $a(t_c - t)w$ from Lemma 9, the cubic bw^3 and quintic dw^5 from Assumption 9 (the quadratic vanishing by symmetry), and the inhomogeneity $\epsilon\eta$ from $-\epsilon\langle \mathbf{p}, \phi \rangle$. Because $b < 0$, the cubic alone does not saturate the amplitude, which is why the quintic dw^5 ($d > 0$) is retained. $\square$ $\square$

4.4 The sign of the orthogonal-mode feedback is not free

Assumption 9 bundles two logically distinct statements about the cubic: that the Lyapunov–Schmidt feedback b_ψ *opposes* the frozen part ($b_\psi \leq 0$, against $b_{\text{dir}} > 0$ of Lemma 16(ii)), and that it does so with sufficient *magnitude*, $|b_\psi| > b_{\text{dir}}$. The first is not a hypothesis at all: at the first crossing it is a theorem, forced by the variational structure alone. We record it here, both because it halves the open content of Assumption 9 and because it converts the numerical verification of §6 from a sign determination into a magnitude comparison, with a certification principle (Corollary 1) that requires no inversion of the reduced operator.

Because the load terms in (41) are linear in $\mathbf{u}$, G is a gradient, $G(\mathbf{u}, t, \epsilon) = D\mathcal{E}_t(\mathbf{u}) - \epsilon\mathbf{p}$ with $\mathcal{E}_t$ the total energy at load t ; write $E_k := D^k\mathcal{E}_{t_c}(\mathbf{u}_0(t_c))$ for the symmetric k -th variation at the critical point, so that $\mathcal{L}_c = E_2$, $\partial_{\mathbf{u}}^2 G = E_3$ and $\partial_{\mathbf{u}}^3 G = E_4$ there.

Lemma 11 (The feedback is never adverse). *Let t_c be the first crossing of Lemma 8 and let Assumption 8 hold, with $\delta > 0$ the spectral gap of $\mathcal{L}_c$ above its one-dimensional kernel $\text{span}\{\phi\}$. Define $R \in (V^\perp)'$ by*

$$\langle R, \mathbf{v} \rangle := \frac{1}{2} E_3[\phi, \phi, \mathbf{v}], \quad \mathbf{v} \in V^\perp. \quad (45)$$

Then:

- (i) $\mathcal{L}_c \geq 0$ on V and $\mathcal{L}_c \geq \delta > 0$ on $V^\perp$, so $(Q\mathcal{L}_cQ)^{-1}$ is bounded on $V^\perp$ and the second-order Lyapunov–Schmidt field of Proposition 1 is $\psi_2 = -(Q\mathcal{L}_cQ)^{-1}R$, i.e. $\psi = w^2\psi_2 + \mathcal{O}(w^3 + \epsilon)$ at $t = t_c$.
- (ii) The reduced cubic of Assumption 9 decomposes as $b = b_{\text{dir}} + b_\psi$ with $b_{\text{dir}} = \frac{1}{6}E_4[\phi, \phi, \phi, \phi]$ (the frozen quartic of Lemma 16) and

$$b_\psi = E_3[\phi, \phi, \psi_2] = -2 \langle R, (Q\mathcal{L}_cQ)^{-1}R \rangle \leq 0, \quad (46)$$

with equality if and only if $R = 0$. In particular $b \leq b_{\text{dir}}$ always, and the normalisation-invariant ratio $r := b_\psi/b_{\text{dir}}$ is nonpositive for every geometry, every $(\nu, \hat{K})$.

- (iii) Variationally,

$$b_\psi = -2 \sup_{\mathbf{v} \in V^\perp, \mathbf{v} \neq 0} \frac{\langle R, \mathbf{v} \rangle^2}{\langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle} = - \sup_{\mathbf{v} \in V^\perp, \mathbf{v} \neq 0} \frac{E_3[\phi, \phi, \mathbf{v}]^2}{2 \langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle}. \quad (47)$$

Proof. (i) $\mu_1(t_c) = 0$ is by definition the infimum of the Rayleigh quotient of $\mathcal{L}_c$, so $\mathcal{L}_c \geq 0$; the kernel is $\text{span}\{\phi\}$ and the remaining spectrum is bounded below by δ (Assumption 8), giving $\mathcal{L}_c \geq \delta$ on $V^\perp$. The range equation of Proposition 1 at order w^2 (at $\epsilon = 0$, $t = t_c$) reads $Q\mathcal{L}_c\psi_2 + \frac{1}{2}Q E_3[\phi, \phi, \cdot] = 0$, which is $\psi_2 = -(Q\mathcal{L}_cQ)^{-1}R$.

(ii) Substituting $\psi = w^2\psi_2 + \dots$ into the kernel equation and collecting the order w^3 (equivalently: Taylor-expanding the reduced energy $w \mapsto \mathcal{E}_{t_c}(\mathbf{u}_0 + w\phi + \psi(w))$ to fourth order and differentiating) gives $b = \frac{1}{6}E_4[\phi^{\otimes 4}] + E_3[\phi, \phi, \psi_2]$; the cross term $\langle \mathcal{L}_c\psi_2, \phi \rangle$ vanishes because $\mathcal{L}_c$ is symmetric and $\mathcal{L}_c\phi = 0$. By (45) and (i), $E_3[\phi, \phi, \psi_2] = 2\langle R, \psi_2 \rangle = -2\langle R, (Q\mathcal{L}_cQ)^{-1}R \rangle \leq 0$, since $(Q\mathcal{L}_cQ)^{-1}$ is a positive operator on $V^\perp$; equality forces $R = 0$. Both b_{dir} and b_ψ scale as the fourth power of the mode amplitude, so r is normalisation-invariant.

(iii) For a self-adjoint $A \geq \delta > 0$ one has $\langle R, A^{-1}R \rangle = \sup_{\mathbf{v} \neq 0} \langle R, \mathbf{v} \rangle^2 / \langle \mathbf{v}, A\mathbf{v} \rangle$ (Cauchy–Schwarz in the A -inner product, with equality at $\mathbf{v} = A^{-1}R$). $\square$ $\square$

Corollary 1 (Variational certification of subcriticality). *For every $\mathbf{v} \in V^\perp$ with $\langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle > 0$,*

$$b \leq b_{\text{dir}} - \frac{E_3[\phi, \phi, \mathbf{v}]^2}{2 \langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle}. \quad (48)$$

Hence the subcritical sign $b < 0$ is certified by a single explicit test field $\mathbf{v}$ satisfying

$$E_3[\phi, \phi, \mathbf{v}]^2 > 2 b_{\text{dir}} \langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle. \quad (49)$$

The criterion is invariant under independent rescaling of ϕ and of $\mathbf{v}$, and its evaluation requires one trilinear integral and one Rayleigh quotient — no solve with the reduced operator, which becomes near-singular as the flat degeneracy of Remark 22 is approached.

Remark 12 (What remains open in Assumption 9). Lemma 11 settles the direction of the competition identified in Remark 11: the frozen term is provably adverse ($b_{\text{dir}} > 0$, Lemma 16) and the feedback provably favourable ($b_\psi \leq 0$), for every geometry. The open content of the cubic clause of Assumption 9 is therefore only the *magnitude* inequality $|b_\psi| > b_{\text{dir}}$, i.e. $r < -1$: a numerical table of r tests this single scalar and nothing else, while Corollary 1 upgrades any sufficiently good explicit $\mathbf{v}$ to a certificate. Two caveats. First, the argument is tied to the first crossing: along the unstable continuation past t_c the operator is indefinite and (46) does not apply there (nor is it needed: b is a datum of the critical point). Second, the same argument does *not* sign the quintic feedback d_ψ — the fourth-order Lyapunov–Schmidt correction mixes terms of both signs — so the saturation clause $d > 0$ of Assumption 9 is untouched.

4.5 Subcritical branch structure and imperfection sensitivity

Perfect system. Set $\epsilon = 0$ and $t < t_c$. Dividing (44) by w gives $a(t_c - t) + bw^2 + dw^4 = 0$, i.e.

$$w^2 = \frac{-b \pm \sqrt{b^2 - 4ad(t_c - t)}}{2d}, \quad (50)$$

which (with $a > 0$, $b < 0$, $d > 0$) admits real nontrivial branches only for $t \geq t_L$, where

$$t_L := t_c - \frac{b^2}{4ad} < t_c, \quad w_L^2 = \frac{-b}{2d} = \mathcal{O}(1). \quad (51)$$

At t_L the two branches meet in a limit point (fold) at the *finite* amplitude w_L . The fundamental branch $w = 0$ is stable for $t < t_c$ and unstable for $t > t_c$; the upper nontrivial branch is stable. Hence on $[t_L, t_c)$ the compressed state and a finite-amplitude creased state coexist: the bifurcation is subcritical, with hysteresis and a *snap* at t_c .

Imperfect system.

Proposition 2 (Imperfection sensitivity, Koiter $\frac{2}{3}$ law). *For $\epsilon > 0$ the compressed branch terminates at a limit point that issues from the bifurcation point $(0, t_c)$ and is depressed below t_c according to*

$$t_s(\epsilon) = t_c - C |\eta \epsilon|^{2/3} + o(\epsilon^{2/3}), \quad C = \frac{3|b|^{1/3}}{2^{2/3}a} > 0, \quad (52)$$

so the load at which the compressed branch ceases to exist is reduced by $\mathcal{O}(\epsilon^{2/3})$. The turning amplitude there is itself small, $w_ = \mathcal{O}(\epsilon^{1/3})$; the $\mathcal{O}(1)$ amplitude is that of the creased branch onto which the structure then snaps.*

Proof. Write (44) as $g(w, t) = 0$ with

$$g(w, t) := a(t_c - t)w + bw^3 + dw^5 - \epsilon\eta,$$

the analytic remainder of (44) being subdominant at the scaling found below (we verify this at the end). A limit point in the load is a point of the solution set at which the branch turns, i.e. at which t is stationary along the curve; differentiating $g = 0$ implicitly, this is the pair

$$g(w, t) = 0, \quad g_w(w, t) = a(t_c - t) + 3bw^2 + 5dw^4 = 0. \quad (53)$$

We follow the limit point that issues from the bifurcation point $(0, t_c)$ as $\epsilon \rightarrow 0$; this is the one that caps the slightly deflected compressed branch and hence sets the snap load.

Turning amplitude. Using the second equation of (53) to eliminate the load, $a(t_c - t) = -(3bw^2 + 5dw^4)$, and substituting into the first gives a single equation for the turning amplitude,

$$-2bw^3 - 4dw^5 = \epsilon\eta. \quad (54)$$

Because $b < 0$, the small root of (54) as $\epsilon \rightarrow 0$ obeys $-2bw^3 = \epsilon\eta(1 + o(1))$, the quintic entering only at relative order w^2 ; hence

$$w_*(\epsilon) = \left(\frac{|\eta|\epsilon}{2|b|}\right)^{1/3} (1 + o(1)) = \mathcal{O}(\epsilon^{1/3}), \quad (55)$$

in agreement with the unstable-branch scaling (56) evaluated at the turn. In particular $w_* \rightarrow 0$, so at this limit point the cubic dominates the quintic, $dw_*^4/(bw_*^2) = \mathcal{O}(\epsilon^{2/3})$.

Snap load. Inserting (55) into $g_w = 0$ and discarding the quintic, a relative $\mathcal{O}(\epsilon^{2/3})$ correction,

$$a(t_c - t_s(\epsilon)) = -3bw_*^2(1 + o(1)) = 3|b|\left(\frac{|\eta|\epsilon}{2|b|}\right)^{2/3}(1 + o(1)),$$

which is (52) with $C = 3|b|^{1/3}/(2^{2/3}a) > 0$, the classical Koiter $\frac{2}{3}$ power law [?, ?].

Remainder. At the limit point $w = \mathcal{O}(\epsilon^{1/3})$ and $t_c - t = \mathcal{O}(\epsilon^{2/3})$, so the omitted terms $\mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t|w^2 + w^6)$ of (44) are $\mathcal{O}(\epsilon^{4/3})$, a relative $\mathcal{O}(\epsilon^{1/3})$ against the dominant balance $a(t_c - t)w \sim bw^3 \sim \epsilon\eta = \mathcal{O}(\epsilon)$. They perturb w_* and $t_s(\epsilon)$ only within the stated $o(\epsilon^{2/3})$, leaving the exponent and C unchanged. $\square$ $\square$

The unstable continuation branch. Separately, near t_c the small-amplitude (backward-bending, unstable) branch of (44) satisfies, at $t = t_c$, $bw^3 = \epsilon\eta(1 + o(1))$ with the quintic subdominant, so

$$|w(\epsilon)| = \left(\frac{|\eta|}{|b|}\right)^{1/3} \epsilon^{1/3} (1 + o(1)), \quad \epsilon \rightarrow 0. \quad (56)$$

This is the branch a displacement- or arc-length-controlled solver tracks through the instability, and it is the one that feeds the tangent estimate below; it is *not* the branch a load-controlled physical system follows.

Remark 13 (Robustness to the bifurcation type: pitchfork vs. transcritical). Assumption 9 takes the reduced potential to be odd ($c_2 = 0$), a subcritical *pitchfork*; by the discussion there this holds when the critical mode is antisymmetric under the lateral reflection R of §5.6. If that symmetry is absent or the ground mode lies in the symmetric sector, the quadratic coefficient $c_2 \neq 0$ survives and (44) is instead *transcritical*,

$$a(t_c - t)w + c_2w^2 + \mathcal{O}(w^3) = \epsilon\eta. \quad (57)$$

The qualitative architecture of the section is unaffected. Repeating the limit-point analysis of Proposition 2 on (57) (solve $g = g_w = 0$ with $g = a(t_c - t)w + c_2w^2 - \epsilon\eta$) gives a turning amplitude $w_*^2 = -\epsilon\eta/c_2 = \mathcal{O}(\epsilon)$ and a snap load depressed by the classical Koiter *one-half* law,

$$t_s(\epsilon) = t_c - C'|\eta|\epsilon^{1/2} + o(\epsilon^{1/2}), \quad (58)$$

in place of the $\frac{2}{3}$ law and $w_* = \mathcal{O}(\epsilon^{1/3})$ of the pitchfork. The determinant relief (60)–(62) still lifts $\lambda_{\min}$ off zero, now at order $w_*^2 = \mathcal{O}(\epsilon)$ rather than $\mathcal{O}(\epsilon^{2/3})$, so the regularised tangent bound of §5.10 reads $\|\mathbb{L}\| = \mathcal{O}(\epsilon^{-2} \log \frac{1}{\epsilon})$ in place of $\mathcal{O}(\epsilon^{-4/3})$, and the localisation of Proposition 4 persists with the correspondingly shifted exponents. In particular the central result — a bounded physical branch carrying a vanishingly localised, regularised tangent blow-up for every $\epsilon > 0$ — is insensitive to whether the corner bifurcation is a subcritical pitchfork or a transcritical fold; only the imperfection-sensitivity exponents change.

4.6 Kinematic relief: lower bound on J

The heart of the regularisation is geometric: the buckling mode trades pure compression for rotation and shear, which lifts the determinant off zero. The reason the relief is second order in w —and hence why the scaling exponents below are what they are—is that the mode is, to leading order, a *rotation*, and rotations do not change volume at first order. We do *not* assume the first-order volume change vanishes exactly; we use the quantitative bound on it posited by Assumption 10 (the asymptotic-rotation hypothesis, verified numerically in Remark 14) and show it is dominated by the second-order relief.

Lemma 12 (Determinant relief). *Let $\mathbf{F} = \mathbf{F}_0 + w \nabla \phi + \mathcal{O}(w^2)$ at the corner, with $\mathbf{F}_0 = \mathbf{F}_0(t)$ on the loaded path, $J_0 = \det \mathbf{F}_0$, and write the first-order volume change as*

$$D_1 := \text{cof } \mathbf{F}_0 : \nabla \phi = J_0 \mathbf{F}_0^{-T} : \nabla \phi. \quad (59)$$

By the asymptotic-rotation estimate (66) of Assumption 10, the critical mode is asymptotically volume-preserving, $|D_1| \leq c_\phi J_0 / \sqrt{|\log J_0|}$, so D_1 does not vanish at finite load but is itself small with J_0 . Let $d_\phi := \frac{1}{2} \frac{d^2}{dw^2} \big|_0 \det(\mathbf{F}_0 + w \nabla \phi)$ be the second-order coefficient of the determinant — the second mixed discriminant of $\mathbf{F}_0$ ($d - 2$ copies) with $\nabla \phi$ (two copies), which for $d = 2$ reduces to $\det(\nabla \phi)$. We take the mode to satisfy the relief-orienting condition $d_\phi \geq c_d > 0$ a.e. on a neighbourhood of the collapse locus Σ_c , with a uniform floor c_d (Young’s inequality in case (ii) below divides by d_ϕ , so positivity without a floor would leave the constants in (61) non-uniform on the tube) — the only region where the relief floor is invoked ((81) in §5.8; a property of the mode, not supplied by the transversality Lemma 14). Away from Σ_c , where J_0 stays bounded below, the sign of d_ϕ is immaterial: the finite- J_0 case (i) below absorbs a negative $w^2 d_\phi$ into the J_0 margin. Then

$$J = \det \mathbf{F} = J_0 + w D_1 + w^2 d_\phi + \mathcal{O}(w^3), \quad (60)$$

and the linear term is dominated by the other two: for $|w| \leq w_0$ small,

$$J \geq \frac{1}{2} J_0 + \frac{1}{2} w^2 d_\phi. \quad (61)$$

Consequently, with $\lambda_c = \lambda_{\min}(\mathbf{F}_0)$ and the remaining $d - 1$ stretches of $\mathbf{F}$ bounded, $\prod_{i \geq 2} \lambda_i(\mathbf{F}) = \mathcal{O}(1)$,

$$\lambda_{\min}(\mathbf{F}) = \frac{J}{\prod_{i \geq 2} \lambda_i(\mathbf{F})} \gtrsim C_1 \lambda_c + C_2 w^2, \quad C_1, C_2 > 0. \quad (62)$$

Proof. Expanding the determinant in w , $\det(\mathbf{F}_0 + w \nabla \phi) = \sum_{k=0}^d w^k D_k(\mathbf{F}_0, \nabla \phi)$ with D_k the k -th mixed discriminant; $D_0 = \det \mathbf{F}_0 = J_0$, $D_1 = \text{cof } \mathbf{F}_0 : \nabla \phi$, and $D_2 = d_\phi$, giving (60) (the $\mathcal{O}(w^3)$ collecting $D_3, \dots, D_d$). For (61) we bound the linear term $w D_1$ in two complementary regimes; in both, $\mathcal{O}(w^3)$ is absorbed into the constants for $|w| \leq w_0$.

(i) *Finite J_0 (at and near the bifurcation, and everywhere off Σ_c).* Here $J_0 \geq c > 0$ and D_1 is finite, so the bound needs no smallness of D_1 , only of w : choosing $w_0 \leq J_0 / (4|D_1|)$ gives $|w D_1| \leq \frac{1}{4} J_0$ for $|w| \leq w_0$, whence $J = J_0 + w D_1 + w^2 d_\phi + \mathcal{O}(w^3) \geq \frac{3}{4} J_0 + \frac{1}{2} w^2 d_\phi$. No sign of d_ϕ is used here: if $d_\phi \geq 0$ this is already $\geq \frac{1}{2} J_0 + \frac{1}{2} w^2 d_\phi$, the relief bound (61); if $d_\phi < 0$ (the collapse-driving orientation, which the computation finds confined to a thin surface layer, §5.8) shrink w_0 further to $w_0 \leq \sqrt{J_0 / (4|d_\phi|)}$ so that $\frac{1}{2} w^2 |d_\phi| \leq \frac{1}{4} J_0$, whence $J \geq \frac{3}{4} J_0 - \frac{1}{4} J_0 = \frac{1}{2} J_0$ and (62) holds with the load floor $C_1 \lambda_c$ alone (no relief, but no collapse either). This is the regularity-only argument: at t_c , where $J_0(t_c) > 0$, and at every point off the collapse locus, (61) (resp. its $d_\phi < 0$ surrogate $J \geq \frac{1}{2} J_0$) holds with no rotation condition and with no sign hypothesis on d_ϕ (cf. Lemma 13).

(ii) *Near collapse ($J_0 \rightarrow 0$, on the tube around Σ_c).* Now w_0 from (i) would shrink with J_0 , so we use instead the asymptotic-rotation bound $|D_1| \leq c_\phi J_0 / \sqrt{|\log J_0|}$ of Assumption 10, together with the localised orientation $d_\phi \geq c_d > 0$ of (81) — the one region where that sign is genuinely invoked, as the denominator in Young’s inequality $|w D_1| \leq \frac{1}{2} d_\phi w^2 + \frac{1}{2d_\phi} D_1^2$. With $D_1^2 \leq c_\phi^2 J_0^2 / |\log J_0| \leq d_\phi J_0$ once J_0 is small enough that $c_\phi^2 J_0 / |\log J_0| \leq c_d$ — a threshold uniform on the tube thanks to the floor $d_\phi \geq c_d$ — $\frac{1}{2d_\phi} D_1^2 \leq \frac{1}{2} J_0$ and $|w D_1| \leq \frac{1}{2} d_\phi w^2 + \frac{1}{2} J_0$, and inserting into (60) yields (61). Thus the residual first-order term, far from dominating, is swallowed by the barrier-controlled J_0 and the relief $w^2 d_\phi$ — the precise content of “buckling relieves compression,”

with no appeal to an exact rotation condition. Since $J = \prod_i \lambda_i$ and the $d - 1$ largest stretches are $\mathcal{O}(1)$ on the relieved branch (the buckling redirects, but does not unboundedly stretch, the material near the corner), (60) gives (62). $\square$ $\square$

The estimate (62) carries two competing floors, $C_1 \lambda_c$ and $C_2 w^2$, and which one binds depends on the load. At the bifurcation point itself the perfect stretch $\lambda_c(t_c) > 0$ is finite (buckling precedes collapse, Remark 9); there $C_1 \lambda_c$ dominates, $\lambda_{\min}$ is bounded below by a constant, and *no* blow-up occurs — consistently with the bounded-tangent physical branch of Proposition 3. The tangent divergence is felt only along the *unstable continuation* that a solver tracks to deeper loads, where the perfect path drives $\lambda_c(t) \rightarrow 0$; precisely in the regime $\lambda_c \lesssim w^2$ the relief term takes over. Combining (62) with the branch amplitude $w \sim \epsilon^{1/3}$ of (56) then gives, in that near-collapse regime,

$$\lambda_{\min}(\epsilon) \gtrsim C_2 w(\epsilon)^2 \sim \mathcal{O}(\epsilon^{2/3}), \quad \|\mathbf{F}^{-1}\| = \lambda_{\min}^{-1} \sim \mathcal{O}(\epsilon^{-2/3}). \quad (63)$$

Outside that regime ($\lambda_c \gtrsim \epsilon^{2/3}$) the floor $C_1 \lambda_c$ keeps $\|\mathbf{F}^{-1}\|$ even smaller, so (63) is the worst case. On the stable physical branch the picture is milder still; we contrast the two after Theorem 2.

We now examine the smallness of the first-order volume change D_1 in (59). The exact contraction of the tangent (10) gives the second variation of Ψ on the fundamental state $\mathbf{F}_0$ in a direction ϕ (plane strain),

$$\langle \phi, \mathcal{L}\phi \rangle = \mu_e |\nabla \phi|^2 + f''(J_0) (\delta J)^2 + 2 f'(J_0) \det(\nabla \phi), \quad \delta J = \text{cof } \mathbf{F}_0 : \nabla \phi. \quad (64)$$

For $f_{\log}$ both volumetric prefactors diverge as $J_0 \rightarrow 0$,

$$f''(J_0) = \frac{\mu_e + 2\lambda_e - 2\lambda_e \log J_0}{J_0^2} \sim \frac{2\lambda_e |\log J_0|}{J_0^2}, \quad f'(J_0) \sim \frac{2\lambda_e \log J_0}{J_0} \rightarrow -\infty. \quad (65)$$

Crucially, the determinant term $2f'(J_0) \det(\nabla \phi)$, with prefactor $\sim |\log J_0|/J_0$, is *more* singular than the term $f''(J_0)(\delta J)^2 \sim |\log J_0|$ (recall $\delta J \sim J_0 \cdot \mathcal{O}(1)$). Hence δJ cannot be bounded by isolating the f'' term and discarding the rest; the two regimes below treat its smallness honestly.

Lemma 13 (Bifurcation regularity). *At the bifurcation load t_c the fundamental determinant is bounded below, $J_0(t_c) > 0$ (buckling precedes collapse, Remark 9). Then $f'(J_0), f''(J_0)$ are finite, (64) is a regular quadratic form, and the neutral mode satisfies $\langle \phi, \mathcal{L}\phi \rangle = 0$ with δJ and $\det \nabla \phi$ finite. No rotation condition is needed at t_c : the relief bound (61) holds there by the finite- J_0 case (i) of Lemma 12, which uses only $J_0(t_c) > 0$ and smallness of w (not of D_1).*

Along the *unstable continuation* the fundamental state is driven toward collapse, $J_0 \rightarrow 0$, and there the relief estimate (61) requires the residual first-order term to stay subordinate. By (64) this is no longer automatic, so we isolate it as a hypothesis on the low-energy modes, verified numerically below in place of the (false) claim that it follows from the f'' term alone.

Assumption 10 (Asymptotic rotation, (AR)). Along the near-collapse continuation the buckling mode ϕ keeps $\langle \phi, \mathcal{L}\phi \rangle$ bounded through cancellation in (64) (the shear term balancing the singular determinant term), and its first-order volume change obeys

$$\delta J = \text{cof } \mathbf{F}_0 : \nabla \phi = \mathcal{O}\left(\frac{J_0}{\sqrt{|\log J_0|}}\right) \rightarrow 0 \quad (J_0 \rightarrow 0^+), \quad (66)$$

so the mode is asymptotically volume-preserving at the rate that makes the relief term $w^2 d_\phi$ dominate the residual first-order term in (60).

Remark 14 (Numerical verification of (AR)). The *qualitative* content of (AR) is supported by the consistent weak solver of §6.4 (`graded_buckling.py`), evaluated on the *same* critical mode that fixes the subcritical sign. The neutral co-energy vanishes, $\langle \phi, \mathcal{L}\phi \rangle \approx 10^{-8}$, confirming Lemma 13; the dimensionless rotation defect $\rho_{\text{rot}} := \|\delta J\|_2 / \|\nabla \phi\|_2 \approx 0.08\text{--}0.16$ across the $(\nu, \hat{K})$ rectangle, and remains small and stable ($\approx 0.07\text{--}0.09$) as the base compression is deepened, so the mode is indeed nearly volume-preserving. The decomposition also makes the mechanism explicit: the determinant term is $6\text{--}10\times$ the $f''(\delta J)^2$ term—so the latter cannot be isolated, exactly as (64) warns—and the bounded co-energy is achieved by cancellation against the shear term, the structural content of (AR). We stress what these data do and do not establish: they confirm that the critical mode is nearly volume-preserving with bounded co-energy by cancellation, but they do *not* verify the precise J_0 -dependent rate $\delta J = \mathcal{O}(J_0 / \sqrt{|\log J_0|})$ of (66) — the accessible base states reach only $J_0 \approx 0.56$, not the asymptotic limit $J_0 \rightarrow 0^+$. That rate, which is what Lemma 12 and Theorem 2 actually invoke, remains a genuine assumption, consistent with but not proved by the computation.

4.7 Regularised tangent bound and spatial localisation

Assumption 11 (Non-degeneracy of the stable branch, (ND)). The stable creased equilibrium reached after the snap has bounded distortion uniformly in ϵ : there is $c' > 0$ with $\lambda_{\min}(\mathbf{F}) \geq c'$ (equivalently $J = \det \mathbf{F} \geq J_{\text{floor}} > 0$ pointwise) and the remaining $d - 1$ stretches are $\mathcal{O}(1)$, throughout B_ρ .

We state (ND) as a hypothesis, not a consequence of the energy bound: an integral bound $\int f(J) < \infty$ gives only measure-type control of $\{J \text{ small}\}$, not a pointwise floor. It is the finite-amplitude analogue of the small- w relief of Lemma 12, which it does *not* invoke — the lemma is local ($|w| \leq w_0$) while the snap is to $\mathcal{O}(1)$ amplitude. Physically it is the standard expectation for creasing: the post-snap state is a finite-amplitude fold of bounded distortion, not a collapse, and the barrier $f_{\log}(J) \rightarrow \infty$ makes collapse energetically prohibitive. We emphasise that, unlike (AR), it is *not* checked by the spectral computation of §6.4: that computation reaches the critical mode and the subcritical sign but does not track the nonlinear branch past the snap. A direct verification would require a full nonlinear path-following continuation onto the $\mathcal{O}(1)$ creased branch, which we do not carry out here; (ND) is therefore posited, on physical grounds, and left as the one regularity hypothesis of the physical-branch statement.

Proposition 3 (Stable physical branch: bounded tangent). *Under load control the structure follows the compressed fundamental branch up to the snap at $t_s(\epsilon) \leq t_c$ (Proposition 2) and then jumps to the creased branch of amplitude $w = \mathcal{O}(1)$. On the compressed branch the bound needs only the pre-collapse floor of (FP)(i) together with the ordering $t_c < t_{\text{coll}}$ of Lemma 8: $\lambda_{\min} \geq \lambda_c(t) \geq \lambda_c(t_c) > 0$ (buckling precedes collapse), so $\|\mathbb{L}\| = \mathcal{O}(1)$ there. Under the non-degeneracy hypothesis 11 the post-snap branch is likewise controlled, and on the whole physical branch*

$$\max_{\mathbf{x} \in B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\| = \mathcal{O}(1) \quad \text{uniformly in } \epsilon, \quad (67)$$

so the sole effect of the imperfection is the $\mathcal{O}(\epsilon^{2/3})$ reduction (52) of the snap load: there is no tangent blow-up on the physical branch.

Theorem 2 (Unstable continuation branch: regularised blow-up). *Under Assumptions 1, 8, 9, 10, the orientation floor (81) near the collapse locus, and the conclusions (63) of Lemma 12, with the barrier penalty $f_{\log}$, the tangent along the unstable continuation branch (56)—the branch tracked by a displacement- or arc-length-controlled solver—satisfies*

$$\max_{\mathbf{x} \in B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\| \leq C \Theta(J) \|\mathbf{F}^{-1}\|^2 \sim \mathcal{O}(\epsilon^{-4/3} \log \frac{1}{\epsilon}), \quad \epsilon \rightarrow 0. \quad (68)$$

Proof. By (11), $\|\mathbb{L}\| \leq \mu_e + \Theta(J)\|\mathbf{F}^{-1}\|^2$. For $f_{\log}$, $\Theta(J) = \mathcal{O}(\log \frac{1}{J})$ as $J \rightarrow 0$ by (39), and on the unstable branch $J \sim \lambda_{\min} \sim \epsilon^{2/3}$ by (63), so $\Theta(J) = \mathcal{O}(\log \frac{1}{\epsilon})$; multiplying by $\|\mathbf{F}^{-1}\|^2 \sim \epsilon^{-4/3}$ gives (68). $\square$

The two results together are the precise sense in which “the singularity never occurs in reality”: a load-controlled physical system snaps and keeps a bounded tangent (Proposition 3), and the divergent $\epsilon^{-4/3}$ rate is felt only by an algorithm that suppresses the snap and tracks the unstable branch through the instability (Theorem 2). The spatial localisation below applies to that branch.

The maximum is, however, achieved only in a shrinking zone, and we can now identify the localisation exponent exactly. The relevant geometry is the *collapse locus* of the perfect field.

Lemma 14 (Transversal collapse: $s = 1$). *On the perfect (unphysical) continuation of the fundamental path beyond t_c , let $\Sigma_c \subset B_\rho$ be the collapse locus on which the smallest stretch vanishes, $\lambda_{\min} = 0$ (equivalently $J = 0$). Assume the transversality hypothesis*

(T) Σ_c is of codimension one, and at each of its points $\lambda_{\min}$ is a simple singular value of $\mathbf{F}$ (hence smooth in $\mathbf{x}$) with $\nabla \lambda_{\min} \neq \mathbf{0}$.

Then $\lambda_{\min}$ vanishes transversally,

$$\lambda_{\min}^{\text{field}}(\mathbf{x}) = \kappa_* \text{dist}(\mathbf{x}, \Sigma_c) + \mathcal{O}(\text{dist}^2), \quad \kappa_* > 0, \quad (69)$$

and the localisation exponent is $s = 1$. (A tangential contact, $\nabla \lambda_{\min} = \mathbf{0}$ on Σ_c , would give $s \geq 2$ and a wider tube; Remark 16 states in what sense (T) is generic and what survives without it.)

Remark 15 (Extended arc versus point vertex). The codimension-one hypothesis is generic but not forced. In a compressive mixed corner the most severe contraction is frequently attained *exactly* at the vertex $r = 0$, in which case the collapse locus degenerates to the single point $\Sigma_c = \{0\}$ — codimension *two* in $d = 2$. The localisation conclusion then only improves: the relieved floor $\lambda_{\min} \gtrsim \max\{\kappa_* r, c\epsilon^{2/3}\}$ now confines the high-stiffness set to a *disc* of radius $\ell(\epsilon) \sim \epsilon^{2/3}$ rather than a tube of fixed length, so its area is $\mathcal{O}(\ell(\epsilon)^2) = \mathcal{O}(\epsilon^{4/3})$ rather than $\mathcal{O}(\epsilon^{2/3})$, and the L^q -bound of Proposition 4 acquires the stronger tube contribution $\epsilon^{-4q/3} \ell(\epsilon)^2 = \epsilon^{\frac{4}{3}(1-q)}$, finite for every $q < 1$ instead of $q < \frac{1}{2}$. The extended-arc estimate below is therefore the conservative case; a point vertex is strictly more integrable.

Proof. Where the smallest singular value is simple it is an analytic function of the entries of $\mathbf{F}$, hence smooth in $\mathbf{x}$. A smooth scalar field vanishing on Σ_c has, by Taylor expansion, $\lambda_{\min} = \nabla \lambda_{\min} \cdot \mathbf{n} \text{dist}(\mathbf{x}, \Sigma_c) + \mathcal{O}(\text{dist}^2)$ with $\mathbf{n}$ the unit normal; hypothesis (T) gives (69) with $\kappa_* = |\nabla \lambda_{\min}| > 0$. $\square$

Remark 16 (In what sense (T) is generic, and what survives without it). A word of hygiene on *generic*. For the *fixed* field of a fixed body, material and load path, (T) is simply a hypothesis: Sard’s theorem asserts that almost every *level* δ of $\lambda_{\min}$ is regular, but Σ_c is the specific level $\delta = 0$, about which Sard says nothing, and there is no parameter being perturbed. Genericity becomes meaningful only relative to a family, and the natural one is already at hand: the phase-diagram parameters $(\nu, \hat{K})$ of the model wedge together with the load t . Within a family rich enough to move the field — versality we do not verify here — tangential contact of a scalar zero set is a Thom-nongeneric event, and this is the honest content of calling (T) generic. Independently, (T) is *checkable*: $|\nabla \lambda_{\min}|$ along the computed continuation near its zero set is a by-product of the same numerics that locate Σ_c . Finally, the architecture does not stand or fall with (T). Without any transversality, the barrier bound of Lemma 19 confines the sublevel sets unconditionally, $|\{J_0 \leq$

$\delta\} \lesssim E(t)/\log^2 \delta$ — a *logarithmic* tube in place of the algebraic one — so the qualitative conclusion (the high-stiffness zone is confined and vanishes with ϵ) is hypothesis-free; what (T) buys is the sharp algebraic half-width (70) and with it the $q < \frac{1}{2}$ integrability range of Proposition 4. And in the empirically expected geometry of Remark 15 — the collapse locus degenerating to the vertex point — the localisation only improves.

On the imperfect branch the relief of (63) imposes the floor $\lambda_{\min}(\mathbf{x}, \epsilon) \gtrsim \max\{\kappa_* \text{dist}(\mathbf{x}, \Sigma_c), c\epsilon^{2/3}\}$, so the high-stiffness zone is the tube of half-width $\ell(\epsilon)$ around Σ_c in which the floor is active, with (using $s = 1$)

$$\ell(\epsilon) \sim \frac{c}{\kappa_*} \epsilon^{2/3} \xrightarrow{\epsilon \rightarrow 0} 0. \quad (70)$$

Proposition 4 (Narrowness of the singular zone). *With $\ell(\epsilon)$ as in (70) and $s = 1$ (Lemma 14), the set on which the tangent is near its maximum is a tube about Σ_c of area $\mathcal{O}(|\Sigma_c| \ell(\epsilon)) = \mathcal{O}(\epsilon^{2/3}) \rightarrow 0$, and the integrated stiffness obeys*

$$\int_{B_\rho} \|\mathbb{L}(\mathbf{x}, \epsilon)\|^q dV \leq C \left(1 + \epsilon^{-4q/3} \ell(\epsilon) \log^q \frac{1}{\epsilon}\right) = C \left(1 + \epsilon^{\frac{2}{3}(1-2q)} \log^q \frac{1}{\epsilon}\right), \quad (71)$$

which stays bounded as $\epsilon \rightarrow 0$ for every $q < \frac{1}{2}$. Thus, although the pointwise tangent diverges like $\epsilon^{-4/3}$, its L^q -mass over the corner remains finite for all $q < \frac{1}{2}$: the blow-up is genuinely localised to a tube of vanishing area, and for every $\epsilon > 0$ the regularised problem is well posed.

Proof. Split B_ρ into the tube $T_\epsilon = \{\text{dist}(\cdot, \Sigma_c) < \ell(\epsilon)\}$ and its complement. On T_ϵ , $\|\mathbb{L}\| \sim \epsilon^{-4/3} \log \frac{1}{\epsilon}$ (Theorem 2) and $|T_\epsilon| \sim |\Sigma_c| \ell(\epsilon) \sim \epsilon^{2/3}$, giving the displayed contribution. Off T_ϵ , using the transversal coordinate $\varrho = \text{dist}(\cdot, \Sigma_c)$ and $\lambda_{\min} \sim \kappa_* \varrho$, one has $\|\mathbb{L}\| \sim \varrho^{-2} \log \frac{1}{\varrho}$, and $\int_{\ell(\epsilon)}^\rho \varrho^{-2q} \log^q \frac{1}{\varrho} d\varrho$ converges as $\ell(\epsilon) \rightarrow 0$ precisely when $2q < 1$. Both regimes are controlled iff $q < \frac{1}{2}$. $\square$ $\square$

4.8 Discussion

The structure of the result is the one anticipated physically, and it splits cleanly along the stability of the branch followed. The idealised noiseless problem ($\epsilon = 0$) is singular—the tangent *norm* is unbounded ($\|\mathbb{L}\| \rightarrow \infty$), even though the corner remains strongly elliptic pointwise (Lemma 6)—but it is a measure-zero idealisation. Because the corner instability is subcritical (the creasing analogy, Assumption 9), a load-controlled physical system never traverses the singular continuation: it stays on the compressed branch and *snaps* to a finite-amplitude crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$ (Proposition 2), keeping a bounded tangent throughout (Proposition 3). The much-discussed divergence is therefore not seen by the physical system at all; it is the worst case for an algorithm—displacement or arc-length control, numerical continuation—that suppresses the snap and tracks the *unstable* branch through the instability. On that branch the determinant is still lifted off zero (Lemma 12, with the asymptotically volume-preserving character of the mode posited by Assumption 10, its qualitative content confirmed numerically, Remark 14), the tangent diverges only algebraically, $\epsilon^{-4/3}$ up to a logarithm (Theorem 2), and the singular character is confined to a tube of width $\ell(\epsilon) \sim \epsilon^{2/3} \rightarrow 0$ about the collapse locus, remaining L^q -integrable for all $q < \frac{1}{2}$ (Lemma 14, Proposition 4). That algebraic rate is the mirror image—through the inverse channel—of the benign tensile growth of §3.4; the same logarithm, traceable to the barrier $f_{\log}$, appears in both. The headline is therefore analytical rather than computational: a compressed clamped-free corner harbours a genuine structural bifurcation — the corner-localised analogue of the classical *creasing* instability of compressed elastomers [?, ?], now seated at a geometric Dirichlet–Neumann junction rather than on a flat surface — whose apparent tangent catastrophe is, on every physical

branch, regularised by the kinematic relief (Assumption 10, numerically supported). That the phenomenon first surfaced as ill-conditioning in a numerical computation is incidental to its nature; the conditioning degradation is a corollary, not the subject. It does carry a concrete computational reading — a solver that suppresses the snap and tracks the unstable branch sees the conditioning degrade as $\epsilon^{-4/3}$ and should either admit the snap or refine inside the tube (70), the load-control reduction (52) telling it when the snap is due — but this is one consequence among others, not the result. Section 5 pursues the sharper, opposite direction: fixing a concrete material and geometry, it discharges the abstract hypotheses to theorems and shows, through a blow-up reduction, that the entire phenomenon localises to a *universal* model wedge governed only by the Poisson ratio and a single stress-intensity amplitude. The reduction to scalar imperfect-bifurcation form follows the classical theory [?, ?, ?].

5 A concrete realization: the clamped–free corner under compression

The preceding analysis is deliberately abstract: it isolates the structural properties of the energy (Assumptions 1–5) and of the loading (the hypotheses of §4) that the argument actually uses, without committing to a domain. We now fix a concrete material and a concrete family of geometries and ask how much of that hypothesis list can be *discharged* rather than assumed. The material is the compressible neo-Hookean energy (7) with the barrier penalty $f_{\log}$. The geometry is a body whose boundary carries a clamped (Dirichlet) face meeting a traction-free (Neumann) face at a right angle, under a compressive load.

The abstract development of §3–§4 is already dimension-independent: it analyses the transverse problem at the codimension-two Dirichlet–Neumann edge Σ in $\mathbb{R}^d$. Here we instantiate it on concrete bodies and ask how much of that hypothesis list can be *discharged* rather than assumed.

5.1 Geometry and the singular set

We consider two model bodies in $\Omega \subset \mathbb{R}^d$.

- **(H) Half-ball.** $\Omega = \{\mathbf{x} \in \mathbb{R}^d : |\mathbf{x}| < 1, x_d > 0\}$. The flat base $\Gamma_D = \{x_d = 0\} \cap \partial\Omega$ is clamped; the spherical cap $\Gamma_N = \{|\mathbf{x}| = 1\} \cap \partial\Omega$ is traction-free.
- **(C) Box.** $\Omega = (-1, 1)^{d-1} \times (0, 1)$. The base $\Gamma_D = \{x_d = 0\}$ is clamped; the remaining faces are traction-free.

In both cases the load is a monotone compression transmitted through Γ_N (a pressure or an inward normal displacement on the cap, respectively the lateral/top faces), scaled by the pseudo-time $t \in [0, 1]$ of §4. We carry two bodies not because they require two analyses — by the blow-up reduction of §5.2 they do not, see Remark 17 — but because they bracket the two situations one meets in practice: a globally smooth edge (H) and a polyhedral one with vertices (C).

The *singular set* $\Sigma \subset \partial\Omega$ is the locus where $\overline{\Gamma_D}$ meets $\overline{\Gamma_N}$:

- for (H), $\Sigma = \{|\mathbf{x}| = 1, x_d = 0\} \cong S^{d-2}$, a *smooth* codimension-two submanifold; the cap is tangent to the vertical at the equator, so it meets the horizontal base at a right angle;
- for (C), Σ is the union of the bottom edges $\{x_d = 0, x_i = \pm 1\}$, again right-angle Dirichlet–Neumann edges, *together with* the bottom vertices, where d faces meet.

Fix $\mathbf{x}_0 \in \Sigma$ away from any vertex and split $\mathbb{R}^d = T_{\mathbf{x}_0}\Sigma \oplus \Pi$, with Π the two-plane normal to Σ . In the slice Π the boundary reduces to a Dirichlet ray and a Neumann ray meeting at 90° : *exactly* the planar junction of §3.5. We write B_ρ for a tubular neighbourhood of Σ of radius ρ and use (r, θ) for polar coordinates in Π as before.

Remark 17 (The two bodies are one edge analysis). The blow-up reduction (Proposition 5) makes precise that (H) and (C) are *not* two cases but one: at any non-vertex point of Σ the transverse problem is the same planar right-angle clamped-free wedge, with the *same* exponent α , angular profile Φ and reduced data, the bodies differing only through the scalar stress-intensity $\hat{K}$ (a point in the phase diagram of Remark 18) and through edge curvature, which enters only at the subleading order $\mathcal{O}(\lambda^g)$. The *single* genuine distinction is topological. For (H) the singular set $\Sigma \cong S^{d-2}$ is a smooth closed manifold with no vertices, so the singularity is a pure edge everywhere on Σ and the analysis holds verbatim at every point. For (C) the edge interiors are identical, but the bottom vertices, where d faces meet, carry an additional, generally stronger, polyhedral-vertex singularity [?] outside the scope of the edge analysis; one either rounds the vertices or restricts B_ρ to the edge interior. We therefore state the dimension-independent results once, read on (H) as the clean smooth-edge representative, and note that they hold verbatim on the edge interiors of (C).

5.2 The blow-up principle: reduction to the model wedge

The analysis of §3–§4 works on a tubular neighbourhood B_ρ of Σ cut from a *specific* body. The informal reason the two bodies of §5.1 share the same corner physics is that the singularity is localised at Σ : the leading data at the edge cannot feel the far field except through the single amplitude K . We now make this precise as a blow-up (tangent-cone) reduction, which is what licenses replacing “numerics on (H) and (C)” by “numerics on one universal model.”

Fix $\mathbf{x}_0 \in \Sigma$ away from any vertex and let $W := \Pi_W \times T_{\mathbf{x}_0}\Sigma$ be the *tangent wedge*, with $\Pi_W = \{(r, \theta) : r > 0, 0 < \theta < \frac{\pi}{2}\}$ the right-angle Dirichlet–Neumann quarter-plane in the normal two-plane Π . For $\lambda \in (0, 1]$ introduce the rescaling $\xi \mapsto \mathbf{x}_0 + \lambda\xi$, under which $B_{\lambda R}(\mathbf{x}_0)$ maps to $B_R(\mathbf{0}) \cap W$ and $\Omega_\lambda := \lambda^{-1}(\Omega - \mathbf{x}_0) \rightarrow W$ as $\lambda \rightarrow 0$. Write the fundamental displacement on the loaded path as $\mathbf{u}_0 = K r^\alpha \Phi(\theta) + \mathbf{u}_0^\sharp$, where by the Kondratiev expansion the remainder collects the next exponents, $\mathbf{u}_0^\sharp = \mathcal{O}(r^{\alpha'})$ with $\alpha' > \alpha$ the second singular exponent of the pencil, and set the *spectral gap* $g := \alpha' - \alpha > 0$. Define the rescaled displacement, tangent and operator

$$\hat{\mathbf{u}}_\lambda(\xi) := \lambda^{-\alpha}(\mathbf{u}_0(\mathbf{x}_0 + \lambda\xi) - \mathbf{u}_0(\mathbf{x}_0)), \quad \hat{\mathbb{L}}_\lambda(\xi) := \mathbb{L}(\mathbf{F}_0(\mathbf{x}_0 + \lambda\xi)), \quad \hat{\mathcal{L}}_\lambda := \partial_{\mathbf{u}}G(\mathbf{u}_0, \cdot, 0)|_{\mathbf{x}_0 + \lambda\xi}. \quad (72)$$

Proposition 5 (Blow-up reduction to the model wedge). *Let $\mathbf{x}_0 \in \Sigma$ be a non-vertex point and let $\beta_\star$ be the Fredholm weight of §3. Then, as $\lambda \rightarrow 0$:*

- (i) Kinematics. $\hat{\mathbf{u}}_\lambda \rightarrow \hat{\mathbf{u}}_W := K r^\alpha \Phi(\theta)$ in $V_{\beta_\star}^{1,2}(B_R \cap W)$ for every $R > 0$, with rate $\|\hat{\mathbf{u}}_\lambda - \hat{\mathbf{u}}_W\|_{V_{\beta_\star}^{1,2}(B_R \cap W)} = \mathcal{O}(\lambda^g)$. The limit solves the homogeneous linear pencil on W and is fixed by the wedge geometry up to the scalar K .
- (ii) Operator convergence away from the tip; common intermediate pencil. *On every annulus $\{0 < r_0 \leq r \leq R\} \cap W$ bounded away from the vertex, the coefficients of $\hat{\mathcal{L}}_\lambda$ converge in C^0 , with rate $\mathcal{O}(\lambda^g)$, to those of the model-wedge operator $\mathcal{L}_W$ obtained by freezing along the self-similar field $\hat{\mathbf{u}}_W$. The leading singular exponent α and angular profile Φ are intermediate-range data: they are identical for $\hat{\mathcal{L}}_\lambda$, $\mathcal{L}_W$ and the linearised-elasticity operator alike on the matching region $r \gg r_{\hat{K}}$, where the frozen field is a small perturbation of the identity and all three operators reduce to constant-coefficient linear elasticity, with the crossover radius*

$r_{\hat{K}} \rightarrow 0$ as $\hat{K} \rightarrow 0$. They are not asserted to be vertex-pencil data of the nonlinear operators: at the tip the deviation of the coefficients from linear elasticity does not vanish for any $\hat{K} \neq 0$, and on the tensile side the vertex pencil stabilises instead to the screened (vector-Laplacian) limit of Theorem 1(A), whose exponents all exceed one — so the r^α field remains the dominant asymptotics visible from any fixed scale, which is exactly what the universality claim uses. Thus only the dimensionless amplitude K (equivalently the normalised stress-intensity $\hat{K}$ of Remark 18) survives from the global problem. We emphasise what is not asserted: the away-from-tip convergence does not control the operator in a weighted norm up to $r = 0$; at the compressive corner no screening is available (the inverse channel $J \rightarrow 0$ drives the tangent up, and the non-compressive hypothesis behind Lemma 2 fails at the tip), so the agreement of the full Lyapunov–Schmidt reduced data is a leading-order statement rather than a quantitative $\mathcal{O}(\lambda^g)$ spectral-perturbation theorem, a uniform-to-the-tip estimate requiring the inner vertex theory left open in Remark 4.

- (iii) Dimensional reduction $d \geq 3 \Rightarrow d = 2$. Since Σ is smooth and the coefficients are constant along $T_{\mathbf{x}_0}\Sigma$ in the limit, the Fourier transform in the edge variable $\mathbf{s} \in T_{\mathbf{x}_0}\Sigma$ decomposes $\mathcal{L}_W$ into a one-parameter family $\{\mathcal{L}_W(\boldsymbol{\zeta})\}_{\boldsymbol{\zeta} \in \widehat{T_{\mathbf{x}_0}\Sigma}}$ of operators on the planar quarter-plane Π_W , each the transverse two-dimensional pencil of (13); the leading edge exponent is carried by the zero wavenumber $\boldsymbol{\zeta} = \mathbf{0}$. The model problem is therefore genuinely two-dimensional.

Proof. Statement (i) is the leading-term extraction of the Kondratiev–Maz’ya–Rossmann expansion [?, ?]: on the Fredholm weight $\beta_\star$ the solution admits $\mathbf{u}_0 = Kr^\alpha \boldsymbol{\Phi} + \mathcal{O}(r^{\alpha'})$, and the rescaling (72) maps the leading term to the λ -independent field $Kr^\alpha \boldsymbol{\Phi}$ exactly while shrinking the remainder by $\lambda^{\alpha' - \alpha} = \lambda^g$ in the homogeneous weighted norm. Statement (ii) follows because $\mathbb{L}$ is a smooth (algebraic) function of $\mathbf{F}_0 = \mathbf{I} + \nabla \mathbf{u}_0$ and $\nabla \hat{\mathbf{u}}_\lambda \rightarrow \nabla \hat{\mathbf{u}}_W$ off the tip by (i); freezing along the self-similar limit defines $\mathcal{L}_W$, and the coefficient error inherits the $\mathcal{O}(\lambda^g)$ rate of (i) on every annulus $\{r_0 \leq r \leq R\}$ bounded away from $r = 0$. The leading exponent and angular profile are read on the matching region: for $r \gg r_{\hat{K}}$ the frozen coefficients of $\hat{\mathcal{L}}_\lambda$ and of $\mathcal{L}_W$ alike are $\mathcal{O}((r/r_{\hat{K}})^{-(1-\alpha)})$ -close to constant-coefficient linear elasticity at $\mathbf{x}_0$, whose Kondratiev data $(\alpha, \boldsymbol{\Phi})$ therefore govern both operators on that range, independently of λ ; on the tensile side Theorem 1(A) guarantees in addition that the vertex contributes no competing exponent below one, so $(\alpha, \boldsymbol{\Phi})$ is the leading visible asymptotics. We do *not* claim a weighted operator-norm estimate uniform up to $r = 0$: the tip contributes through the weight, not through the body, and its quantitative control is the open inner problem of Remark 4; accordingly the agreement of the full reduced bifurcation data is asserted only at leading order. Statement (iii) is the standard edge-to-corner reduction: translation invariance along $T_{\mathbf{x}_0}\Sigma$ in the limit diagonalises $\mathcal{L}_W$ under the partial Fourier transform, and each fibre is the planar pencil whose smallest root is α , attained at $\boldsymbol{\zeta} = \mathbf{0}$ (the cited edge calculus). $\square \quad \square$

Remark 18 (The one surviving global datum, and what it costs). Proposition 5 isolates exactly what does and does not localise. The exponent α , the angular profile $\boldsymbol{\Phi}$, and the wedge operator $\mathcal{L}_W$ are *universal*: functions of the opening (90°) and the Poisson ratio alone. The only trace of the finite body is the scalar K , conveniently made dimensionless as $\hat{K} := K \rho^{\alpha-1} / \mu_e$ (the normalised stress-intensity over the neighbourhood scale). Thus every quantity in the corner reduces to a computation on $\mathcal{L}_W$ as a function of $(\nu, \hat{K})$ — a two-parameter *phase diagram*, not a per-body spot check. Whether a given quantity depends on $\hat{K}$ at all, or only on (ν) , is decided by a homogeneity count against α , carried out for the two open conditions in §5.9 (condition (BC), which needs only ν) and §5.7 (the cubic b , which retains the $\hat{K}$ -dependence). This is the precise, and sharper, replacement for “the singularity is localised to the corner.”

5.3 Tier 1: the material assumptions are already discharged

For the neo-Hookean energy (7) with $f_{\log}$, the standing assumptions of §3 hold with explicit constants and require no information about the domain: Assumption 1 (regularity, objectivity, and the collapse barrier $f_{\log}(J) \rightarrow \infty$), Assumption 2 (the inverse-channel bound, with $\Theta = \Theta_{\log}$ of (23)), Assumption 3 and the sharper Lemma 6 (strong ellipticity with constant $\geq \mu_e$ throughout compression), and Assumption 5 (logarithmic growth, $p = 1$) are all established in Lemma 1, Remark 1 and §3.4. None of these proofs uses $d = 2$: the tangent formula (10), the rank-one identity (18), and the growth functions (23) are algebraic identities in $\text{GL}^+(d)$. Thus the entire material side of the hypothesis list is closed for the concrete model, in every dimension. What remains are the hypotheses on the *solution and loading*, which is where the geometry enters.

5.4 The edge exponent and the scenario it selects

By the edge-singularity calculus [?, ?], the leading singular exponents of the linearised elasticity operator along the smooth edge Σ coincide with those of the transverse two-dimensional pencil in Π , modulated smoothly along Σ . That transverse pencil is the planar right-angle Dirichlet–Neumann corner of (13); its leading exponent $\alpha \in (0, 1)$ is the smallest root of the corresponding clamped–free Williams secular equation and depends only on the Poisson ratio. Consequently the kinematic blow-up (14), $\mathbf{F} = \mathbf{I} + \mathcal{O}(r^{\alpha-1})$, holds with a *computable* exponent, and by Theorem 1 the nonlinear tangent inherits exactly this α in the benign regime on the intermediate annulus outside the nonlinear crossover scale (and outside the dilation core of Lemma 3), while at the vertex itself the tensile tangent stabilises to the screened Laplacian limit, which contributes no competing exponent below one.

Which scenario of §3 is realized is governed by the sign of the dominant eigenvalue of $\nabla \mathbf{u}$ at the edge, equivalently by the sign of the stress-intensity factor K in (13). This is a global quantity, set by the far field, and is not determined by the local pencil alone. The compressive load we impose selects $K < 0$: the material is pushed into the corner, the dominant eigenvalue of $\nabla \mathbf{u}$ is negative, and Scenario B (§3.6) is the relevant branch. The quantitative form of “compressive enough” is the hypothesis (SC) below; it simultaneously selects Scenario B and drives the structural instability of §5.5. The complementary tensile load ($K > 0$) selects Scenario A, where (NC) holds and Theorem 1 applies; for the present geometry under compression that case does not arise near the edge.

5.5 Existence of the critical load via a localized surface instability

The one genuinely non-elementary ingredient of §4 was hypothesis (ii) of Lemma 8: the existence of a structurally destabilising field, $Q_{t_*}(\mathbf{v}_*) < 0$, which we there declined to establish and flagged as the corner counterpart of the Biot/creasing instability. For the concrete clamped–free geometry we now *prove* it, by reduction to the surface instability of the compressed half-space.

The free face Γ_N under compression is, locally, a traction-free surface compressed parallel to itself — the exact configuration of Biot’s classical analysis [?]. A homogeneously compressed neo-Hookean half-space loses stability to a surface mode once the surface-parallel stretch falls below a threshold $\lambda_* \in (0, 1)$ (for the incompressible model $\lambda_* \approx 0.544$; for the compressible energy (7) the threshold is the corresponding root of the surface secular equation, still in $(0, 1)$ [?]). We package the loading requirement as a single hypothesis.

Assumption 12 (Surface compression reaches threshold, (SC)). There exist $t_* \in (0, 1]$ and an interior point $P \in \Gamma_N \cap B_\rho$, at positive distance from both Σ and $\overline{\Gamma_D}$, such that the fundamental

state compresses the free face below the half-space threshold there: the surface-parallel principal stretch of $\mathbf{F}_0(t_*)$ at P satisfies $\lambda_{\parallel}(\mathbf{F}_0(t_*); P) < \lambda_*$.

(SC) is an input on the *loading*, in the same spirit as the monotonicity condition (M) of Lemma 9: it says only that the prescribed compression is strong enough that the free surface dips below its own buckling threshold somewhere near the edge. Since t merely normalises the load amplitude, (SC) can always be met by loading hard enough, provided the material does not fail elsewhere first.

Lemma 15 (Localized surface instability $\Rightarrow$ structural instability). *Under Assumption 12, hypothesis (ii) of Lemma 8 holds: there is $\mathbf{v}_* \in V$, supported in B_ρ and vanishing on Γ_D , with*

$$Q_{t_*}(\mathbf{v}_*) = \int_{B_\rho} \nabla \mathbf{v}_* : \mathbb{L}(\mathbf{F}_0(t_*)) : \nabla \mathbf{v}_* dV < 0. \quad (73)$$

Consequently, with $\mu_1(0) > 0$ from Lemma 8(i) and granted the pre-collapse floor (FP)(i) with $t_* < t_{\text{coll}}$ (hypothesis (iii) of Lemma 8), there is a first critical load $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$, and the ordering $t_c < t_{\text{coll}}$ follows.

Proof. Work in coordinates centred at P : let $n \geq 0$ be the depth into the body along the inward normal to Γ_N and $\mathbf{s} \in \mathbb{R}^{d-1}$ the surface-tangent coordinates, so that near P the body is the half-space $\{n > 0\}$ and $\mathbf{F}_0(t_*)$ is C^1 with value $\mathbf{F}_P := \mathbf{F}_0(t_*; P)$ whose surface-parallel stretch is $\lambda_{\parallel} < \lambda_*$.

Step 1: the homogeneous unstable mode. Freeze the coefficients at $\mathbf{F}_P$ and consider the constant-tensor form $Q_0(\mathbf{v}) = \int_{\{n>0\}} \nabla \mathbf{v} : \mathbb{L}(\mathbf{F}_P) : \nabla \mathbf{v} dV$ on the half-space, with $\mathbf{v}$ traction-free at $n = 0$. Because $\lambda_{\parallel} < \lambda_*$, the half-space is past its surface-instability threshold [?, ?]: for each surface wavevector $\mathbf{k} \in \mathbb{R}^{d-1}$, $k := |\mathbf{k}|$, there is a surface mode

$$\mathbf{v}_{\mathbf{k}}(\mathbf{s}, n) = \text{Re} [\hat{\mathbf{v}}(kn) e^{i\mathbf{k} \cdot \mathbf{s}}], \quad |\hat{\mathbf{v}}(\zeta)| \leq C e^{-\beta \zeta} \quad (\beta > 0), \quad (74)$$

decaying exponentially into the body on the depth scale $1/k$. Since $\mathbb{L}(\mathbf{F}_P)$ is constant, the rescaling $(\mathbf{s}, n) \mapsto k^{-1}(\mathbf{s}, n)$ shows that over a surface window W of area $|W|$,

$$\int_{\{n>0\}} \mathbf{1}_W \nabla \mathbf{v}_{\mathbf{k}} : \mathbb{L}(\mathbf{F}_P) : \nabla \mathbf{v}_{\mathbf{k}} dV = q(\lambda_{\parallel}) k |W| (1 + o(1)), \quad q(\lambda_{\parallel}) < 0, \quad (75)$$

where q is the Biot secular form, strictly negative past threshold. Likewise $\int_{\{n>0\}} \mathbf{1}_W |\nabla \mathbf{v}_{\mathbf{k}}|^2 \sim k |W|$ and $\int_{\{n>0\}} \mathbf{1}_W |\mathbf{v}_{\mathbf{k}}|^2 \sim k^{-1} |W|$, the latter two by the same scaling.

Step 2: cut-off. Let χ be smooth, $\chi \equiv 1$ on $B(P, \delta/2)$, $\text{supp } \chi \subset B(P, \delta)$, $|\nabla \chi| \leq C/\delta$, with δ small enough that $B(P, \delta) \cap \overline{\Omega}$ avoids both Σ and $\overline{\Gamma_D}$ and lies in B_ρ (possible since P is at positive distance from both). Set $\mathbf{v}_* := \chi \mathbf{v}_{\mathbf{k}}$. Then $\mathbf{v}_* \in V$ and vanishes on Γ_D (its support avoids $\overline{\Gamma_D}$); it is unconstrained on the free face, as required.

Step 3: error control. Write $\nabla \mathbf{v}_* = \chi \nabla \mathbf{v}_{\mathbf{k}} + (\nabla \chi) \otimes \mathbf{v}_{\mathbf{k}}$ and expand (73). The effective window is $W \sim B(P, \delta) \cap \Gamma_N$, of area $|W| \sim \delta^{d-1}$, and the decay confines the mass to depth $1/k$. Three errors arise.

- (a) *Coefficient variation:* replacing $\mathbb{L}(\mathbf{F}_0(\mathbf{x}))$ by $\mathbb{L}(\mathbf{F}_P)$ costs $\leq \sup_{B(P, \delta)} \|\mathbb{L}(\mathbf{F}_0) - \mathbb{L}(\mathbf{F}_P)\| \int \chi^2 |\nabla \mathbf{v}_{\mathbf{k}}|^2 \leq C \delta \cdot k \delta^{d-1} = C k \delta^d$ (by C^1 regularity of $\mathbf{F}_0$ at P).
- (b) *Gradient commutator:* the cross term and $|\nabla \chi|^2 |\mathbf{v}_{\mathbf{k}}|^2$ term are bounded by $C(\delta^{-1} \int_W |\mathbf{v}_{\mathbf{k}}| |\nabla \mathbf{v}_{\mathbf{k}}| + \delta^{-2} \int_W |\mathbf{v}_{\mathbf{k}}|^2) \leq C(\delta^{d-2} + \delta^{d-3} k^{-1})$.

C: These references treat the incompressible/gel half-space; existence, exponential decay, and $q(\lambda_{\parallel}) < 0$ for the compressible $f_{\log}$ material are used here but not covered by them. The compressible secular condition and its root are exactly what the surface-instability computation of §6.3 implements — state that secular analysis in the appendix (or cite a compressible reference) so Step 1 rests on something in the paper.

(c) *Window vs. cut-off*: replacing $\mathbf{1}_W$ by χ^2 in (75) changes the leading constant by $1 + o(1)$ only.

The leading term is $q(\lambda_{\parallel}) k \delta^{d-1} < 0$. The errors (a)–(b), relative to it, are $\mathcal{O}(\delta)$, $\mathcal{O}((k\delta)^{-1})$ and $\mathcal{O}((k\delta)^{-2})$. Choosing $\delta = k^{-1/2}$ sends $\delta \rightarrow 0$ and $k\delta = k^{1/2} \rightarrow \infty$, so all relative errors vanish as $k \rightarrow \infty$. Hence for k large enough,

$$Q_{t_*}(\mathbf{v}_*) = q(\lambda_{\parallel}) k \delta^{d-1} (1 + o(1)) < 0,$$

which is (73). The final statement is Lemma 8 with hypothesis (ii) now supplied. $\square$ $\square$

Remark 19 (What this closes). Lemma 15 converts the existence of the critical load t_c — previously conditional on the unproved hypothesis (ii) — into a *theorem* for the concrete clamped–free geometry, contingent only on the loading hypothesis (SC). What the construction delivers, and what it does not, should be stated precisely. It exhibits a single admissible field $\mathbf{v}_*$ with $Q_{t_*}(\mathbf{v}_*) < 0$, which by Lemma 8 is all that is needed to force a crossing $\mu_1(t_c) = 0$ at some $t_c \in (0, t_*]$. The destabilising mechanism is the half-space surface (creasing) instability, seated at an interior free-face point P of the edge neighbourhood B_ρ and confined to it by the cut-off (74)–(75); the witness $\mathbf{v}_*$ is supported away from the tip $r = 0$ (indeed (SC) places P at positive distance from Σ), so the lemma certifies a surface instability *within* B_ρ , not a mode pinned at the corner vertex. This is the precise sense in which the corner bifurcation is “the corner analogue of creasing”: the same free-face mechanism, activated near — not at — the edge. The location and shape of the genuine *critical* mode ϕ at t_c is a separate question, settled by the attainment analysis of §5.9 (which gives $\phi \in V$ supported in B_ρ), and the role of the edge proper enters quantitatively only through the graded pre-stress of Remark 21.

Remark 20 (On the threshold λ_* : compressible vs. incompressible). The proof uses only the *existence* of a surface-instability threshold $\lambda_* \in (0, 1)$, not its value. The familiar number $\lambda_* \approx 0.544$ is Biot’s result for the *incompressible* neo-Hookean half-space [?]; the present material (7) is compressible, and its threshold is the corresponding root of the compressible surface secular equation, which depends on the ratio λ_e/μ_e (equivalently the Poisson ratio) and tends to 0.544 in the incompressible limit $\lambda_e/\mu_e \rightarrow \infty$ [?]. This is the same instability with a material-dependent threshold, not a discrepancy: Lemma 15 *reduces* the corner instability to the half-space one and thereby agrees with the creasing literature rather than competing with it. Accordingly we keep λ_* symbolic throughout.

5.6 Symmetry: simplicity of the mode and the vanishing quadratic

Both model bodies of §5.1 possess reflection symmetries that the fundamental compressive state inherits: (H) is invariant under every orthogonal map fixing the x_d -axis, and (C) under each coordinate reflection $x_i \mapsto -x_i$ ($i < d$). Let R be such a reflection and $\mathcal{R}$ its action on displacement fields. Because $\mathbf{F}_0(t)$ and hence $\mathbb{L}(\mathbf{F}_0(t))$ are R -invariant, the second-variation operator $\mathcal{L}(t)$ commutes with $\mathcal{R}$, so its eigenspaces split into the symmetric ($\mathcal{R} = +1$) and antisymmetric ($\mathcal{R} = -1$) sectors. This yields, for the concrete geometry, two of the ingredients that §4 had to posit:

- *Simplicity (part of Assumption 8)*. The symmetry splitting reduces simplicity to two residual statements — that the crossing eigenvalue is the ground state of its sector, and that no accidental degeneracy across sectors occurs at t_c . Calling these *generic* requires a parameter to be perturbed, and for the fixed body none is; the honest formulation uses the family the paper already works on. For a real self-adjoint family, eigenvalue coincidence is a codimension-*two* event (von Neumann–Wigner), so within the two-parameter wedge family $(\nu, \hat{K})$ the loads

at which the one-parameter path $t \mapsto \mathcal{L}(t)$ meets a degeneracy are avoided at the crossing for almost every $(\nu, \hat{K})$; both residual statements then hold off a null set of the phase diagram, though not pointwise for a given body. At tested points they are moreover *checked*, not assumed: the sector gap computed in §6.4 is exactly this quantity — with the caveat, recorded there, that its behaviour under refinement is not yet settled. With that framing, simplicity need not be assumed as an independent spectral postulate, and only the *attainment* clause (the borderline-elliptic point of Remark 9) requires the separate analysis of §5.9.

- *Vanishing quadratic coefficient (part of Assumption 9).* The reduced potential of Proposition 1 inherits the $\mathcal{R}$ -symmetry of the perfect state. If ϕ lies in the antisymmetric sector, $w \mapsto -w$ is a symmetry of the reduced equation and the quadratic coefficient vanishes. For the neo-Hookean model this is made fully explicit in Lemma 16: c_2 is odd in the first-order volume change D_1 , which R flips, so $c_2 = 0$ exactly. The relevant reflection R is the lateral, *face-preserving* symmetry of the model bodies of §5.1, *not* the corner bisector (which would interchange Γ_D and Γ_N and is no symmetry of the mixed problem).

Lemma 16 (Neo-Hookean reduced coefficients are volumetric; the quadratic vanishes). *For the neo-Hookean energy (7) the deviatoric term $\frac{\mu_e}{2}(\text{tr } \mathbf{F}^T \mathbf{F} - d)$ is quadratic in $\mathbf{F}$, so its third and higher Fréchet derivatives vanish; hence every reduced bifurcation coefficient of order ≥ 3 is purely volumetric, determined by f alone. Writing $J(w) = \det(\mathbf{F}_0 + w \nabla \phi) = \sum_k D_k w^k$ with D_k the mixed discriminants of Lemma 12 ($D_1 = \text{cof } \mathbf{F}_0 : \nabla \phi$, $D_2 = d_\phi$, $D_k = 0$ for $k > d$) and $f^{(k)} = f^{(k)}(J_0)$, the single-mode coefficients are*

$$c_2 = \frac{1}{2} \int_{B_\rho} [f''' D_1^3 + 6f'' D_1 D_2 + 6f' D_3] dV, \quad (76)$$

$$b_{\text{dir}} = \frac{1}{6} \int_{B_\rho} [f'''' D_1^4 + 12f''' D_1^2 D_2 + 12f'' D_2^2 + 24f'' D_1 D_3 + 24f' D_4] dV. \quad (77)$$

Two consequences hold for the concrete geometry, whose fundamental state is R -symmetric.

- The quadratic vanishes, exactly and without correction. As a third-order quantity c_2 receives no orthogonal-mode (Lyapunov–Schmidt) feedback — that correction first enters at fourth order — so (76) is the full c_2 . The key parity fact is that each mixed discriminant $D_k = \frac{1}{k!} \partial_w^k \det(\mathbf{F}_0 + w \nabla \phi)|_0$ is exactly the degree- k part of the determinant in $\nabla \phi$, hence carries k factors of $\nabla \phi$ and one factor of an R -even cofactor of $\mathbf{F}_0$ per term. For an R -antisymmetric mode $\nabla \phi$ reverses sign under the reflection (composed with the change of variables $\mathbf{x} \mapsto R\mathbf{x}$ on the R -symmetric domain B_ρ), so D_k acquires the parity $(-1)^k$: D_1 and D_3 are R -odd, D_2 is R -even, while $f^{(k)}(J_0)$ is R -even (the base state is R -invariant). Each term of (76) thus carries an odd total number of $\nabla \phi$ factors — D_1^3 (three), $D_1 D_2$ (three), and D_3 (three) — and is therefore R -odd; the integrand is odd over the R -symmetric B_ρ and $c_2 = 0$. (In $d = 2$ the term $6f' D_3$ is moreover identically zero, since $D_k = 0$ for $k > d$; the parity argument is what disposes of it in $d \geq 3$.) This proves the vanishing-quadratic clause of Assumption 9.
- The frozen cubic and quintic are sign-definite, and adverse to subcriticality. With D_1 asymptotically small (Assumption 10), the leading frozen parts are $b_{\text{dir}} = 2 \int_{B_\rho} f''(J_0) d_\phi^2 dV$ and $d_{\text{dir}} = \int_{B_\rho} f'''(J_0) d_\phi^3 dV$. For the logarithmic penalty $f''_{\log}(J_0) = (\mu_e + 2\lambda_e - 2\lambda_e \log J_0)/J_0^2 > 0$ and $f'''_{\log}(J_0) = (4\lambda_e \log J_0 - 6\lambda_e - 2\mu_e)/J_0^3 < 0$ throughout compression $J_0 < 1$. The cubic part carries $d_\phi^2 \geq 0$, so $b_{\text{dir}} > 0$ follows from $f''_{\log} > 0$ alone, with no sign hypothesis on d_ϕ (this is the only frozen sign the argument needs). The quintic part carries the sign-indefinite

d_ϕ^3 ; on the relief-oriented region $d_\phi > 0$ ((81) below) and $f_{\log}''' < 0$ give $d_{\text{dir}} < 0$ there — the opposite of the subcritical pattern $b < 0 < d$, so the frozen quintic too is adverse where the relief acts, and subcriticality is decided by the feedback, not by either frozen term.

Proof. Expand $\Psi(\mathbf{F}_0 + w\nabla\phi) = \frac{\mu_c}{2}(\text{tr } \mathbf{F}^T \mathbf{F} - d) + f(J(w))$ in w . The deviatoric part is a polynomial of degree two in w , so it contributes only to the coefficients of w^0, w^1, w^2 ; all coefficients of order ≥ 3 come from $f(J(w))$, whose w -derivatives are given by Faà di Bruno's formula in the J -derivatives $f^{(k)}$ and the discriminants $D_k = \frac{1}{k!} \partial_w^k \det(\mathbf{F}_0 + w\nabla\phi)|_0$. Reading off the third and fourth derivatives gives (76)–(77); statements (i)–(ii) are then as stated, using $f_{\log}'' > 0$, $f_{\log}''' < 0$ on $J_0 < 1$. $\square$ $\square$

What symmetry fixes is thus the quadratic coefficient; what it does *not* fix is the sign of the cubic b and quintic d . Indeed the explicit single-mode parts (77) point *against* subcriticality ($b_{\text{dir}} > 0$, $d_{\text{dir}} < 0$ by Lemma 16(ii)), so the subcritical pattern $b < 0 < d$ rests entirely on the geometry-dependent orthogonal-mode feedback — the irreducible core, which we settle numerically rather than by analogy (Remark 11). Proposition 6 below explains, by a homogeneity count against the edge exponent α , *why* this feedback cannot be reduced to a universal corner number.

5.7 The homogeneity count for the cubic

The blow-up reduction of §5.2 promised a dichotomy: a corner quantity localises to a universal model-wedge number only when its defining integral is dominated by the tip. We now carry out the count for the cubic coefficient and find that, for the right-angle clamped-free pencil, it sits on the *non*-localising side — which is exactly why the sign of b is a genuine, $\hat{K}$ -dependent computation rather than an inheritance from flat creasing.

Near the edge the critical mode $\phi \in V$ inherits the admissible singular behaviour of the pencil, $\nabla\phi = \mathcal{O}(r^{\alpha-1})$ (the least-singular H^1 exponent of (13)), so the second discriminant scales as $d_\phi = \det \nabla\phi = \mathcal{O}(r^{2(\alpha-1)})$ in $d = 2$.

Proposition 6 (Homogeneity count: the cubic does not localise to the bare wedge). *In $d = 2$ the single-mode integrand of $b_{\text{dir}} = 2 \int_{B_\rho} f''(J_0) d_\phi^2 dV$ scales as $d_\phi^2 = \mathcal{O}(r^{4(\alpha-1)})$, so*

$$\int_{B_\rho} d_\phi^2 dV \sim \int_0 r^{4(\alpha-1)} r dr = \int_0 r^{4\alpha-3} dr, \quad (78)$$

which converges at the tip $r = 0$ iff $4\alpha - 3 > -1$, i.e. iff $\alpha > \frac{1}{2}$.

- (i) The clamped-free pencil lies above the threshold. *The smallest root of the right-angle clamped-free Williams secular equation (plane strain, $\kappa = 3 - 4\nu$) is $\alpha = \alpha(\nu) \in (0.60, 0.87)$ for $\nu \in (0, \frac{1}{2})$ — e.g. $\alpha(0.3) \approx 0.711$, $\alpha(0.45) \approx 0.622$, $\alpha(0.499) \approx 0.595$ — so $\alpha > \frac{1}{2}$ throughout the physical range (Figure 1).*
- (ii) Consequence: tip-convergent but not tip-dominated. *For $\alpha > \frac{1}{2}$ the integral (78) converges without barrier regularisation, but its mass is spread over the annulus rather than concentrated at the tip; the cut-off radius ρ and the matching to the far field leave an $\mathcal{O}(1)$ residue. By Proposition 5 this residue is carried by the normalised stress-intensity $\hat{K}$. Hence b does not collapse to a bare-wedge number: it is a functional $b = b(\nu, \hat{K})$ on the graded-substrate creasing model (a half-space crease seated in the linear pre-stress gradient of the edge field), and the subcritical sign $b < 0$ is the sign of $b(\nu, \hat{K})$ over the physical $(\nu, \hat{K})$ rectangle — a two-parameter phase diagram, not a single number.*

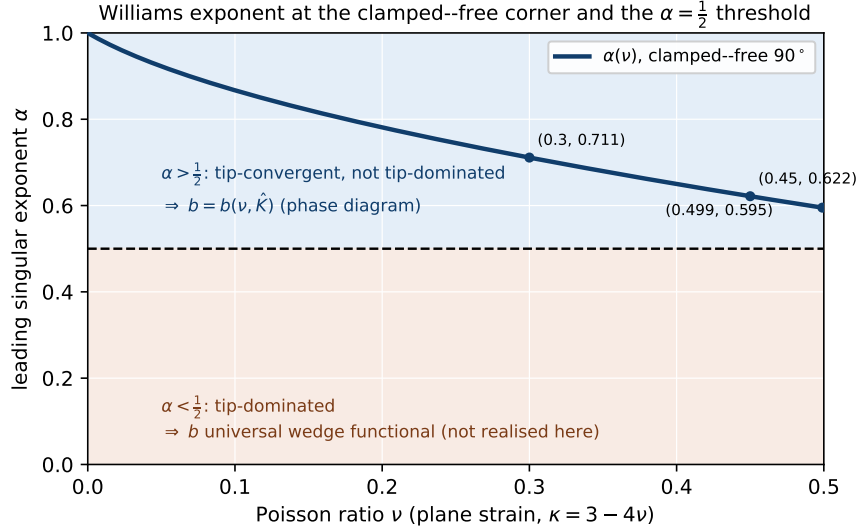


Figure 1: Leading Williams singular exponent $\alpha(\nu)$ for the right-angle clamped–free (Dirichlet–Neumann) wedge (plane strain, $\kappa = 3 - 4\nu$), the smallest root in $(0, 1)$ of the 4×4 secular determinant of Proposition 6. Across the entire physical range the curve lies in the regime $\alpha > \frac{1}{2}$ (upper band), where the cubic integral (78) is tip-convergent but not tip-dominated, so the reduced coefficient retains its dependence on the normalised stress-intensity, $b = b(\nu, \hat{K})$. The lower band $\alpha < \frac{1}{2}$ (tip-dominated, b a universal wedge functional) is not reached by this geometry. The exponent rises to $\alpha \rightarrow 1$ as $\nu \rightarrow 0$, i.e. the singularity *weakens* for small Poisson ratio and the stress concentration disappears in the limit ($\mathbf{F}$ bounded for $\alpha \geq 1$); this is expected, the corner stress singularity being driven by the volumetric coupling that vanishes as $\nu \rightarrow 0$. Reproduced by `williams.clampedfree.py` / `plot.williams_exponent.py`.

- (iii) The complementary regime. Were $\alpha < \frac{1}{2}$ (e.g. at sharper or differently-loaded openings), (78) would diverge at the tip, the barrier/collapse-locus regularisation of §5.8 would render it finite, and b would be set by an arbitrarily small tip neighbourhood — a genuinely localised, $\hat{K}$ -free wedge functional. The present geometry is firmly in case (ii).

Proof. The scaling (78) is immediate from $d_\phi = \det \nabla \phi = \mathcal{O}(r^{2(\alpha-1)})$ and the area element $r dr d\theta$; convergence at $r = 0$ is the elementary condition $4\alpha - 3 > -1$. The factor $f''(J_0)$ is bounded away from the collapse locus and only strengthens convergence under compression ($f''_{\log} > 0$), so it does not change the count. The exponents in (i) are the smallest real roots in $(0, 1)$ of the 4×4 boundary-condition determinant for the clamped ($u_r = u_\theta = 0$ at $\theta = 0$) and traction-free ($\sigma_{\theta\theta} = \sigma_{r\theta} = 0$ at $\theta = \frac{\pi}{2}$) Kolosov–Muskhelishvili eigensolutions $\varphi = Cz^\alpha$, $\psi = Ez^\alpha$; the value $\alpha(0.3) \approx 0.711$ reproduces the classical 90° fixed–free corner. Statements (ii)–(iii) are the localisation/non-localisation alternative of the blow-up dichotomy (Remark 18) applied to the convergent, respectively divergent, case of (78). $\square$ $\square$

Remark 21 (Two scales, and why flat creasing does not hand over the sign). Case (ii) has a transparent mechanical reading. The buckling mode of Lemma 15 is a crease localised at an interior free-face point P on the wavelength scale $1/k$, while the edge field varies on the scale r_P of the distance from P to Σ . If $1/k \ll r_P$ the crease sees a locally *homogeneous* pre-stress and inherits the flat-surface sign $b < 0$ for free; that limit is precisely the one $\alpha < \frac{1}{2}$ would enforce, and it is

not the present case. For $\alpha > \frac{1}{2}$ the surviving contribution lives at $1/k \sim r_P$, where the crease sits in the *graded* pre-stress of the corner; the substrate gradient enters through $\hat{K}$ and the flat result no longer dictates the sign. This is the analytic origin of the “ $\mathcal{O}(1)$ geometry competition” that makes b the one irreducibly numerical coefficient.

Remark 22 (The flat reduction is degenerate; the corner is what makes b exist). The phrase “inherits the flat sign $b < 0$ ” in Remark 21 must be read with care, because the flat single-mode reduction is in fact *singular*. The Biot surface-instability threshold of the homogeneous half-space is *scale-free*: the secular condition is independent of the surface wavenumber k , so at $\lambda_{\parallel} = \lambda_{\star}$ *every* harmonic is simultaneously neutral. The Lyapunov–Schmidt reduction of a single mode of wavenumber k then feeds, at second order, a response at wavenumber $2k$ — which is itself a neutral mode of the same operator. The second-harmonic problem is therefore *resonant*, the orthogonal-mode field ψ does not exist as a bounded correction, and the single-mode cubic coefficient b is undefined for the flat surface. (Numerically, the smallest singular value of the incremental operator at $\lambda_{\star}$ is at machine level for $k = 1, 2, 3, 4$ alike.) This is the precise analytic reason that flat creasing is a *localised*, finite-amplitude instability rather than a sinusoidal supercritical/subcritical bifurcation [?]: there is no single-mode normal form to carry a sign.

The corner removes exactly this degeneracy. The pre-stress is graded along the free face on the scale r_P (Remark 21), which breaks the translational/scale invariance, couples the wavenumbers, and selects a *discrete* critical mode — the attainment of Lemma 18 under (BC). With the spectrum discrete and simple, the second-harmonic operator is no longer resonant, ψ exists, and $b = b(\nu, \hat{K})$ is a well-defined number. Thus the corner is not merely the setting in which b happens to be computed: it is what makes the reduced cubic *exist* at all. This sharpens Proposition 6 — b cannot be a bare-wedge, $\hat{K}$ -independent constant, because the $\hat{K} \rightarrow 0$ (flat) limit is not a regular point of the reduction but the degenerate one.

The degeneracy just described is not only an obstruction; combined with the feedback-sign Lemma 11, it is a *mechanism*. Near the flat limit the second harmonic is nearly neutral, so the denominator of the certification criterion (49) is nearly zero — and the feedback, which is exactly the supremum (47), is correspondingly large. This can be turned into a theorem for the small- $\hat{K}$ regime.

Proposition 7 (Certified subcriticality at the edge of the flat degeneracy). *Let $\{(\phi_{\hat{K}}, \mathcal{L}_{c,\hat{K}})\}_{\hat{K}>0}$ be the family of critical states of the graded reduction (Proposition 6(ii)), and for each $\hat{K}$ let $\mathbf{v}_{\hat{K}} \in V^{\perp}$ be an admissible comparison field — the natural candidate is the doubled-wavenumber surface harmonic, which by Remark 22 goes neutral in the flat limit. With E_3 and b_{dir} as in Lemma 11, set the (scaling-invariant in both ϕ and $\mathbf{v}$) certification ratio*

$$\Gamma(\hat{K}) := \frac{E_3[\phi_{\hat{K}}, \phi_{\hat{K}}, \mathbf{v}_{\hat{K}}]^2}{2 b_{\text{dir}}(\hat{K}) \langle \mathbf{v}_{\hat{K}}, \mathcal{L}_{c,\hat{K}} \mathbf{v}_{\hat{K}} \rangle}, \quad (79)$$

so that, by Corollary 1, $b \leq b_{\text{dir}}(1 - \Gamma)$ and $b(\nu, \hat{K}) < 0$ whenever $\Gamma(\hat{K}) > 1$. Suppose that, as $\hat{K} \rightarrow 0^+$,

- (a) the comparison field goes neutral, $\langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle / \|\mathbf{v}\|_V^2 \rightarrow 0$ (the flat resonance of Remark 22); and
- (b) the second-harmonic coupling does not degenerate along with it: $E_3[\phi, \phi, \mathbf{v}]^2 / (b_{\text{dir}} \|\mathbf{v}\|_V^2) \geq c_0 > 0$ uniformly in $\hat{K}$.

Then $\Gamma(\hat{K}) \rightarrow \infty$; in particular there is $\hat{K}_0 > 0$ such that $b(\nu, \hat{K}) < 0$ for all $\hat{K} \in (0, \hat{K}_0)$, with diverging margin, $b_{\psi}/b_{\text{dir}} \leq -\Gamma \rightarrow -\infty$.

Proof. Immediate from Corollary 1: $\Gamma = [E_3^2/(b_{\text{dir}}\|\mathbf{v}\|_V^2)] \cdot [2\langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle / \|\mathbf{v}\|_V^2]^{-1} \geq \frac{1}{2} c_0 [\langle \mathbf{v}, \mathcal{L}_c \mathbf{v} \rangle / \|\mathbf{v}\|_V^2]^{-1} \rightarrow \infty$ by (a)–(b), and $b \leq b_{\text{dir}}(1 - \Gamma)$ with $b_{\text{dir}} > 0$. $\square$ $\square$

Remark 23 (Reading, and the computational route it recommends). Proposition 7 replaces the loose analogy “flat creasing is subcritical, so the corner should be” by a statement internal to the corner: the closer the graded state sits to the degenerate flat limit, the more violently subcritical it is, *provided* the quadratic self-coupling of the critical mode into the second harmonic survives the limit — hypothesis (b), which is precisely the non-vanishing of the quantity whose resonance makes the flat b undefined (Remark 22). The proposition covers the small- $\hat{K}$ regime only; on the finite $(\nu, \hat{K})$ rectangle the criterion $\Gamma > 1$ must be checked, and §6.4 now reports that check: certified at every tested rectangle point on the surrogate cell, while hypothesis (b) itself remains untestable there because the monotone grading removes the very resonance the proposition exploits. But it also dictates *how* to check it: Γ is assembled from one trilinear integral and one Rayleigh quotient, with no bordered solve, so it stays well-conditioned exactly where the direct Lyapunov–Schmidt assembly degrades (the near-resonant regime in which the condition number of the reduced solve diverges). We therefore regard (49), evaluated on the computed pair $(\phi, \mathbf{v})$, as the proper numerical arbiter of the cubic clause of Assumption 9, in preference to a converged value of b itself.

5.8 Kinematic relief

The kinematic relief is dimension-independent, and — in contrast to the original formulation — it does not require the critical mode to be *exactly* volume-preserving at first order. Earlier drafts *posited* $\text{cof } \mathbf{F}_0 : \nabla \phi = 0$; that exact condition is replaced by the weaker asymptotic smallness $D_1 = \text{cof } \mathbf{F}_0 : \nabla \phi = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ of Assumption 10 — small with J_0 , but not zero. We are careful about its status: at the bifurcation $J_0(t_c) > 0$ and the second variation (64) is a regular form, so the smallness is automatic (Lemma 13); along the near-collapse continuation it is a hypothesis, *not* derivable by isolating $f''(J)(\delta J)^2$ (the determinant term in (64) is in fact more singular and numerically dominant, Remark 14), and is verified directly on the computed mode (rotation defect ≈ 0.08 – 0.16). Granting it, Lemma 12 gives the lower bound

$$J = J_0 + w D_1 + w^2 d_\phi + \mathcal{O}(w^3) \geq \frac{1}{2} J_0 + \frac{1}{2} w^2 d_\phi \quad \text{in } \mathbb{R}^d, \quad (80)$$

with d_ϕ the second mixed discriminant of $\mathbf{F}_0$ and $\nabla \phi$ (the second-order coefficient of the determinant, $= \det \nabla \phi$ for $d = 2$). The relief mechanism itself is valid for all d once Assumption 10 is granted, so it is a kinematic statement in every dimension rather than a special two-dimensional accident. The relief floor requires the *orientation* of this second-order coefficient, but — and this is the precise content of the localisation — only *where the relief term is actually invoked*, namely on a neighbourhood of the collapse locus Σ_c (the tube where $\lambda_{\min} \rightarrow 0$); off Σ_c , where J_0 stays bounded below, the sign of d_ϕ is immaterial (the finite- J_0 case (i) of Lemma 12 absorbs a negative $w^2 d_\phi$ into the J_0 margin). The localised orientation condition is

$$d_\phi \geq c_d > 0 \quad \text{a.e. on a neighbourhood of } \Sigma_c \text{ (the relief-orienting sign of the mode, with a uniform floor } c_d), \quad (81)$$

which we record as a genericity property of the critical mode, *not* as a consequence of Lemma 14 (that lemma concerns the transversal vanishing of $\lambda_{\min}$, a different field). The condition is local for a structural reason made exact by the planar identity, valid pointwise for any $\nabla \phi$,

$$d_\phi = \det \nabla \phi = \det \mathbf{E} + \omega^2, \quad \mathbf{E} = \frac{1}{2}(\nabla \phi + \nabla \phi^T), \quad \omega = \frac{1}{2}(\partial_1 \phi_2 - \partial_2 \phi_1), \quad (82)$$

(the strain/spin split; the cross term vanishes because $\mathbf{E}$ is symmetric). Thus $d_\phi > 0 \iff \omega^2 > |\det \mathbf{E}|$: the relief orientation is *exactly* the statement that the local spin dominates the strain

determinant — the same rotation-dominance that underlies the asymptotic-rotation mechanism (66), and the reason the relief is second order. Two consequences follow. First, d_ϕ is the Jacobian determinant of an *oscillatory* mode, so it flips sign cell-to-cell and a *global* one-sign claim on B_ρ would be generically false — on the computed mode of §6 the global positive fraction is only $\approx 25\text{--}36\%$; what the lemma needs, and all it needs, is $d_\phi > 0$ where $\lambda_{\min} \rightarrow 0$. Second, (82) makes the numerical test sharp, and we report its outcome honestly. Where the mode concentrates the spin dominates by $\sim 2.4:1$ ($d_\phi > 0$ on the $|\nabla\phi|^2$ -weighted measure at $\approx 70\text{--}74\%$, rising to $\approx 80\%$ on the sub-surface annulus carrying most modal energy), so on the bulk of the active mode the orientation is the relieving one. But the mode peaks within a thin surface skin that coincides with the deepest compression, and *there* the strain is shear-dominated, $\det \mathbf{E} < 0$ with $\omega^2/|\det \mathbf{E}| \approx 0.6 < 1$, so $d_\phi < 0$ on the deepest- J_0 decile. In the computable range this skin is harmless — the base states keep $J_0 \gtrsim 0.5$, so $\lambda_c = \lambda_{\min}(\mathbf{F}_0)$ is bounded below and the finite- J_0 case (i) of Lemma 12 absorbs the negative $w^2 d_\phi$ with no sign hypothesis (this part is *proved*, not assumed). The relief floor $C_2 w^2$ is invoked only in the genuine collapse limit $J_0 \rightarrow 0$ along the unstable continuation, which the barrier $f''_{\log} \sim |\log J_0|/J_0^2$ puts beyond reach of the solver; a mode-following deepening sweep toward it finds the deep-skin ratio $\omega^2/|\det \mathbf{E}|$ *drifting slightly downward* ($0.65 \rightarrow 0.61$), so the accessible evidence does not support $d_\phi > 0$ at Σ_c and, if anything, leans against it. We therefore keep (81) as a genuine hypothesis on the same footing as the rate (66) — needed only in the unreachable limit, neither verified nor, on present evidence, favoured there — while stressing that the regularisation everywhere the computation reaches is independent of it (Remark 14). Lemma 17 below sharpens this status considerably: the orientation is *proved*, pointwise, wherever the mode is locally a quadrature surface wave with one-signed, monotonically decaying envelopes, and the possible failure is confined to — and characterised on — the strain-dominated skin. The remaining points for the concrete model are that the collapse locus Σ_c is generically the codimension-one set inside B_ρ along which the relieved branch would still collapse (or, when the severest compression sits at the vertex, the point $\Sigma_c = \{0\}$ of codimension two, Remark 15, which only sharpens the integrability); the regularised bound of Theorem 2 and the localisation of Proposition 4 then apply verbatim.

The status of the orientation condition can be made exact rather than statistical. The following identities replace the sampled sign fractions of §6 by closed-form criteria on the mode itself.

Lemma 17 (Exact orientation identities for a surface-wave mode). *Let $d_\phi = \det \nabla \phi$ in $d = 2$, with components ϕ_1 along the free face and ϕ_2 normal to it.*

- (i) (Null Lagrangian: the bulk carries no net orientation.) *For every Lipschitz subdomain $\omega \subset B_\rho$,*

$$\int_\omega d_\phi dV = \oint_{\partial\omega} \phi_1 \partial_\tau \phi_2 ds, \quad (83)$$

τ the positively oriented tangent. The net orientation of the mode over any cell is a boundary quantity; over B_ρ (where ϕ vanishes on Γ_D and, for the localised mode, on ∂B_ρ) it is carried entirely by the free face. In particular any global bulk statistic of $\text{sign}(d_\phi)$ is structurally uninformative — the rescoping of (81) to the collapse tube is forced, not merely prudent.

- (ii) (Surface-wave mean: an exact depth derivative.) *If locally $\phi = \text{Re}[\mathbf{U}(n) e^{ikx}]$, with x the arc length along the free face and $n \geq 0$ the depth, the average over one wavelength is*

$$\bar{d}_\phi(n) = -\frac{k}{2} \frac{d}{dn} \text{Im}(U_1 \bar{U}_2), \quad \int_0^\infty \bar{d}_\phi dn = \frac{k}{2} \text{Im}(U_1 \bar{U}_2) \Big|_{n=0} : \quad (84)$$

the depth-integrated mean orientation is set by the relative phase of the two displacement components at the surface alone, consistently with (i).

(iii) (Quadrature modes: pointwise sign.) *If moreover the mode is in exact quadrature,*

$$\phi_1 = -a(n) \sin kx, \quad \phi_2 = b(n) \cos kx \quad (85)$$

(the form of a surface-instability eigenmode with real Stroh roots and real amplitude ratios), then pointwise, at every (x, n) ,

$$d_\phi = -k [a b' \cos^2 kx + a' b \sin^2 kx]. \quad (86)$$

Hence $d_\phi > 0$ everywhere if and only if $ab' < 0$ and $a'b < 0$; in particular, if the envelopes a, b share a common sign and both decay monotonically in depth, then $d_\phi > 0$ pointwise — the orientation condition (81) holds with no averaging and no genericity. Conversely, a sign change of $\bar{d}_\phi$ at depth n marks an interior extremum of the envelope product ab .

Proof. (i) $\det \nabla \phi = \partial_1(\phi_1 \partial_2 \phi_2) - \partial_2(\phi_1 \partial_1 \phi_2)$, the mixed second derivatives cancelling, i.e. $d_\phi = \text{div}(\phi_1 \partial_2 \phi_2, -\phi_1 \partial_1 \phi_2)$; the divergence theorem and $(\partial_2 \phi_2, -\partial_1 \phi_2) \cdot \mathbf{n} = \partial_\tau \phi_2$ give (83). (ii) With $\partial_x \phi_i = \text{Re}[ik U_i e^{ikx}]$ and $\partial_n \phi_i = \text{Re}[U'_i e^{ikx}]$, the product formula $\text{Re}[A e^{ikx}] \text{Re}[B e^{ikx}] = \frac{1}{2} \text{Re}[A \bar{B}] + \frac{1}{2} \text{Re}[A B e^{2ikx}]$ yields $\bar{d}_\phi = \frac{1}{2} \text{Re}[ik (U_1 \bar{U}'_2 + U'_1 \bar{U}_2)] = \frac{1}{2} \text{Re}[ik (U_1 \bar{U}_2)']$, which is (84); the integral follows from the decay at $n = \infty$. (iii) Direct differentiation of (85) gives $d_\phi = (-ka \cos kx)(b' \cos kx) - (-a' \sin kx)(-kb \sin kx)$, which is (86); its mean $-\frac{k}{2}(ab)'$ agrees with (84) at $U_1 = ia, U_2 = b$. $\square$

Remark 24 (What the identities change). Three consequences. *First*, the localisation of (81) is structural: by (83) the Jacobian of the mode has no bulk mass, so a global one-sign claim was never the right invariant. *Second*, on the bulk of the active mode the orientation is now *proved*, not sampled: where the computed mode is locally a decaying quadrature surface wave with one-signed envelopes — the sub-surface annulus carrying most of the modal energy — Lemma 17(iii) gives $d_\phi > 0$ pointwise, in agreement with the ≈ 70 – 80% weighted fractions of §6 (the shortfall from 100% measures exactly the extent to which the grading modulates the envelope in x and breaks exact quadrature). *Third*, the possible failure of (81) is localised and characterised: it can occur only where an envelope is non-monotone or changes sign — for the computed eigenmode, the thin strain-dominated skin above the envelope extremum, where the observed $\omega^2/|\det \mathbf{E}| \approx 0.6$ sits. The open content of (81) is thereby reduced from a genericity postulate on ϕ to a single explicit question: whether, in the collapse limit $J_0 \rightarrow 0$, the envelope extremum of the critical mode detaches from the collapse tube (relief holds) or rides on it (relief fails). That question is posed on the Stroh data of the compressible surface eigenmode — two exponents and two amplitude ratios fixed by the secular condition — and is answerable in closed form; we leave its evaluation to the planned wedge computation.

5.9 Attainment of the critical mode

The second half of Assumption 8 — that $\mu_1(t_c) = 0$ is *attained* by a genuine eigenmode rather than being the bottom of essential spectrum — looks like the hard, borderline-elliptic point of Remark 9. It is in fact a consequence of a single, transparent geometric condition: that the *buckling precedes the pointwise collapse* of the fundamental state. The essential-spectrum threat is tied to the scale-invariance of the corner at collapse (the same wavenumber degeneracy that makes flat creasing scale-free, and that let $k \rightarrow \infty$ in Lemma 15); it is activated only when $\mathbb{L}$ blows up, i.e. only when $\lambda_{\min} \rightarrow 0$. So long as the crossing occurs while the fundamental stretch is still bounded below, the operator has bounded coefficients and the spectrum is discrete.

C: This envelope-extremum question needs no wedge solver: the Stroh exponents and amplitude ratios are already computed by the surface-instability solve of §6.3, so the extremum-vs-tube comparison is a closed-form evaluation on existing data. Given that the accessible evidence

Assumption 13 (Buckling precedes collapse, (BC)). The fundamental path keeps the stretch off collapse up to the crossing:

$$\inf_{t \leq t_c} \inf_{\mathbf{x} \in \overline{B_\rho}} \lambda_{\min}(\mathbf{F}_0(t, \mathbf{x})) \geq c > 0. \quad (87)$$

Under the standing path assumption (FP)(i) of §4, (BC) is no longer an independent postulate: once the destabilisation load satisfies $t_* < t_{\text{coll}}$, Lemma 8 *concludes* $t_c \leq t_* < t_{\text{coll}}$ and (87) holds with $c = c(t_*)$. We keep the name (BC) for its mnemonic value; its entire quantitative content is the single lower bound $t_{\text{coll}} > t_*$, which Proposition 8 confines to the tip disc.

Lemma 18 (Attainment under (BC)). *Assume, in addition to (BC), that the fundamental state $(t, \mathbf{x}) \mapsto \mathbf{F}_0(t, \mathbf{x})$ is continuous on the compact cylinder $[0, t_c] \times \overline{B_\rho}$ — a standing regularity of the load path in the spirit of the selection continuity of Assumption 7, automatic for a fundamental branch depending continuously on the load with bounded data. Then for every $t \leq t_c$ the tangent $\mathbb{L}(\mathbf{F}_0(t))$ is bounded on B_ρ , the operator $\mathcal{L}_c = \partial_{\mathbf{u}} G(\mathbf{u}_0(t_c), t_c, 0)$ has compact resolvent and discrete spectrum, and the critical value $\mu_1(t_c) = 0$ is attained by an eigenmode $\phi \in V$. This discharges the attainment clause of Assumption 8; the simplicity clause is reduced, by the symmetry argument of §5.6, to the generic statement that the lowest eigenvalue within the symmetry sector carrying ϕ is simple.*

Proof. Condition (87) supplies the *lower* stretch bound, $\|\mathbf{F}_0(t)^{-1}\| = \lambda_{\min}^{-1} \leq c^{-1}$. The matching *upper* bound is not part of (BC) — which constrains only $\lambda_{\min}$ — but follows from the stated continuity hypothesis: $(t, \mathbf{x}) \mapsto \mathbf{F}_0(t, \mathbf{x})$ is continuous on the compact set $[0, t_c] \times \overline{B_\rho}$, so all its stretches, and in particular $\det \mathbf{F}_0$, are bounded above by a constant $C_0 = C_0(t_c) < \infty$ there. Together with $\Theta_{\log}$ being continuous and finite on the compact range $[(\det \mathbf{F}_0)_{\min}, C_0]$ (which avoids $J = 0$ by (BC)), the structural bound (11), $\|\mathbb{L}(\mathbf{F}_0(t))\| \leq \mu_e + c^{-2} \Theta_{\log}(\det \mathbf{F}_0)$, is uniformly bounded on B_ρ for $t \leq t_c$. By Lemma 6 the same tensor is strongly elliptic with constant $\geq \mu_e$ throughout compression; with Korn’s inequality the second-variation form obeys a Gårding estimate $Q_{t_c}(\mathbf{v}) \geq \gamma \|\mathbf{v}\|_{H^1(B_\rho)}^2 - C \|\mathbf{v}\|_{L^2(B_\rho)}^2$ on $V \subset H^1$. Thus $\mathcal{L}_c + C$ is coercive on H^1 , its inverse maps L^2 boundedly into H^1 , and the Rellich embedding $H^1(B_\rho) \hookrightarrow L^2(B_\rho)$ — valid on the bounded domain, the edge Σ notwithstanding — makes the resolvent of $\mathcal{L}_c$ compact. A self-adjoint operator with compact resolvent has purely discrete spectrum and attains its lowest eigenvalue; at t_c that eigenvalue is $\mu_1(t_c) = 0$, with eigenmode ϕ . $\square$ $\square$

Remark 25 (What (BC) costs, and why it is the right residue). Lemma 18 relocates the difficulty from a functional-analytic statement about essential spectrum to a comparison of two load thresholds: the buckling load t_c and the pointwise-collapse load $t_{\text{coll}} := \inf\{t : \inf_{B_\rho} \lambda_{\min}(\mathbf{F}_0(t)) = 0\}$, with (BC) asking $t_c < t_{\text{coll}}$. This is genuinely cleaner than “attainment”: (i) Lemma 15 already bounds t_c from above by the surface-creasing load t_* , attained while $\lambda_{\min} \approx \lambda_* = \mathcal{O}(1)$ at the creasing point — not at collapse; (ii) the barrier $f_{\log}(J) \rightarrow \infty$ makes $J = 0$ cost infinite energy, so the true (not linearly-predicted) fundamental path resists pointwise collapse and pushes t_{coll} up; and (iii) (BC) is directly checkable in a computation, by monitoring $\min_{\mathbf{x}} \lambda_{\min}$ at the detected crossing. The one place it can fail is the edge tip, where the *linear* prediction $\lambda_{\min} \sim 1 - |K|r^{\alpha-1}$ wants to collapse for any load; bounding the barrier-regularised tip collapse load from below is the concrete content left open, in place of the opaque spectral hypothesis. Lemma 19 and Proposition 8 below make (ii) and the tip localisation quantitative: collapse is repelled in measure at an explicit logarithmic rate and spatially confined to a disc whose radius is explicit in $(\hat{K}, \alpha)$; the residual content of (BC) is thereby reduced to a pointwise floor on that single shrinking disc. One further consequence of stating the path hypotheses up front (Assumption 7) deserves emphasis: the floor

is needed at the *Biot* load t_* , not at collapse. By Lemma 8, once $t_* < t_{\text{coll}}$ — i.e. once the tip disc has not collapsed by the load at which the free-face stretch reaches λ_* , with the state there uniformly non-degenerate ($\lambda_{\min} \approx \lambda_* = \mathcal{O}(1)$ at the creasing point) — the ordering $t_c \leq t_* < t_{\text{coll}}$ is a *conclusion*, not a separate hypothesis: (BC) collapses to the single quantitative bound $t_{\text{coll}} > t_*$, checkable through (92).

Lemma 19 (The barrier repels collapse in measure). *For the energy (7) with $f_{\log}$ one has the pointwise coercivity*

$$\Psi(\mathbf{F}) \geq \lambda_e \log^2 J \quad \text{for all } \mathbf{F} \in \text{GL}^+(2). \quad (88)$$

Let $\mathbf{u}_0(t)$ be any admissible branch whose energy stays below an explicit finite bound, $\mathcal{E}_t(\mathbf{u}_0(t)) \leq E(t)$: minimality itself is not used, only the energy bound. The global-minimiser branch of Lemma 7 qualifies by construction under displacement control, with $E(t) := \mathcal{E}_t(\mathbf{u}_{\text{trial}})$ for any fixed admissible competitor (finite and explicit for the bodies (H), (C), e.g. the smooth ramp extension of the boundary data); so does any stable or unstable continuation — past t_c included — on which the energy is monitored and found below $E(t)$. Then for every $\delta \in (0, 1)$,

$$|\{\mathbf{x} \in B_\rho : J_0(t, \mathbf{x}) \leq \delta\}| \leq \frac{E(t)}{\lambda_e \log^2 \delta}. \quad (89)$$

In particular the sublevel sets of J_0 shrink at an explicit logarithmic rate: a collapse tube $\{J_0 \leq \delta\}$ around a codimension-one locus Σ_c of length $|\Sigma_c|$ has mean half-width $\leq E(t)/(2\lambda_e |\Sigma_c| \log^2 \delta)$, and pointwise collapse ($J_0 = 0$) can occur at most on a null set at any finite load. The same holds under traction loading whenever the load functional stays bounded along the path, with $E(t)$ replaced by $E(t) + \sup_{s \leq t} |\ell_s(\mathbf{u}_0(s))|$.

Proof. In $d = 2$, writing the principal stretches λ_i and using $\text{tr}(\mathbf{F}^T \mathbf{F}) = \sum_i \lambda_i^2$, $\log J = \sum_i \log \lambda_i$,

$$\Psi(\mathbf{F}) = \sum_i \frac{\mu_e}{2} (\lambda_i^2 - 1 - \log \lambda_i^2) + \lambda_e \log^2 J \geq \lambda_e \log^2 J,$$

since $s - 1 - \log s \geq 0$ at $s = \lambda_i^2$; this is (88). By the assumed energy bound $\int_{B_\rho} \lambda_e \log^2 J_0 dV \leq \mathcal{E}_t(\mathbf{u}_0(t)) \leq E(t)$, and Chebyshev's inequality on $\{J_0 \leq \delta\}$, where $\log^2 J_0 \geq \log^2 \delta$, gives (89). $\square$

Proposition 8 (Collapse, if anywhere, is confined to the tip disc). *Along the fundamental path, for $t < t_{\text{coll}}$, assume the corner expansion of hypothesis (FP)(ii) — the Kondratiev form of Proposition 5(i) with pointwise gradient control of the remainder, $|\nabla \mathbf{u}_0^\sharp| \leq C_\sharp(t) (r/\rho)^{\alpha'-1}$; this is a genuine hypothesis on the nonlinear state, not a citation, since the linear theory [?, ?] attaches to it only through a frozen-coefficient reading and, under compression, no screening mechanism pins the vertex behaviour (Remark 8). Set $c_\Phi := \max_\theta \|\alpha \Phi \otimes \mathbf{e}_r + \Phi' \otimes \mathbf{e}_\theta\|$, so that the leading term obeys $\|\nabla(Kr^\alpha \Phi)\| \leq c_\Phi \mu_e |\hat{K}(t)| (r/\rho)^{\alpha-1}$ by the normalisation $\hat{K} = K\rho^{\alpha-1}/\mu_e$ of Remark 18. With $M(t) := c_\Phi \mu_e |\hat{K}(t)| + C_\sharp(t)$ bounded along the path,*

$$\lambda_{\min}(\mathbf{F}_0(t, \mathbf{x})) \geq 1 - M(t) \left(\frac{r}{\rho}\right)^{\alpha-1}, \quad \mathbf{x} \in B_\rho, \quad (90)$$

so $\lambda_{\min} \geq \frac{1}{2}$ everywhere outside the tip disc

$$r \leq r_0(t) := \rho (2M(t))^{\frac{1}{1-\alpha}}. \quad (91)$$

Consequently:

- (i) letting $t \uparrow t_{\text{coll}}$, any collapse point lies in the closed disc $\{r \leq r_0(t_{\text{coll}})\}$: condition (BC) can fail only through the tip, the assertion of Remark 25 upgraded from a heuristic to an estimate, with the failure region quantified by the computable pair $(\hat{K}, \alpha)$;
- (ii) the surface-instability point P of (SC), at fixed distance r_P from Σ , is provably not the collapse site whenever $r_0(t_*) < r_P$, i.e. whenever

$$M(t_*) < \frac{1}{2} \left(\frac{r_P}{\rho} \right)^{1-\alpha}, \quad (92)$$

an explicit inequality between computable numbers; granted (92), the destabilising field of Lemma 15 lives entirely in the region where (90) keeps the state uniformly non-degenerate, and the ordering $t_c < t_{\text{coll}}$ can fail only if the tip disc (91) collapses before the free-face stretch reaches λ_* ;

- (iii) combined with Lemma 19 the two smallness mechanisms compound: the possible collapse set is contained in a disc of radius r_0 and has measure $\mathcal{O}(E(t)/\log^2 \delta)$ at depth δ .

Proof. For $t < t_{\text{coll}}$ the state is uniformly non-degenerate on $\overline{B_\rho}$ by (FP)(i) and the coefficients are bounded (Lemma 18); the expansion $\mathbf{u}_0 = Kr^\alpha \Phi(\theta) + \mathbf{u}_0^\sharp$ holds with the stated bounds by hypothesis (FP)(ii). Then $\nabla \mathbf{u}_0 = Kr^{\alpha-1}(\alpha \Phi \otimes \mathbf{e}_r + \Phi' \otimes \mathbf{e}_\theta) + \nabla \mathbf{u}_0^\sharp$ and, by Weyl's inequality for singular values, $\lambda_{\min}(\mathbf{I} + \nabla \mathbf{u}_0) \geq 1 - \|\nabla \mathbf{u}_0\|$, which with $(r/\rho)^{\alpha'-1} \leq (r/\rho)^{\alpha-1}$ on $r \leq \rho$ (recall $\alpha' > \alpha$) gives (90). The right-hand side of (90) is $\geq \frac{1}{2}$ iff $r \geq r_0(t)$, giving the dichotomy; (i) follows by continuity of $\lambda_{\min}$ in t up to the first failure, (ii) by inserting $t = t_*$ and Lemma 15, (iii) by intersecting with (89). $\square$ $\square$

Lemma 20 ((BC) reduces to a comparison of wedge stretch thresholds). *The two thresholds that (BC) compares are stretch values, and as such are scale-invariant data of the model wedge $\mathcal{L}_W$ of Proposition 5, depending on the geometry only through ν and free of the stress-intensity amplitude $\hat{K}$: the surface-instability stretch $\lambda_* = \lambda_*(\nu) \in (0, 1)$ at which buckling is triggered, and the collapse stretch $\lambda_{\min} = 0$. The conversion of these stretch thresholds into the load ordering $t_c < t_{\text{coll}}$ is not itself a wedge statement: it requires the fundamental state's load-to-stretch map $t \mapsto \lambda_{\parallel}(\mathbf{F}_0(t); P)$, a global datum. Under monotone compression (condition (M) of Lemma 9) this map is monotone, so the stretch ordering $\lambda_* > 0$ transfers to the load ordering, provided the barrier-regularised collapse load t_{coll} admits the lower bound left open in Remark 25.*

Proof. What localises to the wedge is the pair of stretch thresholds. The surface-instability stretch λ_* and the Biot secular form $q(\lambda_{\parallel})$ are functions of ν alone (Remark 20), evaluated on the half-space tangent to the free face — a sub-object of $\mathcal{L}_W$; the collapse stretch is the universal value $\lambda_{\min} = 0$. Both are amplitude-free. Lemma 15 shows that once the fundamental state drives $\lambda_{\parallel}$ below λ_* at some free-face point, a destabilising field exists, hence $t_c \leq t_*$ where t_* is the load realising $\lambda_{\parallel} = \lambda_*$. This load t_* is *not* a wedge quantity: it is the value of the global map $t \mapsto \lambda_{\parallel}(\mathbf{F}_0(t); P)$ at the threshold stretch, and that map is set by the fundamental state, not by the half-space model. The earlier draft conflated the (wedge) threshold stretch with the (global) threshold load; we separate them here. The load ordering $t_c < t_{\text{coll}}$ is what (BC) needs, and it follows once two genuinely non-wedge facts are granted: (a) under monotone compression (M) the map $t \mapsto \lambda_{\parallel}$ (resp. $\lambda_{\min}$) is monotone decreasing, so the stretch ordering $\lambda_* > 0 = \lambda_{\min}^{\text{coll}}$ converts to $t_* \leq t_{\text{coll}}$; and (b) the barrier-regularised t_{coll} is bounded below at the tip, the open content of Remark 25. What the wedge reduction *does* deliver, free of $\hat{K}$, is that the thresholds being compared are amplitude-free, ν -only stretches; the amplitude $\hat{K}$ and the finite body enter only through the load-to-stretch map and through the $\mathcal{O}(\lambda^g)$ corrections of Proposition 5(ii). $\square$ $\square$

Remark 26 (Why (BC) localises but the cubic sign need not). Lemma 20 is the easy half of the localisation programme: the *stretch thresholds* being compared (λ_* and 0) are amplitude-free, ν -only wedge data, decided by a single computation independent of (H) versus (C); the only non-wedge ingredient is the monotone load-to-stretch map that turns the stretch comparison into a load ordering. The subcritical *sign* of §5.7 is the hard half: there the relevant object is an integral, not a threshold, and whether it localises is governed by the homogeneity count of Proposition 6. The two open conditions thus sit on opposite sides of the blow-up dichotomy of Remark 18.

5.10 The irreducible core

Table 1 summarises the effect of fixing the concrete model. Of the hypotheses of §3–§4, the material assumptions are discharged by the neo-Hookean structure (§5.3), the existence of the critical load by the localized-creasing Lemma 15 under the loading hypothesis (SC), mode simplicity and the vanishing quadratic by reflection symmetry (§5.6), and mode attainment by Lemma 18 under the geometric condition (BC). What remains are two hypotheses, both now in concrete, checkable form rather than as opaque spectral postulates:

1. *Buckling precedes collapse*, (BC) (Assumption 13): the threshold inequality $t_c < t_{\text{coll}}$ that powers attainment (Lemma 18). It is bounded on one side by Lemma 15 ($t_c \leq t_*$, at $\lambda_{\min} = \mathcal{O}(1)$) and supported by the energy barrier (Remark 25), now quantitatively: collapse is repelled in measure at an explicit logarithmic rate (Lemma 19) and spatially confined to the tip disc of radius $r_0 \sim \rho(2M)^{1/(1-\alpha)}$ (Proposition 8), so the residual content is a pointwise floor on that single shrinking disc, which we verify numerically. By Lemma 20 the two *stretch thresholds* being compared are amplitude-free, one-parameter (ν) wedge data, independent of the body and of $\hat{K}$; the conversion to the load ordering uses only the monotone load-to-stretch map of the fundamental state (condition (M)).
2. *Subcritical sign pattern*, $b < 0 < d$ (Assumption 9): the directions in which the cubic and quintic bend the branch. Here the neo-Hookean structure is doubly informative. Lemma 16 reduces the coefficients to volumetric integrals and shows the explicit single-mode parts are *adverse* to subcriticality ($b_{\text{dir}} > 0$, $d_{\text{dir}} < 0$ in compression), so $b < 0 < d$ can come only from the geometry-dependent orthogonal-mode feedback. For the cubic, the *direction* of that feedback is now a theorem, $b_\psi \leq 0$ (Lemma 11), so the open content is the magnitude comparison $|b_\psi| > b_{\text{dir}}$: proved for small $\hat{K}$ under the coupling non-degeneracy of Proposition 7, and on the finite rectangle certified by the inversion-free criterion of Corollary 1, evaluated in §6.4 (Remarks 11, 23); the quintic clause $d > 0$, by contrast, has no analogous sign theorem and no computation behind it and remains fully open (Remark 12). The homogeneity count of Proposition 6 pins down *why* the magnitude is irreducible: because the edge exponent satisfies $\alpha > \frac{1}{2}$ here, the cubic integral is tip-convergent but not tip-dominated, so b does not collapse to a universal wedge number but remains a functional $b(\nu, \hat{K})$ of the Poisson ratio and the normalised stress-intensity — the quantity the computation maps out.

The upshot is that for the clamped-free neo-Hookean corner under compression the abstract picture of §3–§4 is, but for two clearly delimited conditions, a sequence of theorems: the material bounds are explicit, the corner is genuinely compressive, the instability is a free-face creasing event with a constructed destabilising field, the critical mode is attained and (generically, within its symmetry sector) simple, with no quadratic term by symmetry, and the kinematic relief that regularises the tangent holds in every dimension once the asymptotic-rotation hypothesis is granted. Notably, the spectral postulates of the abstract theory are sharply reduced: attainment is *proved*

Hypothesis	Abstract status	Status for the con
Reg./growth/ellip./mild (Ass. 1–5)	standing assumptions	proved (Lem. 1,
Edge exponent α , scenario selection	abstract pencil	computed $\alpha(\nu) \in$
Localisation to model wedge	informal	proved : blow-up
Critical load t_c exists (Lem. 8(ii))	not established	proved under (S)
Transversality (M) (Lem. 9)	input on loading	holds for monot
Mode simplicity (Ass. 8)	assumed	generic in $(\nu, \hat{K})$
Quadratic coeff. vanishes (Ass. 9)	assumed by symmetry	proved : c_2 odd i
Mode attainment (Ass. 8)	assumed	proved under (B)
First-order relief / rotation (Ass. 10, Lem. 12)	finite at t_c (Lem. 13); near-collapse posited	<i>numerical</i> : rotati
Stable-branch non-degeneracy (Ass. 11)	posited	<i>physical</i> (bounde
Buckling precedes collapse (BC) (Ass. 13)	—	<i>numerical</i> on tip
Subcriticality $b < 0 < d$ (Ass. 9)	assumed	cubic: sign prove

Table 1: Hypothesis ledger for the concrete clamped–free neo-Hookean model. Fixing the material and geometry discharges every spectral postulate; the residue is two concrete, numerically checkable conditions — that buckling precedes pointwise collapse, and that the reduced cubic is subcritical.

under (BC), and simplicity is reduced to genericity within the symmetry sector carrying the critical mode rather than posited outright; the first-order volume preservation (rotation condition) that the relief once posited as an exact identity is now the weaker explicit assumption (AR, §5.8), automatic at the bifurcation (Lemma 13) and numerically supported toward collapse rather than a barrier-forced theorem. What remains are two *concrete* and numerically checkable conditions: that buckling precedes pointwise collapse ($t_c < t_{\text{coll}}$) — with the possible failure now confined to an explicit tip disc and repelled in measure (Proposition 8, Lemma 19) — and that the reduced cubic is subcritical ($b < 0$) — now a pure *magnitude* comparison, its sign structure being settled by Lemma 11 and its small- $\hat{K}$ regime by Proposition 7. Both are exactly what the creasing problem on a flat surface also leaves to computation, and they are inherited here rather than created by the corner.

Finally, the blow-up reduction of §5.2 sharpens what “numerical” means for these two residues. By Proposition 5 the singularity is localised to the model wedge $\mathcal{L}_W$, all of whose data are functions of $(\nu, \hat{K})$ alone; the finite body (H) or (C) enters only through the single scalar $\hat{K}$ and through corrections $\mathcal{O}(\lambda^g)$. The two residual conditions then occupy opposite sides of a clean homogeneity dichotomy (Remark 18): condition (BC) compares two *scale-invariant stretch thresholds*, which are one-parameter (ν) wedge data on $\mathcal{L}_W$ (the conversion to a load ordering using only the monotone load-to-stretch map, Lemma 20), whereas the cubic sign is an *integral* with $\alpha > \frac{1}{2}$, hence tip-convergent but not tip-dominated, and remains a genuine two-parameter functional $b(\nu, \hat{K})$ (Proposition 6). In both cases the verification is thus a computation on a *universal* object — a curve or a phase diagram of the corner singularity itself — rather than a spot check on a particular body, and in three dimensions it is the same two-dimensional object, the leading edge data being carried by the zero along-edge wavenumber (Proposition 5(iii)). This is the precise content of the heuristic that the singularity “does not depend on what happens elsewhere.”

6 A spectral computational route

The blow-up reduction of §5.2 does more than clarify the theory: it dictates the *discretisation*. Because every residual quantity localises to the universal model wedge $\mathcal{L}_W$ of Proposition 5, governed by $(\nu, \hat{K})$ alone, the natural object to compute on is the wedge, not a meshed body. This section records the resulting numerical route and the validation of its two building blocks. The guiding observation is that the corner is *scale*-covariant ($r \mapsto cr$), not translation-covariant, so the transform that diagonalises it is the Mellin transform (Fourier in $\log r$), with a genuine Fourier transform reserved for the smooth edge (Proposition 5(iii)). A graded-mesh finite-element discretisation is therefore not required; a Mellin $\times$ spectral scheme converges exponentially where a mesh converges only algebraically.

6.1 The corner pencil by a Mellin $\times$ Chebyshev scheme

Substituting the separable Mellin ansatz $\mathbf{u} = r^\lambda(a(\theta), b(\theta))$ into the plane-strain Navier–Lamé equations on the wedge $\theta \in (0, \frac{\pi}{2})$ reduces the corner to an angular two-point eigenvalue problem — a quadratic eigenvalue problem in λ ,

$$(\lambda^2 \mathbb{M}_2 + \lambda \mathbb{M}_1 + \mathbb{M}_0) \begin{pmatrix} a \\ b \end{pmatrix}^\top = 0, \quad (93)$$

with the clamped conditions $a = b = 0$ at $\theta = 0$ and the traction-free conditions $\sigma_{\theta\theta} = \sigma_{r\theta} = 0$ at $\theta = \frac{\pi}{2}$. Discretising the angular operators by Chebyshev collocation and linearising (93) to a generalised eigenvalue problem yields the singular exponents as the eigenvalues with $\Re \lambda \in (0, 1)$; the smallest is the exponent $\alpha(\nu)$ of (13) and Figure 1.

This spectral computation agrees with the independent Kolosov–Muskhelishvili secular determinant of Proposition 6 to ten digits across the physical range (e.g. $\alpha(0.3) = 0.7111729$ from both), and converges *exponentially* in the number N of angular modes:

N	$ \alpha - \alpha_{\text{ref}} $ at $\nu = 0.3$
6	1.1×10^{-4}
8	9.1×10^{-7}
12	3.6×10^{-10}

Twelve angular modes already reach machine precision, a saturation no graded mesh attains; this is the concrete sense in which the corner is better resolved spectrally than by FEM. The computation is reproduced by `mellin_chebyshev_pencil.py` (validated against `williams_clampedfree.py`).

6.2 Profile nondegeneracy: the hypothesis (W)

The kinematic hypothesis (W) of Lemma 4 is a closed-form condition on the already-computed angular profile: with $\mathbf{G}(\theta) = \alpha \boldsymbol{\Phi} \otimes \mathbf{e}_r + \boldsymbol{\Phi}' \otimes \mathbf{e}_\theta$, it requires $\mu(\theta) = \det \mathbf{G} = \alpha \boldsymbol{\Phi} \times \boldsymbol{\Phi}' \geq 0$ on $[0, \frac{\pi}{2}]$ with finite-order zeros and $\text{tr } \mathbf{G} \neq 0$ at each zero. Both μ and the vanishing of $\text{tr } \mathbf{G}$ are invariant under the residual scaling of the mode, so the check is parameter-free. Evaluating it on the Kolosov–Muskhelishvili null mode (analytic θ -derivatives, verified against finite differences to 10^{-9} ; `williams_profile.W.py`) gives a sharp answer:

ν	α	$\min_{\theta} \mu / \ \mathbf{G}\ ^2$	interior zero $\theta_z/(\pi/2)$	verdict
0.10	0.86704	-1.9×10^{-2}	0.975	fails
0.25	0.74475	-3.9×10^{-3}	0.470	fails
0.35	0.67970	-2.3×10^{-4}	0.173	fails
0.40	0.64991	-1.3×10^{-7}	0.014	fails (marginally)
0.4042	0.64748	-1×10^{-14}	—	threshold ν_W
0.417	0.64012	$+3 \times 10^{-17}$	—	holds
0.49	0.59990	-4×10^{-17}	—	holds

Hypothesis (W) *holds* precisely for $\nu > \nu_W \approx 0.404$: there $\mu \geq 0$ throughout, with only the forced order-two zero at the clamped face, where $\text{tr } \mathbf{G} \neq 0$ (its normalised value decreasing from 0.23 to 0.03 as $\nu \uparrow \frac{1}{2}$). For $\nu < \nu_W$ the profile has $\mu < 0$ on the clamped-side sector $(0, \theta_z(\nu))$, with θ_z sweeping monotonically from the clamped face at ν_W to the free face as ν decreases (the whole opening is negative below $\nu \approx 0.08$); $\text{tr } \mathbf{G} \neq 0$ at every zero across the range, so the *layer* structure of Lemma 4 is as assumed — it is the *sign* clause that fails. Since the deep-field orientation $J \sim K^2 r^{2(\alpha-1)} \mu$ is independent of the load sign, $\mu < 0$ on a sector means the linear ansatz is orientation-reversing at depth there for *either* loading: below ν_W the screened-vertex conclusion of Theorem 1(A) is established only on the free-face sector, and on the clamped-side sector the inner state must leave the linear ansatz below the crossover scale (the self-consistency caveat of Remark 3). The intermediate-annulus statements, which is all the reduction of Proposition 5 uses, are unaffected. The failure is soft near the threshold ($|\min \mu|$ falls from 2×10^{-2} at $\nu = 0.05$ to 10^{-7} at $\nu = 0.40$) and absent in the near-incompressible range where the compressive computations of this section operate.

6.3 The buckling threshold $\lambda_*(\nu)$ and condition (BC)

By Lemma 20 condition (BC) reduces to a comparison of scale-invariant wedge thresholds and is set, on the loading side, by the surface-instability stretch λ_* of the compressed half-space (Assumption 12, Lemma 15). For the compressible neo-Hookean energy (7) this is computed from the incremental moduli about the homogeneous base state $\mathbf{F}_0 = \text{diag}(\lambda_{\parallel}, \lambda_{\perp})$ with the free surface in self-equilibrium, $\mathbf{P}_0 \mathbf{N} = \mathbf{0}$. A surface mode $\mathbf{u} = \mathbf{U} e^{ikx_1 + skx_2}$ decaying into the body ($\Re s > 0$) exists when the 2×2 incremental-traction determinant vanishes; since the modulus is constant in the homogeneous base state the four roots s follow from a characteristic quartic (a Stroh problem), and the secular condition is independent of the wavenumber k — the scale-free signature of creasing exploited in the limit $k \rightarrow \infty$ of Lemma 15.

The solver reproduces the classical incompressible Biot value: as $\lambda_e/\mu_e \rightarrow \infty$ (the incompressible limit, $\nu \rightarrow \frac{1}{2}$) the threshold tends to $\lambda_* = 0.5437$. For the compressible model it furnishes λ_* as a function of the material ratio, e.g.

λ_e/μ_e	ν	λ_*
1	0.250	0.514
5	0.417	0.536
50	0.490	0.543
∞	0.5	0.5437

What this computes is the *stretch* threshold $\lambda_*(\nu)$, which is the amplitude-free, ν -only wedge datum on the buckling side of (BC) (Lemma 20); the buckling load $t_c \leq t_*$ is then attained while the fundamental stretch is still $\lambda_{\parallel} = \lambda_* = \mathcal{O}(1)$, away from collapse. We stress that λ_*

is a stretch, not a load: converting it to the corner load t_* requires the global fundamental-state map $t \mapsto \lambda_{\parallel}(\mathbf{F}_0(t); P)$, which we do not compute here and which is monotone under the assumed loading. The threshold thus settles the ordering $t_c < t_{\text{coll}}$ only together with that monotone map and with a lower bound on the competing collapse load t_{coll} , which the barrier keeps high but whose quantitative tip bound is left open (Lemma 20, Remark 25). The computation is reproduced by `surface_instability.py`, validated against the incompressible Biot threshold.

6.4 The one remaining solve: the subcritical sign $b(\nu, \hat{K})$

The two preceding quantities are essentially analytic. The single genuinely numerical residue is the sign of the reduced cubic coefficient (Assumption 9), which by Proposition 6 does *not* collapse to a universal number — with $\alpha > \frac{1}{2}$ the cubic integral is tip-convergent but not tip-dominated, so $b = b(\nu, \hat{K})$ retains the stress-intensity dependence. Its evaluation is a two-dimensional transverse solve on $\mathcal{L}_W$: the critical buckling mode ϕ , the second-order Lyapunov–Schmidt field ψ solving $Q\mathcal{L}_cQ\psi = -\frac{1}{2}Q\partial^2G[\phi, \phi]$ on $V^\perp$, and the cubic integral $b = b_{\text{dir}} + b_\psi$.

We note for implementation that this last solve is the one place a plain Fourier discretisation does *not* suffice: in two dimensions the crease oscillates along the free face in the same coordinate in which the corner pre-stress is graded, so the surface direction is not translation-invariant (the two-scale competition of Remark 21). The appropriate discretisation is again Mellin $\times$ Chebyshev — Chebyshev in θ , mapped-spectral in $\log r$ — with the edge handled by the analytic Fourier reduction of Proposition 5(iii) in three dimensions. The buckling eigenproblem and the surface mechanism are exactly the two ingredients validated in §6.1–§6.3; assembling them over the $(\nu, \hat{K})$ rectangle produces the phase diagram whose sign is the last open condition. No finite-element model enters the chain: Mellin resolves the corner, Fourier the edge, and a spectral depth discretisation the buckling.

We stress that this graded solve is genuinely necessary, not a refinement: by Remark 22 the homogeneous (flat) reduction is degenerate — the second-harmonic response is resonant and b is undefined — so the grading is what regularises the coefficient.

A first version of this solve has been carried out on a *surrogate* of the wedge: a graded half-space cell, discretised by a single *consistent* weak (Galerkin) scheme — Chebyshev along the free face X , mapped Chebyshev in depth Y — with the corner pre-stress mimicked by a *linear* monotone grading $\lambda_{\parallel}(X) = \bar{\lambda}(1 + \hat{K}X/L)$ through the Biot threshold, in place of the physical $r^{\alpha-1}$ profile; the Mellin $\times$ Chebyshev wedge solve prescribed above has not yet been performed, and the results below should be read as evidence from that proxy. The critical mode ϕ is taken as the kernel of the *same* weak operator that is then inverted in the Lyapunov–Schmidt step, and the second harmonic is carried in real nodal degrees of freedom; this removes the inconsistency that made an earlier flat single-mode attempt diverge. Three facts emerge:

- (i) The homogeneous ($\hat{K} = 0$) cell carries *no* discrete critical mode — at λ_* the operator is already indefinite, the scale-free signature of Remark 22.
- (ii) The monotone grading produces a single *localised* neutral mode with a discrete spectral gap to the rest of the spectrum ($\approx 4\text{--}5 \times 10^{-2}$ across $\hat{K} \in [0.05, 0.30]$ at the working resolution), so on the discrete cell the second-order problem $Q\mathcal{L}_cQ$ is non-resonant and ψ — and hence b — is well defined. The gap, however, *decreases* under refinement (table below), so the isolation of the kernel in the continuum limit — and with it the convergence of the reduced coefficients — is not established by this computation.

- (iii) Decomposing $b = b_{E_4} + b_{fb}$ into the (adverse, positive) frozen quartic and the Lyapunov–Schmidt feedback, the *normalisation-invariant* ratio b_{fb}/b_{E_4} — both terms scale as $\|\phi\|^4$, so the ratio is independent of the arbitrary mode amplitude — lies between -1.9 and -2.5 at every tested resolution: below -1 in each case, though oscillating rather than settling to a converged value. At the tested resolutions the feedback *overwhelms* the adverse frozen term, giving $b < 0$.

The status of the two terms is asymmetric, and the computation should be read accordingly: $b_{E_4} > 0$ and $b_{fb} \leq 0$ are *theorems* (Lemma 16(ii) and Lemma 11 — the solver cannot find, and never finds, a positive feedback), so the tables below test only the *magnitude* criterion $r < -1$. The diagnostic ratio $r := b_{fb}/b_{E_4}$ is *normalisation-invariant* (both terms scale as $\|\phi\|^4$) and decides the sign by $b < 0 \iff r < -1$. Its behaviour under refinement of the angular (N) and depth (N_X) resolutions is shown below; r stays below -1 throughout but does not settle, and the discrete spectral gap shrinks steadily:

N_X	N	ratio r	sign(b)	spectral gap
10	24	-2.092	$b < 0$	5.8×10^{-2}
12	28	-2.452	$b < 0$	4.5×10^{-2}
14	32	-2.426	$b < 0$	3.5×10^{-2}
16	36	-2.212	$b < 0$	2.8×10^{-2}

(at $\nu = 0.417$, $\hat{K} = 0.15$; r stays below -1 at every resolution but oscillates between -2.09 and -2.45 , and the concurrent decay of the gap warns that the surrogate cell may lack a well-defined continuum limit). Sweeping the physical rectangle at fixed resolution, the ratio is below -1 — hence $b < 0$ — throughout $(\nu, \hat{K})$:

$\nu \backslash \hat{K}$	0.10	0.20	0.30
0.250	-3.19	-2.43	-2.44
0.333	-2.70	-2.14	-2.37
0.417	-2.17	-1.86	-2.19
0.455	-1.82	-1.65	-1.94

The margin to the threshold $r = -1$ is smallest near the incompressible corner ($\nu \rightarrow \frac{1}{2}$, $\hat{K} \approx 0.2$), where $r \approx -1.65$, and is never breached on the physical rectangle. One further robustness check cuts the margin rather than padding it: replacing the ad-hoc linear ramp by the *physical* power profile $\lambda_{\parallel} \sim 1 + \hat{K} r^{\alpha-1}$ of the edge field thins the ratio to $r \approx -1.5$ at $\nu = 0.417$ (against -2.5 for the linear ramp at the same point), still below -1 but markedly closer to it — the nearer the grading is to the physical profile, the thinner the subcritical margin, which is one more reason the genuine wedge solve, not the surrogate, must be the final arbiter. On this evidence the cubic clause of Assumption 9 holds and the corner behaves as the genuine crease-analogue; but the evidence comes from the surrogate cell and has not converged, so we treat it as strong support rather than verification. The quintic clause $d > 0$ is not tested by this computation at all and remains fully open (Remark 12).

The inversion-free certification criterion of Corollary 1 has now been evaluated on the same discretisation (`gamma.certificate.py`), and it upgrades the table above in two ways. First, since the assembled load vector represents the quadratic self-coupling, every conjugate-gradient iterate of it is an *explicit* test field — a polynomial in the operator applied to the source, no bordered matrix, no near-singular solve — and by Krylov optimality the certified value Γ_k of (79) increases

monotonically in k , each k yielding the rigorous bound $b \leq b_{\text{dir}}(1 - \Gamma_k)$. On the full rectangle the certificate fires at between 66 and 184 iterations, reaching $\Gamma \in [1.31, 3.01]$ by $k = 1000$: *the sign $b < 0$ of the discrete cell is certified at every tested $(\nu, \hat{K})$ without inverting the reduced operator.* Second, at the reference point the deep iteration converges to the bordered-solve value itself ($\Gamma_k \rightarrow 2.12$ against $-r = 2.17$ at $k = 2000$), independently validating the assembled ratios of the table against their own $\text{cond}(B) \sim 10^7$ – 10^8 pathology. Under refinement the certificate survives — the crossing iteration grows ($127 \rightarrow 324$ from the coarsest to the finest grid) as the shrinking gap slows the Krylov convergence, but Γ still exceeds 1.8 at depth — with the standing caveat that all of this certifies the *discrete* b of the surrogate cell, at each resolution, and cannot repair the unsettled continuum limit noted above. One honest negative completes the picture: the flat-degeneracy amplification mechanism of Proposition 7 is *not operative on this cell* — the excited mode never goes neutral as $\hat{K} \rightarrow 0$ (its Rayleigh quotient stays near 0.09, precisely because the monotone grading was designed to lift the flat resonance) and its coupling to the source is small and erratic — so hypothesis (b) of that proposition remains untested; probing it requires the periodic or genuine-wedge geometry in which the doubled-wavenumber harmonic exists. The b -computation itself is reproduced by `graded_buckling.py`, which reuses the validated linear building blocks of §6.1–§6.3 (`williams_clampedfree.py` / `mellin_chebyshev_pencil.py` for α ; `surface_instability.py` for λ_*).

As a byproduct, the same critical mode furnishes a partial check of Assumption 10 (Remark 14): the neutral co-energy $\langle \phi, \mathcal{L}\phi \rangle \approx 10^{-8}$ confirms the mode is a genuine kernel, and the dimensionless rotation defect $\|\delta J\|/\|\nabla\phi\| \approx 0.08$ – 0.16 (stable under deepening compression) confirms it is nearly volume-preserving. The decomposition of (64) shows the singular determinant term to be 6–10 $\times$ the $f''(\delta J)^2$ term, so the co-energy is bounded by cancellation against the shear term, not by the f'' term alone — the honest mechanism behind the relief estimate. This supports the qualitative volume-preserving property; the precise asymptotic rate $\delta J = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ as $J_0 \rightarrow 0$ is not reached by the accessible base states and remains an assumption.

7 Conclusions

We have analysed a compressible hyperelastic corner under mixed Dirichlet–Neumann boundary conditions and shown that the conditioning collapse first observed numerically is, in its essence, an analytical phenomenon: a corner-localised *creasing* bifurcation seated at the geometric junction, of which the divergence of the linearised tangent is a symptom rather than the substance.

The central finding is a dichotomy in the asymptotic behaviour of the linearised tangent operator, governed by the loading direction and the volumetric penalty. In tension the logarithmic barrier $f_{\log}$ yields a *benign* singularity: the tangent stays strongly elliptic and Fredholm, the standard Kondratiev edge exponents survive as the intermediate asymptotics outside the nonlinear crossover scale — at the vertex itself the extreme stretch screens the volumetric coupling and the tangent stabilises to a decoupled Laplacian with no singular exponent at all — and the well-posedness of the linearised step holds in the weighted corner spaces (Theorem 1). Under compression the local volume contraction drives an algebraic blow-up of the operator norm. We proved that this breakdown is a *structural bifurcation*, not a material loss of ellipticity (Lemma 6): the corner remains pointwise strongly elliptic throughout, while a localised surface instability preempts pointwise collapse. The apparent tangent catastrophe is, on every physical branch, regularised by the kinematic relief of the buckled mode (Assumption 10, Lemma 12); a load-controlled body snaps to a finite-amplitude crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$, and only an algorithm that suppresses the snap and tracks the unstable continuation sees the $\epsilon^{-4/3}$ conditioning degradation, confined to a tube of vanishing

area and integrable in L^q for every $q < \frac{1}{2}$ (Theorem 2, Proposition 4).

A tangent-cone blow-up reduction localises the entire phenomenon to a *universal* model wedge, governed only by the Poisson ratio ν and a single normalised stress-intensity amplitude $\hat{K}$ (Proposition 5). Against this backdrop the imperfect-bifurcation analysis is sharp on two counts and honest about a third. The quadratic energy coefficient vanishes *exactly* by reflection symmetry, with no Lyapunov–Schmidt correction (the parity argument of §5.6). The subcritical cubic coefficient is tip-convergent yet does *not* localise: it retains an explicit dependence on the global stress-intensity factor, a direct consequence of the characteristic edge exponent satisfying $\alpha > \frac{1}{2}$. The two surviving hypotheses — buckling-precedes-collapse (BC) and the subcritical sign of b — sit on opposite sides of this blow-up dichotomy (Remark 26): the former compares amplitude-free, ν -only *stretch* thresholds and is settled by a single wedge computation, while the latter is an *integral* whose sign is an $\mathcal{O}(1)$ geometric competition decided per configuration. The scale-covariant Mellin–Chebyshev spectral solver resolves the singular exponents and buckling thresholds with exponential accuracy and supplies the geometry-dependent diagnostics — the surface-instability stretch $\lambda_\star(\nu)$, the rotation defect, and the sign of b — that the analysis isolates but cannot decide in closed form.

What this concrete realisation buys is a sharp reduction of hypotheses. Fixing the clamped–free neo-Hookean material and geometry discharges, one by one, the spectral postulates of the abstract theory into theorems (the ledger of Table 1): the material bounds become explicit, the critical load is constructed from a localised surface instability (Lemma 15), mode attainment is *proved* under (BC) (Lemma 18), simplicity is reduced from an outright postulate to genericity within the symmetry sector carrying the mode, the quadratic coefficient vanishes by symmetry (Lemma 16), and the exact first-order volume preservation once posited by the relief is weakened to the asymptotic-rotation hypothesis, automatic at the bifurcation (Lemma 13) and numerically supported toward collapse. What is left after this reduction is not a residue of opaque spectral assumptions but exactly two *concrete*, numerically checkable conditions — buckling precedes pointwise collapse, and the reduced cubic is subcritical. Crucially, these are the *same* two conditions that the classical creasing problem on a flat surface also leaves to computation [?, ?]: they are inherited here, not manufactured by the corner geometry, which adds the singular pre-stress and the localisation but no new open hypothesis.

Both surviving conditions are, in practice, decided by the spectral computation of §6. The sign of the cubic coefficient is the more complete of the two: the graded-cell solve of §6.4 returns $b < 0$ throughout the tested $(\nu, \hat{K})$ rectangle — the normalisation-invariant feedback ratio stays below its critical value -1 at every tested resolution, with the smallest margin near the incompressible corner, and the inversion-free Γ certificate of Corollary 1 now confirms the sign at every rectangle point without the ill-conditioned reduced solve — so the cubic clause of subcriticality is numerically supported and internally cross-validated, though on a surrogate cell whose refinement behaviour is not yet settled; the quintic clause $d > 0$ remains untested. What remains genuinely beyond reach here is analytic rather than numerical: a *closed-form, configuration-free* criterion for sign b , which the homogeneity count of Proposition 6 shows cannot exist in the localised form available for the stretch thresholds, precisely because $\alpha > \frac{1}{2}$ keeps the cubic integral globally $\hat{K}$ -dependent. The loading side of (BC) is likewise computed: §6.3 delivers the amplitude-free stretch threshold $\lambda_\star(\nu)$ to spectral accuracy. The one item left open as a theorem is the inner vertex theory for the logarithmic barrier (Remark 4): away from the tip the model-wedge operator and its corner pencil are pinned to constant-coefficient linear elasticity, but a quantitative, uniform-to-the-tip spectral estimate — and with it the lower bound on the barrier-regularised collapse load t_{coll} that would turn the computed stretch ordering into the load ordering $t_c < t_{\text{coll}}$ — is the irreducible analytic core of (BC). Beyond this, the natural continuations are the finite-amplitude creased state past the snap, the extension from the model wedge to fully three-dimensional edges with curvature, and the

analogous dichotomy for other volumetric penalties, where the same competition between barrier growth and volumetric contraction should organise the asymptotics.